HKUSPACE
香港大學專業進修學院
HKU School of Professional and Continuing Education

Business Process Automation with VBA and Python

Mr. Eddie Chow / 7 November 2026

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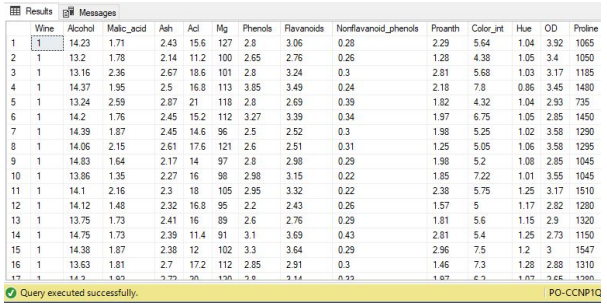
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Concept of Structured, Unstructured Data

Structured Data

- Fixed schema (rows/columns, E.g. Tables, CSV, Excel)
- Easy to query/search (names, prices)
- Use VBA, SQL or ML techniques



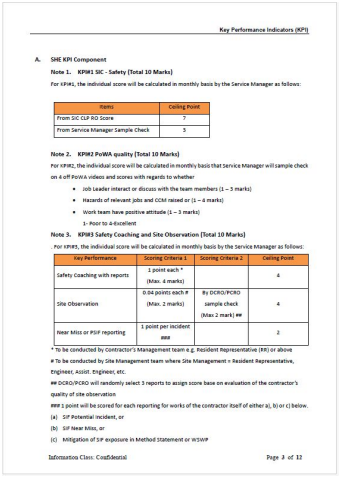
Results | Messages

	Wine	Alcohol	Malic_acid	Ash	Acl	Mg	Phenols	Flavanoids	Nonflavanoid_phenols	Proanth	Color_int	Hue	OD	Proline
1	1	14.23	1.71	2.43	15.6	127	2.8	3.06	0.28	2.29	5.64	1.04	3.92	1065
2	1	13.2	1.78	2.14	11.2	100	2.65	2.76	0.26	1.28	4.38	1.05	3.4	1050
3	1	13.16	2.36	2.67	18.6	101	2.8	3.24	0.3	2.81	5.68	1.03	3.17	1185
4	1	14.37	1.95	2.5	16.8	113	3.85	3.49	0.24	2.18	7.8	0.86	3.45	1480
5	1	13.24	2.59	2.87	21	118	2.8	2.69	0.39	1.82	4.32	1.04	2.93	735
6	1	14.2	1.76	2.45	15.2	112	3.27	3.39	0.34	1.97	6.75	1.05	2.85	1450
7	1	14.39	1.87	2.45	14.6	96	2.5	2.52	0.3	1.98	5.25	1.02	3.58	1290
8	1	14.06	2.15	2.61	17.6	121	2.6	2.51	0.31	1.25	5.05	1.06	3.58	1295
9	1	14.83	1.64	2.17	14	97	2.8	2.98	0.29	1.98	5.2	1.08	2.85	1045
10	1	13.86	1.35	2.27	16	98	2.98	3.15	0.22	1.85	7.22	1.01	3.55	1045
11	1	14.1	2.16	2.3	18	105	2.95	3.32	0.22	2.38	5.75	1.25	3.17	1510
12	1	14.12	1.48	2.32	16.8	95	2.2	2.43	0.26	1.57	5	1.17	2.82	1280
13	1	13.75	1.73	2.41	16	89	2.6	2.76	0.29	1.81	5.6	1.15	2.9	1320
14	1	14.75	1.73	2.39	11.4	91	3.1	3.69	0.43	2.81	5.4	1.25	2.73	1150
15	1	14.38	1.87	2.38	12	102	3.3	3.64	0.29	2.96	7.5	1.2	3	1547
16	1	13.63	1.81	2.7	17.2	112	2.85	2.91	0.3	1.46	7.3	1.28	2.88	1310

Query executed successfully. PO-CCNP1C

Unstructured Data

- No fixed format (text, images, videos, documents)
- Needs deep learning vs NLP/ML techniques
- Or use LLM / Foundation Model



Key Performance Indicators (KPI)

A. SHE KPI Component

Note 1: KPIM1 SIC - Safety (Total 10 Marks)

For KPIM1, the individual score will be calculated in monthly basis by the Service Manager as follows:

Item	Ceiling Point
From SIC CUP RO Score	7
From Service Manager Sample Check	3

Note 2: KPIM2 PoWA quality (Total 10 Marks)

For KPIM2, the individual score will be calculated in monthly basis that Service Manager will sample check on 4 off PoWA videos and scores with regards to whether

- Job Leader interact or discuss with the team members (1 – 3 marks)
- Hazards of relevant jobs and CCM raised or (1 – 4 marks)
- Work team have positive attitude (1 – 3 marks)

1- Poor to 4-Excellent

Note 3: KPIM3 Safety Coaching and Site Observation (Total 10 Marks)

For KPIM3, the individual score will be calculated in monthly basis by the Service Manager as follows:

Key Performance	Scoring Criteria 1	Scoring Criteria 2	Ceiling Point
Safety Coaching with reports	1 point each * (Max. 4 marks)		4
Site Observation	0.04 points each # (Max. 2 marks)	By DCRO/PCRO sample check (Max 2 mark) ##	4
Near Miss or PSIF reporting	1 point per incident ###		2

* To be conducted by Contractor's Management team e.g. Resident Representative (RR) or above

To be conducted by Site Management team where Site Management = Resident Representative, Engineer, Asstd. Engineer, etc.

DCRO/PCRO will randomly select 3 reports to assign score base on evaluation of the contractor's quality of site observation

1 point will be scored for each reporting for works of the contractor itself of either a), b) or c) below.

(a) SIF Potential Incident, or

(b) SIF Near Miss, or

(c) Mitigation of SIF exposure in Method Statement or WSWP

Information Class: Confidential Page 3 of 12



Concept of Structured, Unstructured, Discrete, Continuous Data

Structured Data

- Fixed schema (rows/columns, E.g. Tables, CSV, Excel, SQL)
- Easy to query/search (names, prices)
- Use VBA, SQL or traditional ML techniques (XGBoost, Random Forests techniques)

Unstructured Data

- No fixed format (text, images, videos, documents)
- needs deep learning vs NLP/ML techniques
- Or use LLM / Foundation Model

Discrete Data

- Countable, distinct values (e.g., 0,1,2 items)
- needs traditional ML classification models like Logistic Regression or Decision Trees

Continuous Data

- Any value in range (e.g., 1.75m height)
- No gaps; line/histograms
- needs traditional ML regression models such as Linear Regression or Gradient Boosting

Data Processing & Analysis
Practise 2

Which Data Type is Structured/Unstructured, Discrete/Continuous?

E-mails	
Digital Images	
Stock Market Logs	
Historical Gold Prices	
Credit Approval Records	
Social Media Friend Relationships	
Tweets & Trending Topics	
Sales Records	

Source: Practical Data Analysis - Second Edition Hector Cuesta, Dr. Sampath Kumar OACKT Publishing BIRMINGHAM - MUMBAI

What is LLM/SLM/Foundation Model?

Large Language Models (LLMs)

LLMs are designed for complex natural-language processing tasks, including translation, summarization, question answering, and conversational reasoning. They are trained on extensive text corpora, enabling them to understand context and generate fluent, human-like text.

Small Language Models (SLMs)

SLMs prioritize efficiency through compact architectures and optimized attention mechanisms, such as sparse or grouped-query attention, to reduce computational and memory requirements. This makes them well suited for edge devices, on-premises systems, and other resource-constrained environments while retaining strong task performance.

Foundation Models

Foundation models act as reusable building blocks for a broad range of AI applications across text, vision, speech, and multimodal domains. They are pretrained on large, diverse datasets and can then be adapted through prompting, fine-tuning, or lightweight adaptation methods for domain-specific tasks.

What is LLM/SLM/Foundation Model?

Foundation Model & Embedding Model

LLM Foundation & Vector Embeddings — Explained

What is a Foundation Model?

- -Large-scale pre-trained model learned from massive diverse data (text, code, images, etc.)
- Serves as the general-purpose foundation for many AI tasks
- Example: Large Language Model (LLM) understands & generates human-like text
- Capabilities: reasoning, summarization, Q&A, content generation, code

What is an Embedding Model?

- Converts text (or other data) into numerical vectors/embeddings
- Each embedding is a high-dimensional vector capturing semantic meaning
- Used for similarity search, clustering, retrieval (RAG), recommendations
- Example: "cat' and "kitten" have similar vectors, close in vector space

How Text Becomes Vectors — LLM Foundation & Embeddings Workflow

Input Text
"The cat sat on the mat"

→

LLM Foundation / Tokenization
Text is tokenized into tokens and processed by the Foundation Model

→

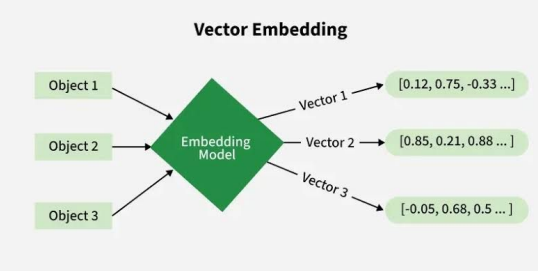
Embedding Model
Transforms tokens into dense numerical vector representations

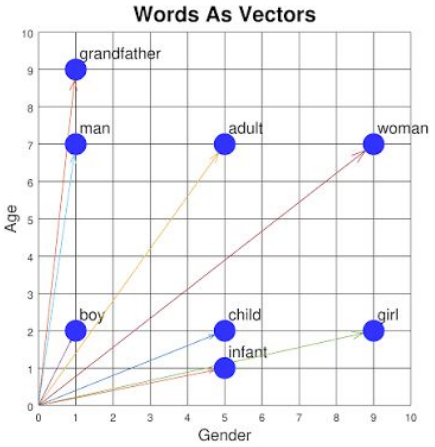
→

Vector Embedding
[0.12, -0.45, 0.78, ... 0.33]
Vectors stored in Vector Database / Vector Space

Key Idea: Foundation Models provide broad intelligence/understanding, while Embedding Models map text into vectors to enable semantic search & retrieval.

Applications: RAG • Semantic Search • Chatbots • Recommendation Systems • Clustering





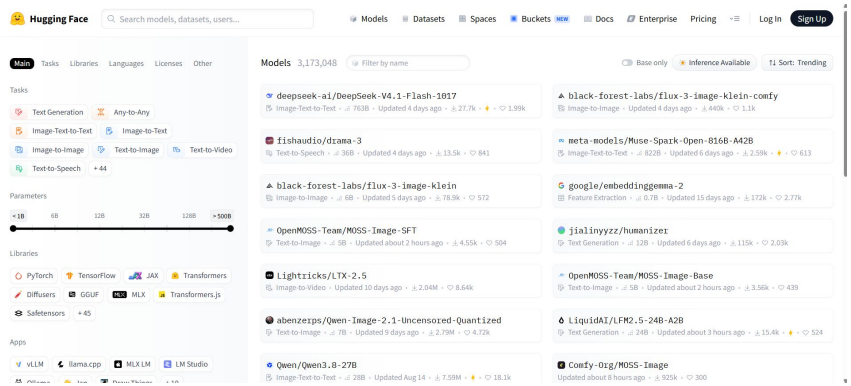
LLM Model List

Hugging Face

Open Source Developers can upload their own LLM Models.
Website: https://huggingface.co/models

Category:

1. Computer Vision
2. Text Generation

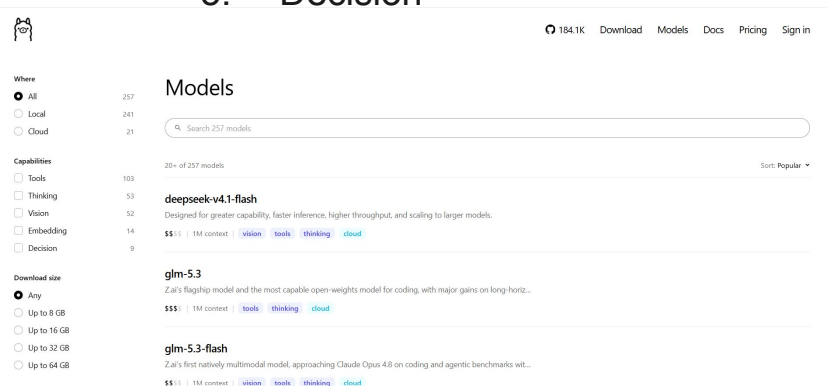


Ollama

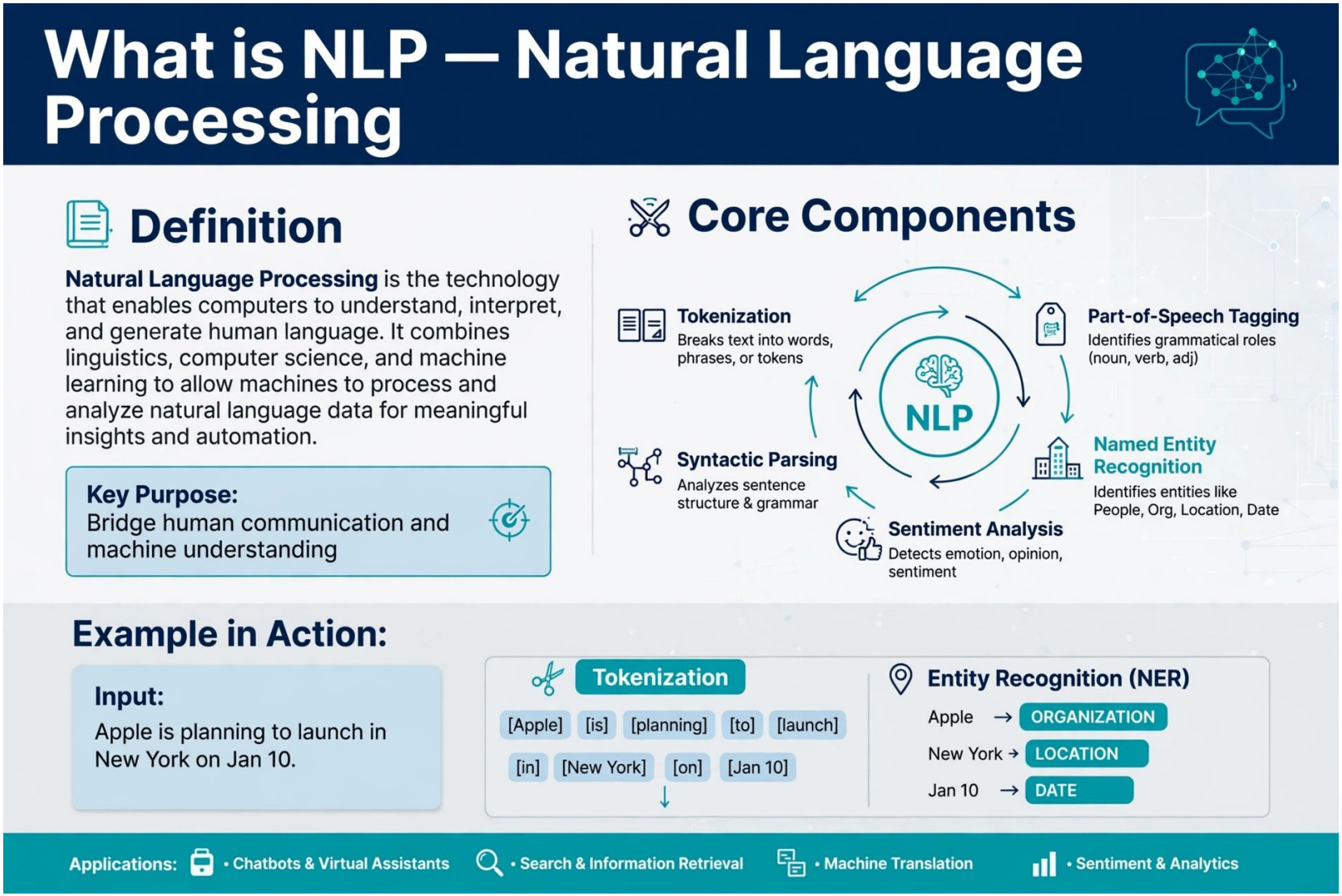
Developers can directly use locally.
Website: https://ollama.com/search

Category:

1. Tools
2. Thinking
3. Vision
4. Embedding
5. Decision



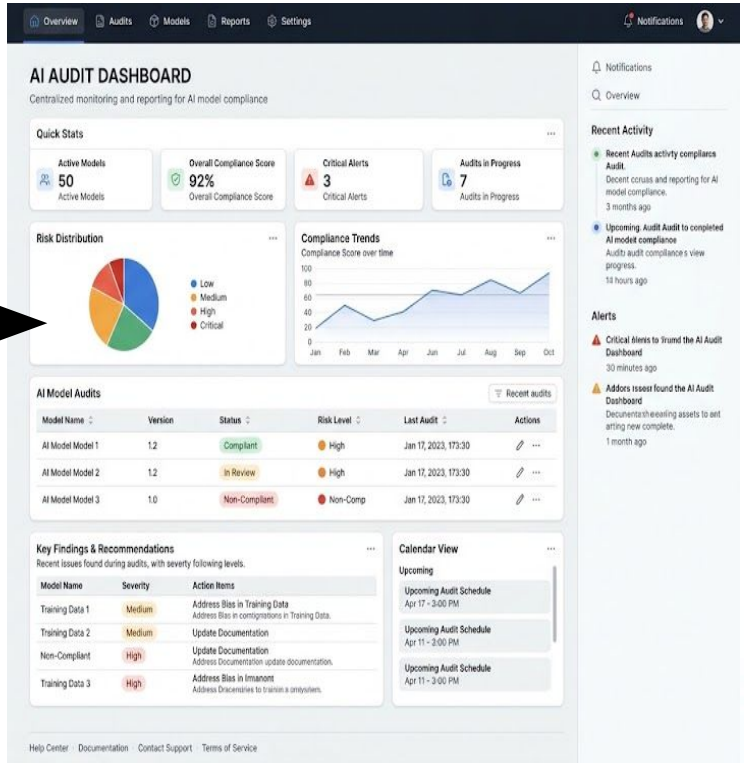
What is NLP(Natural Language Processing)?



Data-Driven Business Strategy

A **data-driven business strategy** is a systematic plan that guides business decisions, operational improvements, and growth using verified data analysis rather than habit or intuition.





Data-Driven vs Scenario-Driven Decision Making

Choosing the right approach for planning, execution, and risk management.

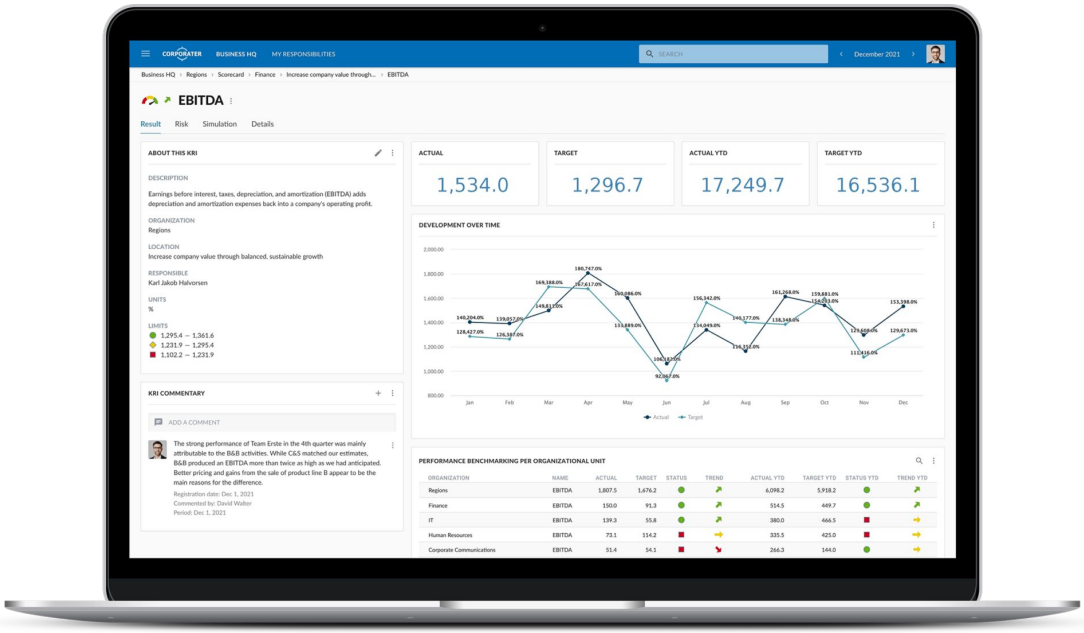
Aspect	Data-driven	Scenario-driven
Starting point	Observed data (logs, metrics, events)	Hypothetical or expected situations / “stories”
Main driver	Statistical patterns and KPIs	Business or operational logic of a situation
Best for	Optimization, performance tracking, and evidence-based management.	Strategic planning, experimentation, and uncertain environments.
Typical questions	“What is happening now?” What does the data predict?”	“What could happen?” “What should we do if it does?”

Use Case - Investment portfolio monitoring

Portfolio monitoring is used by VC, PE, hedge fund, wealth, and asset-management teams to follow performance, health, and risk after investment, while AI/ML helps analyze large financial datasets, detect patterns, and improve risk prediction.

Typical use case

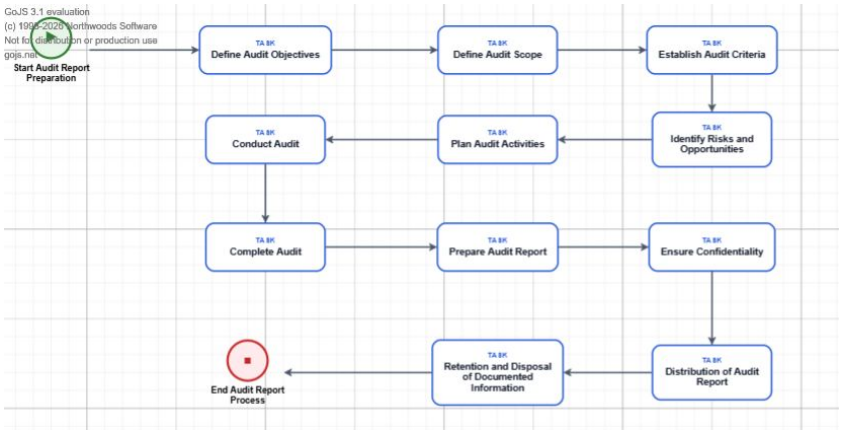
A portfolio monitoring system ingests holdings, prices, cash flows, sector exposure, and company KPIs, then checks them against policy targets and benchmarks. It can automatically flag issues like concentration risk, style drift, liquidity problems, covenant breaches, or underperforming positions.

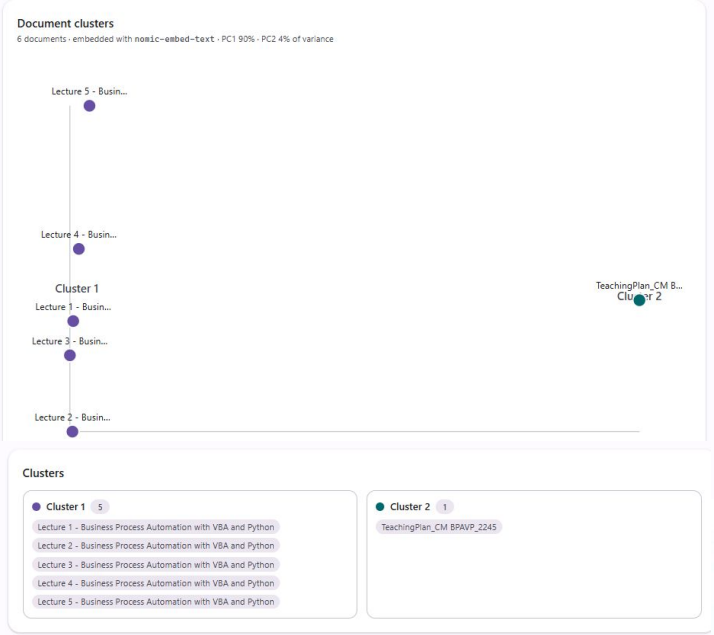


Use Case - Investment portfolio monitoring

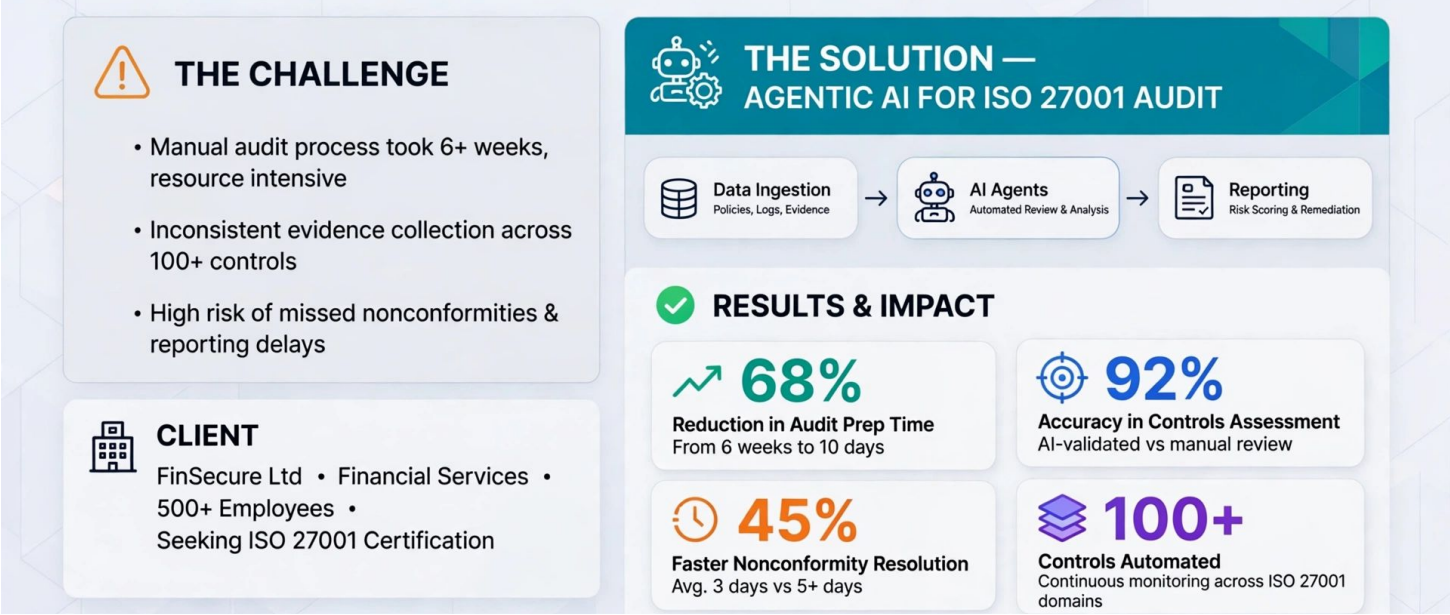
How ML and algorithms work together

- Algorithms handle deterministic logic: threshold alerts, rebalancing rules, benchmark comparisons, and risk-limit checks.
- Machine learning handles pattern detection: anomaly detection, predictive risk signals, and scoring which holdings are likely to deteriorate or outperform.
- The output is usually an alert, dashboard, or action recommendation for an investor, analyst, or portfolio manager.



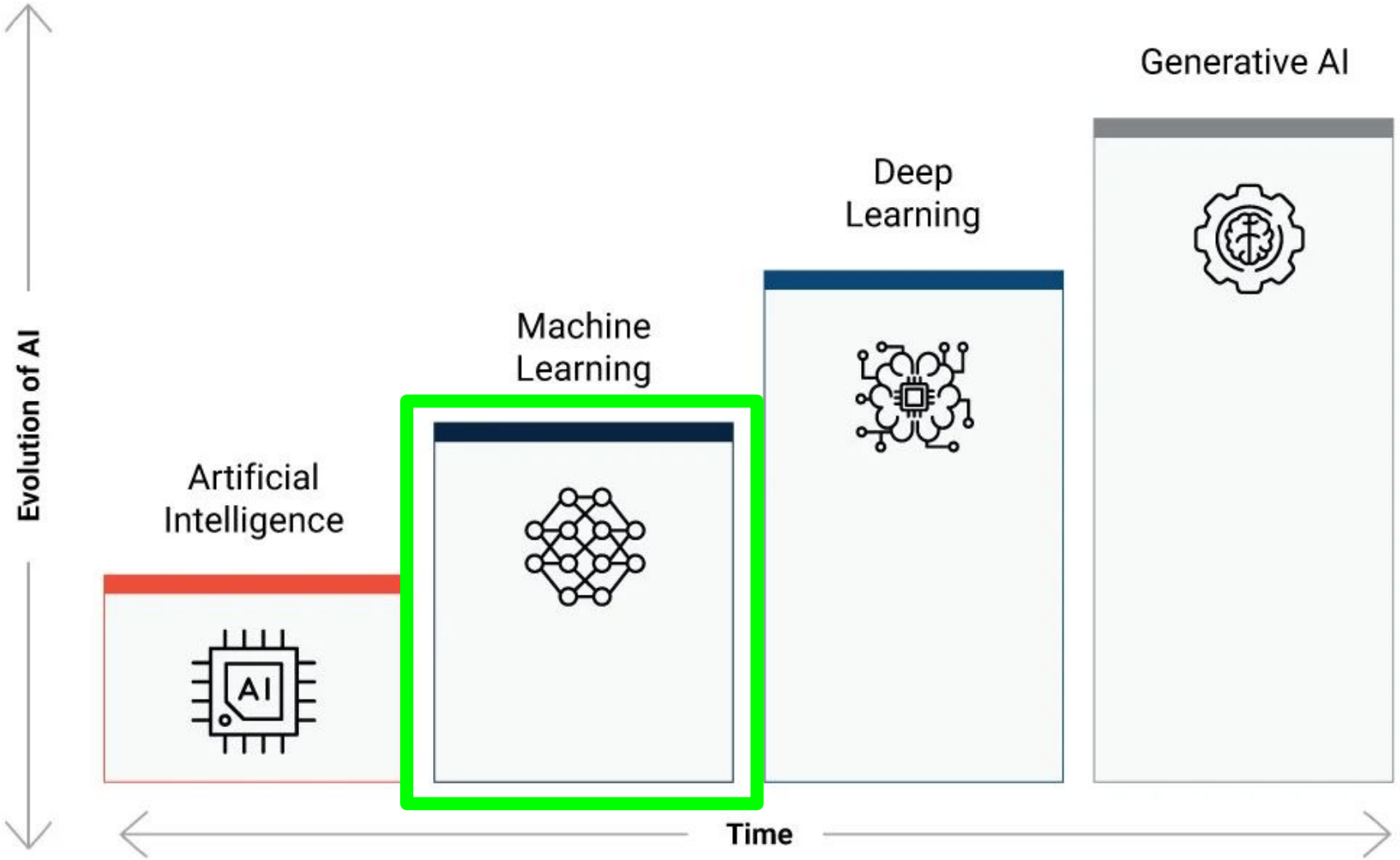


Use Case - ISO Audit with Agentic AI



Finding ID	Classification	Risk Category (Q/S/E)	Process Step	Activity	Hazard/Failure Mode	S (1-5)	L (1-5)	RPS (SxL)	Risk Level (per Matrix)
BP-01	Best Practice (Conformity)	Q	3.4	Continuous Thermal Envelope Surveillance	Chiller drift / loss of control (prevented by design)	5	1	5	Low
BP-02	Best Practice (Conformity)	Q	3.1-3.6	End-to-end process	Incomplete evidence / traceability	4	1	4	Low
OFI-01	Opportunity for Improvement	Q	3.4-3.5	Long dwell monitoring	Slow drift / delayed trend detection	4	2	8	Medium
OFI-02	Opportunity for Improvement	S	3.2	Transfer operations	Cold stress during door-open peaks	3	2	6	Medium
NC-IMPLIED-01	No NC Identified (Sampled)	Q/S/E	All	Audited scope	N/A	1	1	1	Low

Overall Artificial Intelligence Journey



Machine Learning Techniques

Machine Learning

An approach to achieve artificial intelligence through '*software*' systems that can ***learn*** from ***experience*** to find ***patterns*** in a ***set of data***.

Write a computer program with **explicit rules** to follow

```
if email contains V!agrå
  then mark is-spam;
if email contains …
if email contains …
```

Traditional Programming

Write a computer program to **learn from examples**

```
try to classify some emails;
change self to reduce errors;
repeat;
```

Machine Learning Programs

Data Processing & Analysis - What Is Machine Learning?

Supervised Learning	Unsupervised Learning
Used as an advanced form of *predictive modeling*	Used either as a form of automated data analysis or automated signal extraction
Each observation must be labeled with a "correct answer"	Unlabeled data has no predetermined "correct answer"
Only then can you build a predictive model because you must tell the algorithm what's "correct" while training it (hence, "supervising" it)	You'll allow the algorithm to directly learn patterns from the data (without "supervision")
Regression is the task for modeling continuous target variables	Clustering is the most common unsupervised learning task, and it's for *finding groups* within your data
Classification is the task for modeling categorical (a.k.a. "class") target variables	





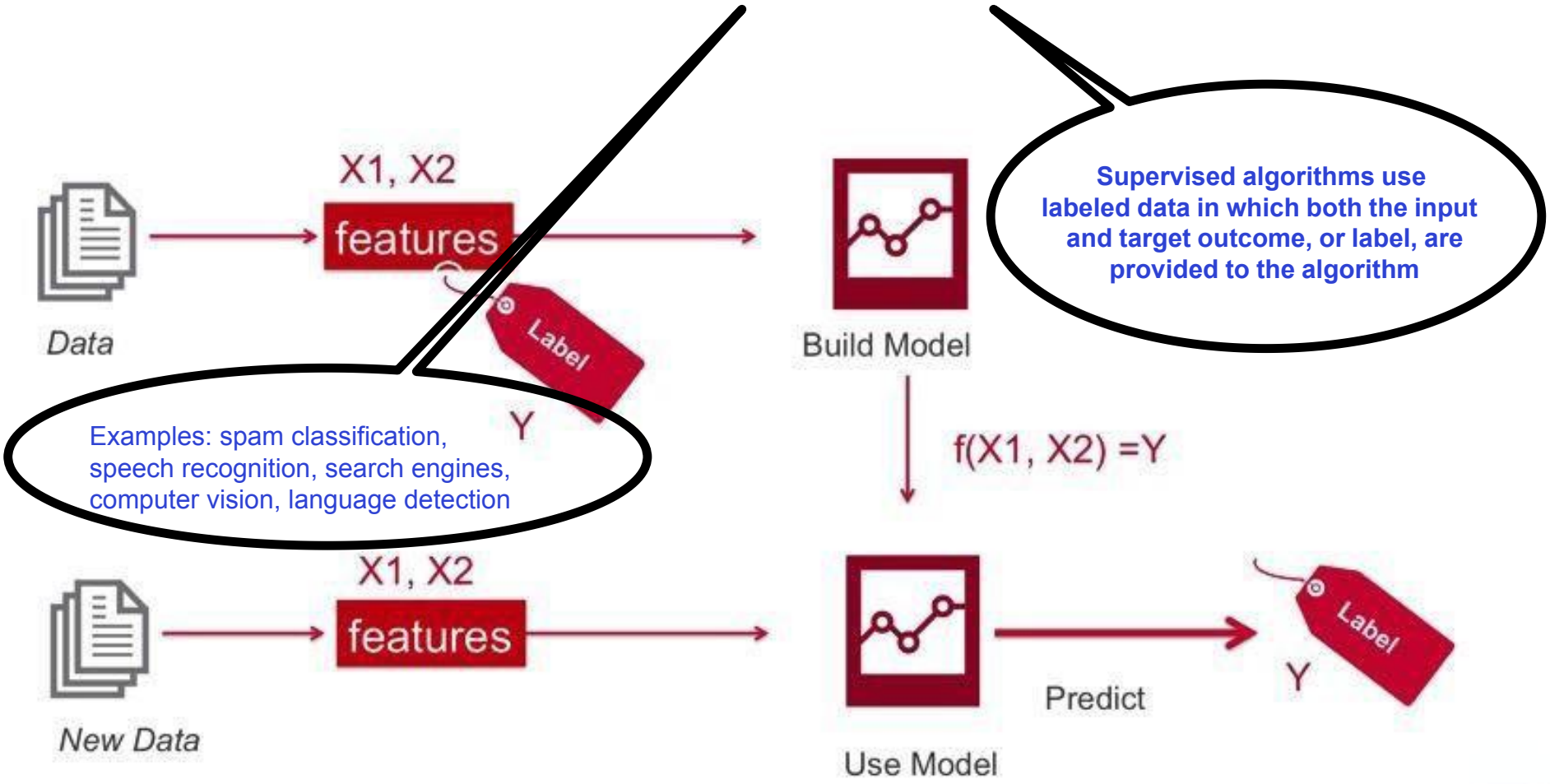
Source: https://dzone.com/articles/demystifying-ai-machine-learning-and-deep-learning

Data Processing & Analysis
Supervised Learning: Classification For Debit Card Fraud



Source: https://dzone.com/articles/demystifying-ai-machine-learning-and-deep-learning

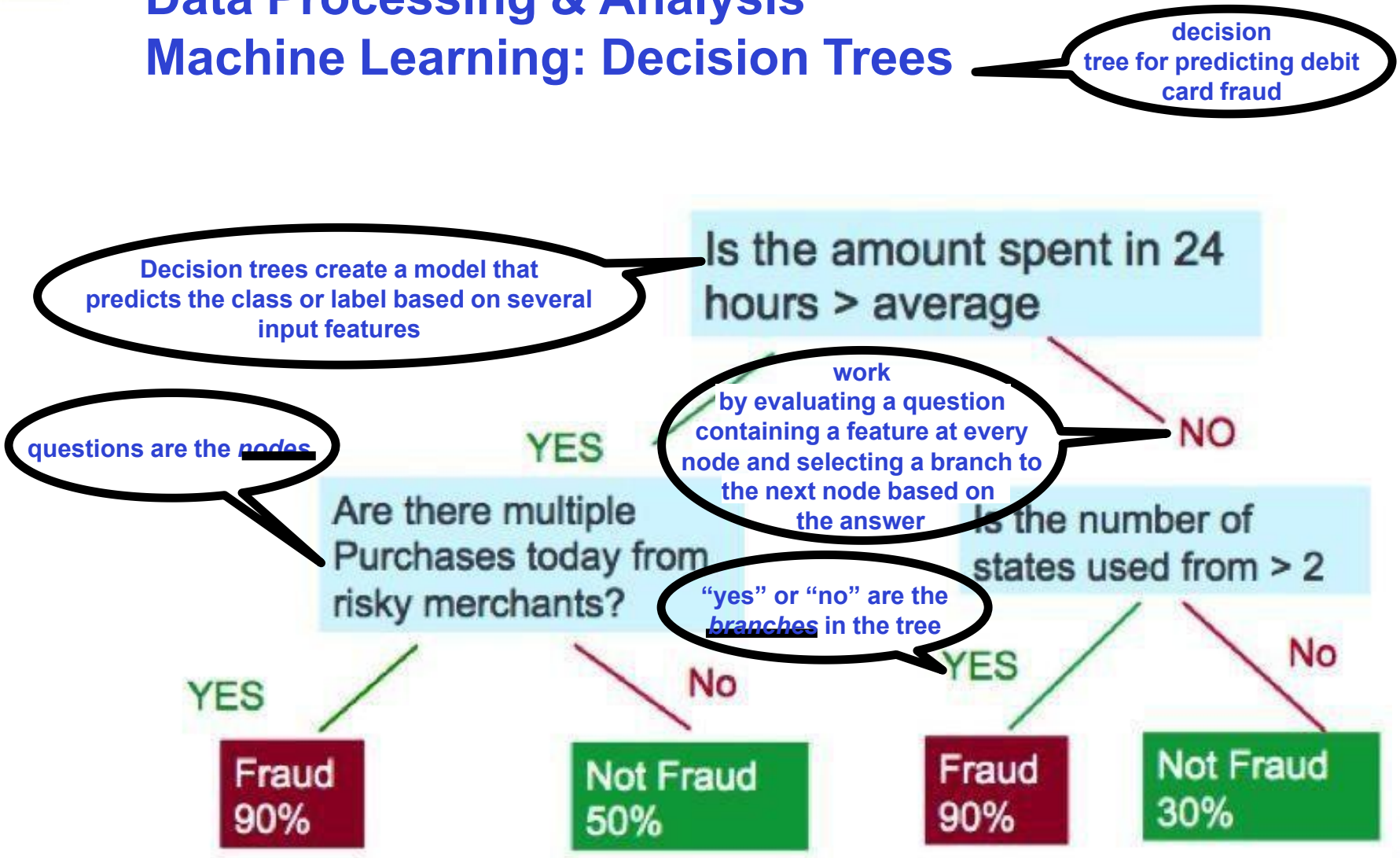
Data Processing & Analysis
Machine Learning: Supervised Learning



Source: https://dzone.com/articles/demystifying-ai-machine-learning-and-deep-learning

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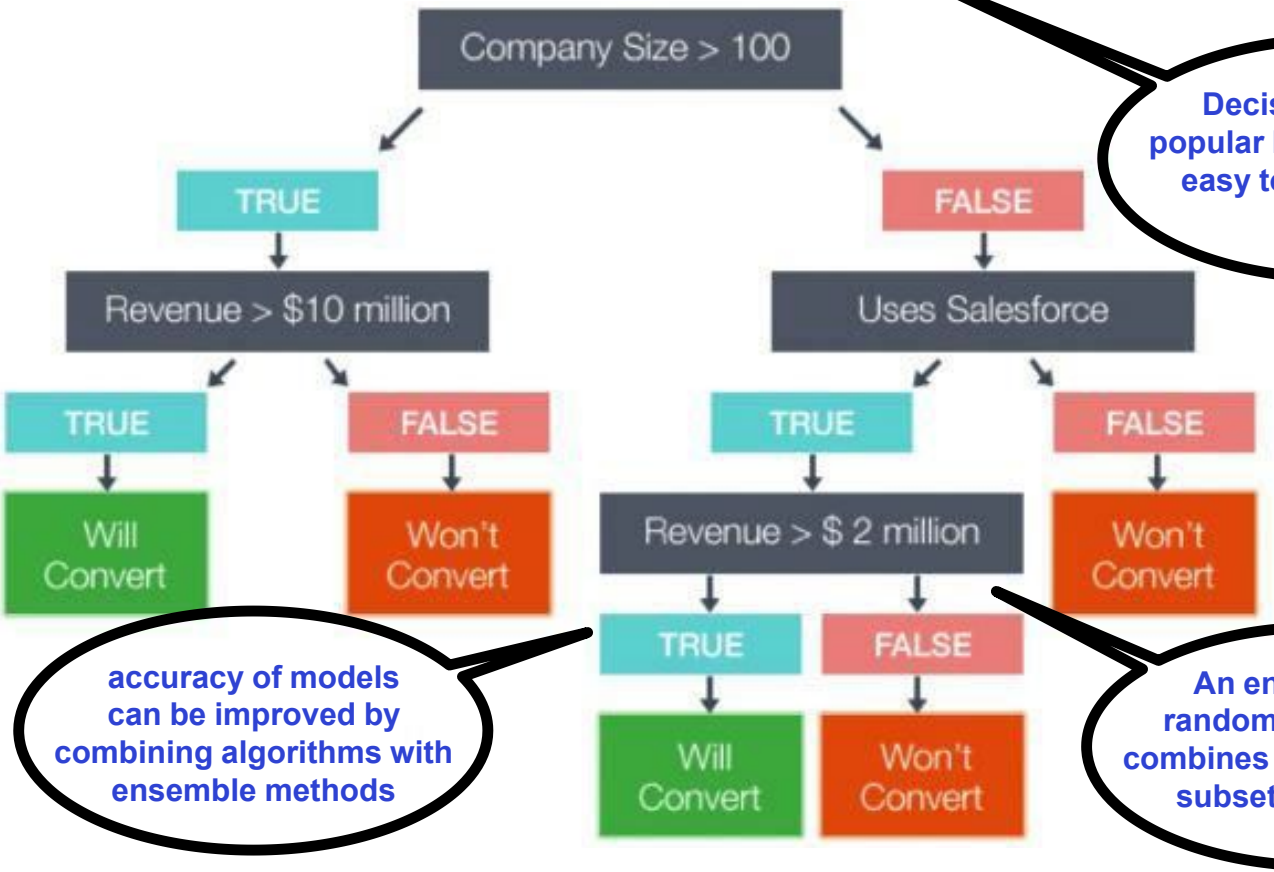
Data Processing & Analysis
Machine Learning: Decision Trees



Source: https://dzone.com/articles/demystifying-ai-machine-learning-and-deep-learning

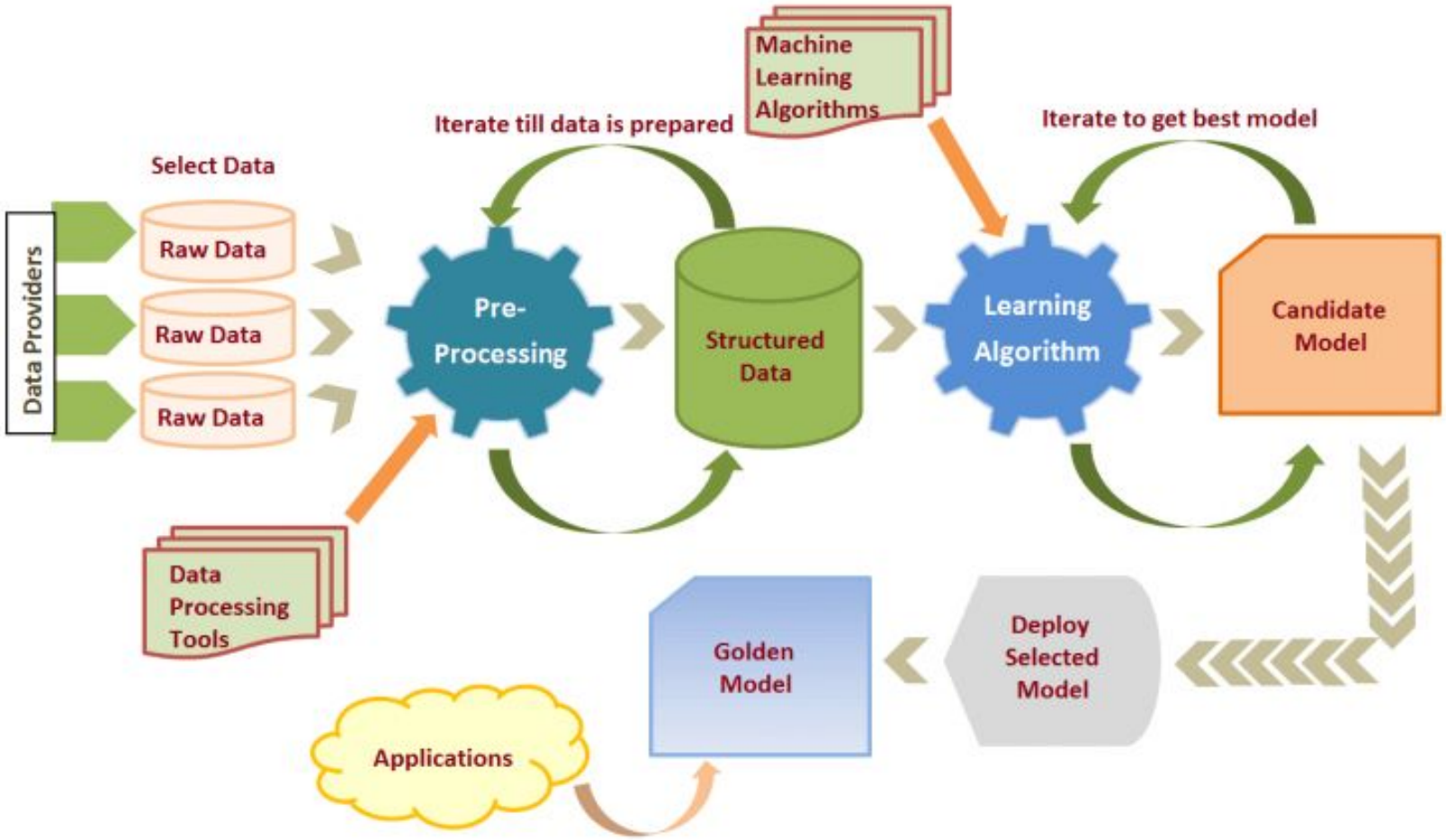
Data Processing & Analysis

Decision Trees: Ensemble Random Forests

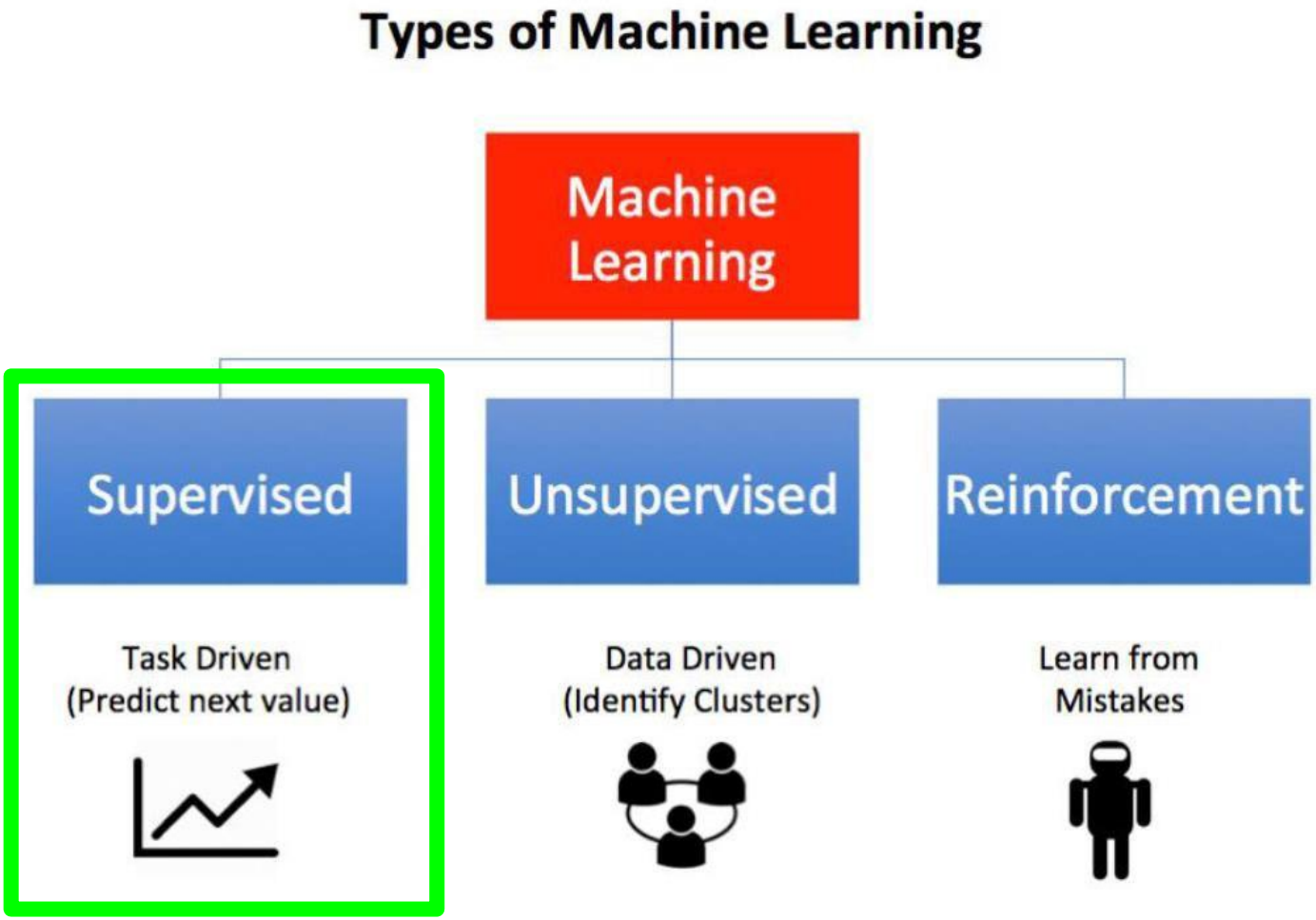


Source: http://dataconomy.com/2015/05/predictive-machine-learning-behind-the-scenes-at-fliptop-and-predictions-for-the-future-of-martech/

Applied Machine Learning Pipeline Overview



What is Business Analytics & Machine Learning?



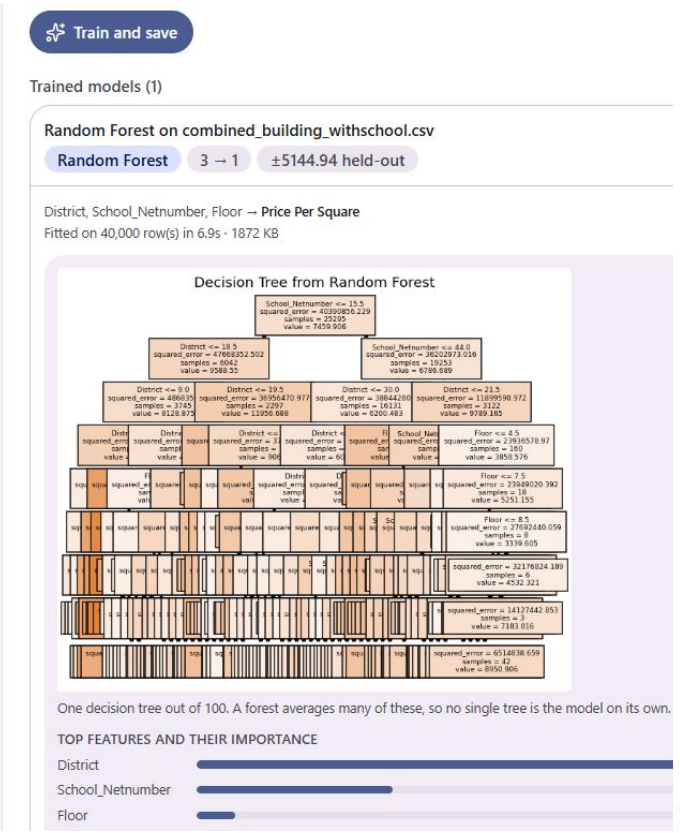
Applied Machine Learning Pipeline Overview

Structured Data Set (Excel, CSV)

Dataset profile | 20,000 rows | 11 numeric | 16 categorical | 20000 missing cells

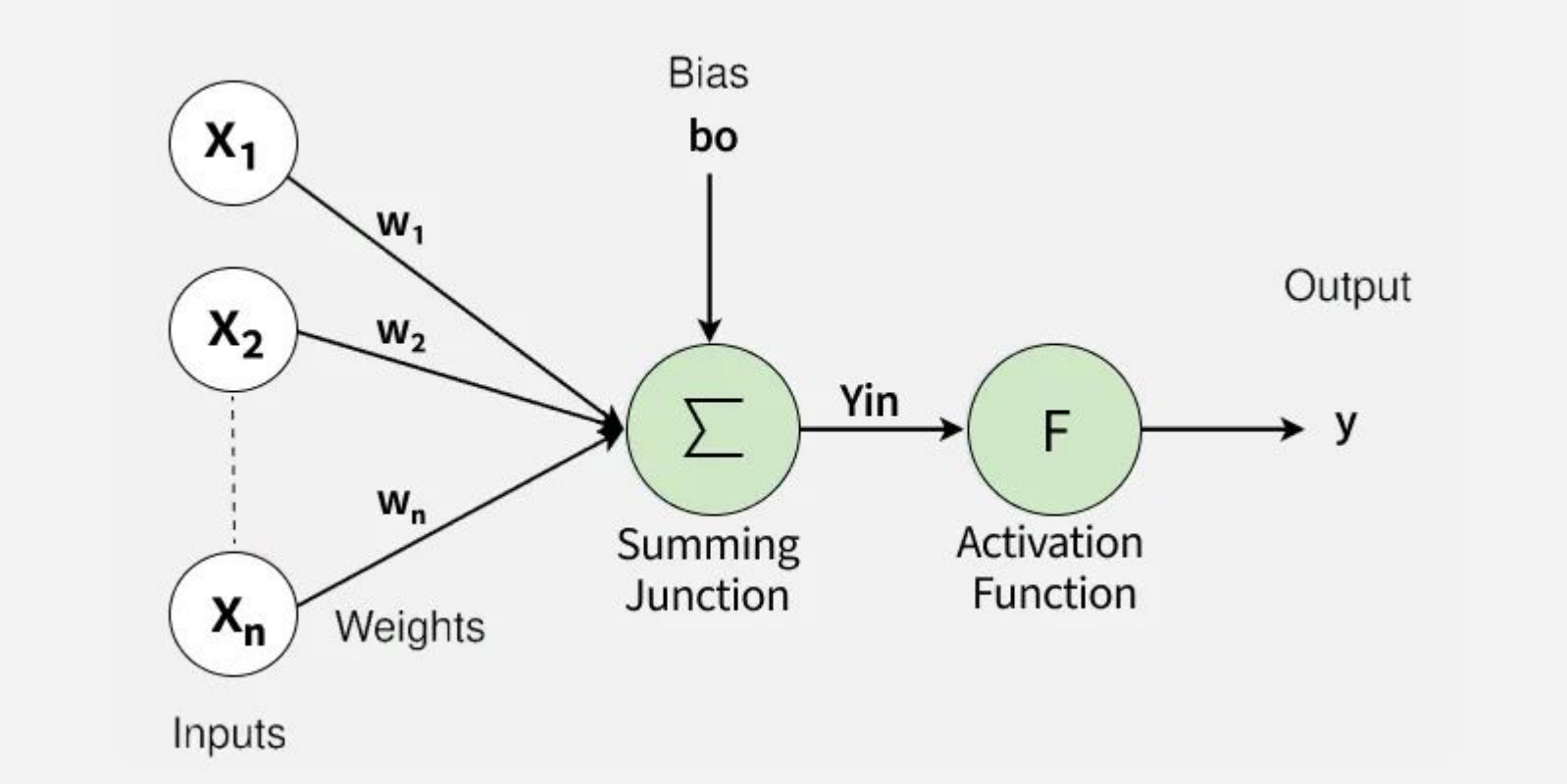
Column	Kind	Missing	Distinct	Summary
date	categorical	0	425	2020-05-27 (101), 2019-03-01 (95), 2019-03-05 (93)
id	numeric	0	20000	mean 668764.79355 · sd 9270.39635 · 652462 to 683339
Building Name	categorical	0	1732	Upper East (735), Mei Foo Sun Chuen (691), Whampoa Garden (561)
District	categorical	0	32	hung hom (2140), to kwa wan (1282), kowloon bay (1182)
url_father	categorical	0	32	https://data.28hse.com/en/kl/hung-hom (2140), https://data.28hse.com/en/kl/to-kwa-wan (1282), https://data.28hse.com/en/kl/kowloon-bay (1182)
url_cat	categorical	0	1820	https://data.28hse.com/en/kl/hung-hom/8639_upper-east (735), https://data.28hse.com/en/kl/mei-foo/2520_mei-foo-sun-chuen (691), https://data.28hse.com/en/kl/whampoa/3060_whampoa-garden (561)
contract	categorical	0	2	Agreement (17200), Assignment (2800)
memo	numeric	0	20000	mean 19588314673410.906 · sd 498818990737.454 · 19022800150022 to 20112702480015
price	categorical	0	1909	$6M (455), $4M (376), $10M (273)
price_value	numeric	0	3016	mean 8092319.18945 · sd 7909575.369751 · 150000 to 180000000
holddate	categorical	0	6489	-- (3032), 3 years 142 days (25), 3 years 145 days (23)
winloss_flag	numeric	0	3	mean 0.60005 · sd 0.779757 · -1 to 1
winloss	categorical	0	861	-- (3680), 30% (161), 33% (154)
act_area	categorical	0	911	-- (10721), 517ft² (294), 194ft² (112)
area	categorical	0	1154	-- (5936), 639ft² (185), 680ft² (165)

Model Training (Random Forest)



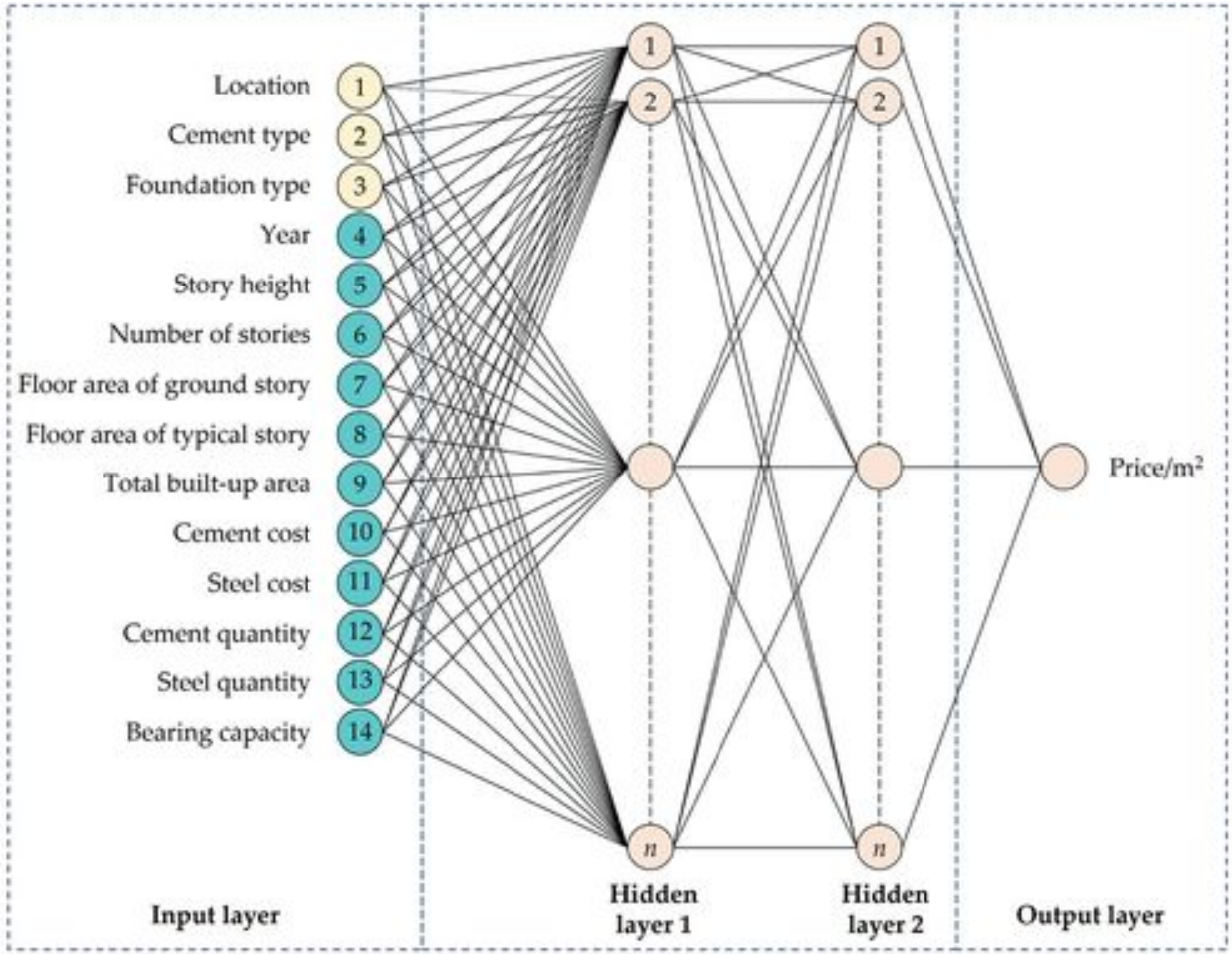
Neural Network: Random Forest

$$\sum_{i=1}^{n} W^i * X^i$$



Random Forest Example

Price prediction of residential buildings using random forest



https://www.tandfonline.com/doi/full/10.1080/16874048.2023.2296036

Housing Prediction using Machine Learning Techniques

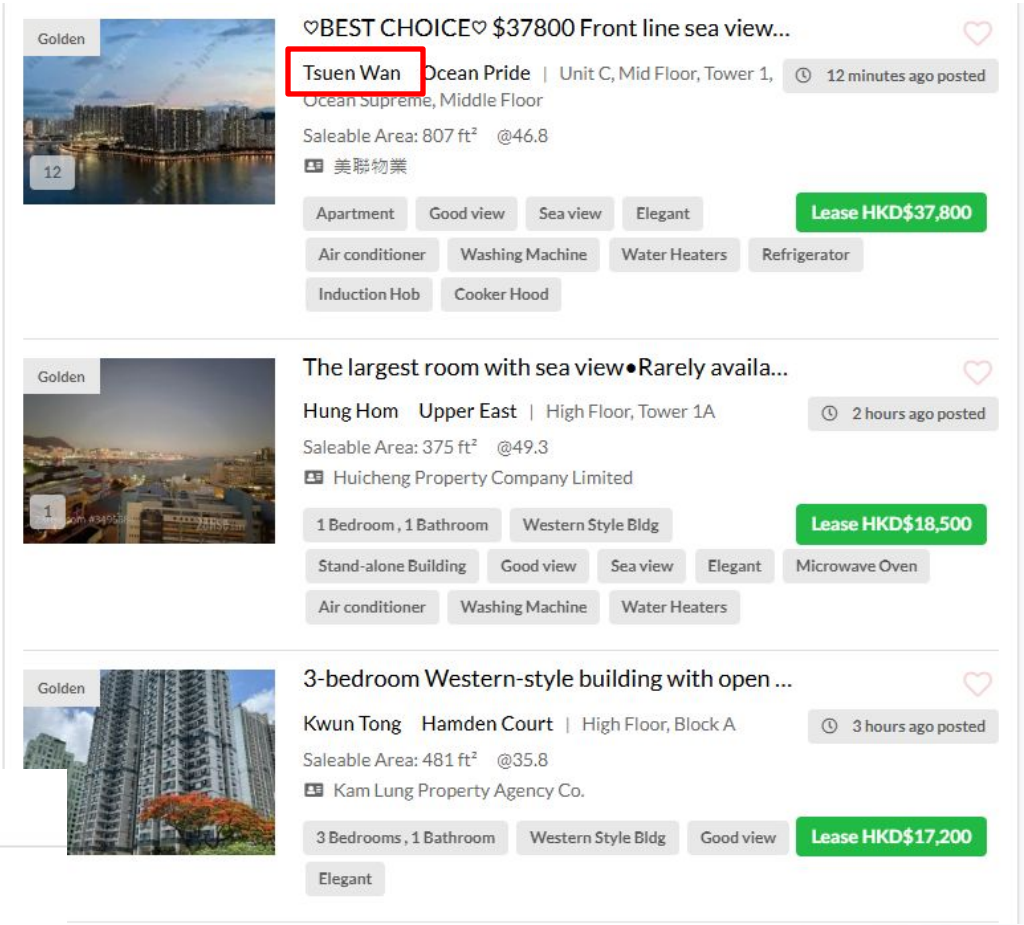
Features / Attributes

1. District
2. Floor

Outcome

1. Price Per Square

https://www.28hse.com/en/rent/apartment



https://www.28hse.com/en/rent/apartment/property-3496859



Machine Learning Pipeline



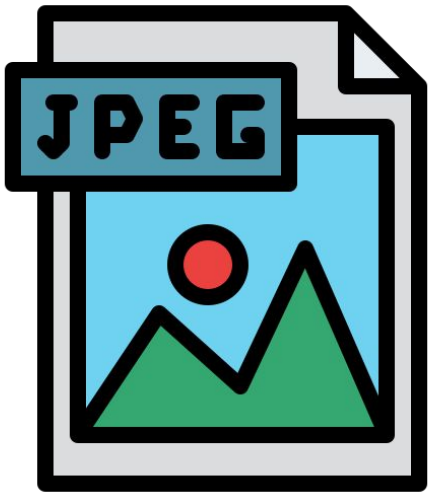
Unstructured Data Type



PDF Document



Paper Form



Image, Photo

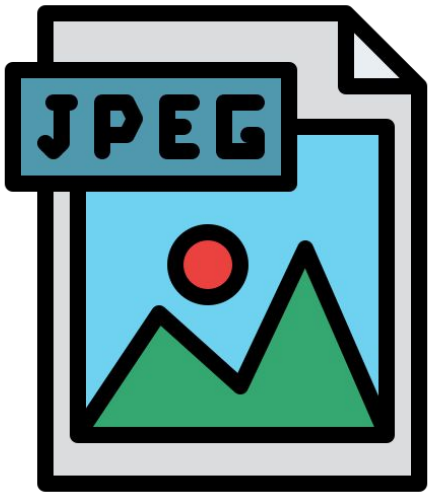
Methodology of Converting Unstructured(Qualitative) to Structured(Quantitative) Data



PDF Document
- pypdf
- pymupdf
- ollama model



Paper Form
- easyocr
- terrasect
- paddleocr



Image, Photo
- easyocr
- terrasect
- paddleocr

Processed Unstructured Data - PDF Document

Step 1 : Open Anaconda Prompt and enter “pip install pymupdf”
Step 2 : Open Jupyter notebook and enter following code

```
# RACAS Application Form Conversion (PDF to form)
def download_pdf(url, save_path):
    response = requests.get(url)
    if response.status_code == 200:
        with open(save_path, 'wb') as f:
            f.write(response.content)
        print(f"PDF downloaded and saved to {save_path}")
    else:
        print(f"Failed to download PDF: {response.status_code}")

import requests
pdf_url =    '
# Usage example
pdf_file_path = "RACAS_applicationform_14012026.pdf"
# Step 1: Download the PDF
download_pdf(pdf_url, pdf_file_path)
```

Processed Unstructured Data - PDF Document

```
import fitz  # PyMuPDF

pdf_file_path = "RACAS_applicationform_14012026.pdf"
pdf_document = fitz.open(pdf_file_path)

# Initialize a list to hold the table data
table_data = []

# Iterate through each page of the PDF
for page_num in range(len(pdf_document)):
    page = pdf_document[page_num]
    text = page.get_text("text")  # Extract text from the page

    # Split the text into lines
    lines = text.splitlines()

    # Process lines to find tables (you may need to adjust this logic)
    for line in lines:
        # Example logic: Split lines by whitespace (this may need
adjustment)
        row = line.split()
        if row:  # If the row is not empty
            table_data.append(row)

pdf_document.close()
# Print extracted table data
for row in table_data:
    print(row)
```

Raw PDF Document


HKAGAS001

香港安老院舍評審計劃申請表格

院舍名稱：_____（中文）
_____（英文）
院舍地址：_____
電話：_____ 傳真：_____
電郵：_____
院舍經營模式： 資助 / 自負盈虧 / 私營 / 合約院舍 (請刪去不適用者)
院舍類型： 安老院 / 護理安老院 / 護養院 / 安老院暨護理安老院/ 護養院暨護理安老院/ 其他類型請註明 _____(請刪去不適用者)
社會福利署/衞生署發出牌照院舍名額：_____
買位後院舍床位數目：_____(請附上社會福利署買位協議書副本)
院舍開始運作日期：_____年_____月_____日

申請費：一千元以劃線支票形式（支票抬頭：**香港老年學會**）寄回下列地址：

九龍尖沙咀金巴利道35號
金巴利中心一樓全層
香港老年學會「香港安老院舍評審計劃」。

團體/院舍(蓋印)：

負責人簽署：_____
姓名(正楷)：_____（先生/女士）
職銜：_____
電話：_____
傳真：_____
電郵：_____
日期：_____

初版日期: 04/2005　上次修定日期：6/2015　最新修定日期：1/2020

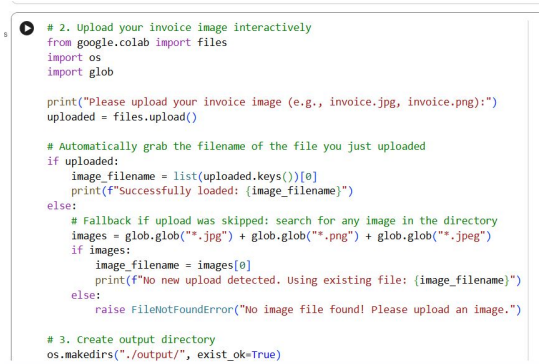
Structured Data

```
['HKAGAS001']
['初版日期:', '04/2005', '上次修定日期：6/2015',
'最新修定日期：1/2020']
['香港老年學會']
['香港安老院舍評審計劃']
['九龍尖沙咀金巴利道35', '號金巴利中心1', '樓/']
['新界葵涌石籬(一)邨石興樓地下18-20', '號']
['電話號碼:', '27273250/', '37052383', '傳真號碼:
', '37052507']
['電郵:', 'racas.adm@hkag.org', '網頁:', 'http://
www.hkag.org']
['香港安老院舍評審計劃申請表格']
['院舍名稱：', '_____
_____（中文）']
```

Python colab code to convert Image to Excel File

Procedure

1. Open https://colab.research.google.com/drive/1QiVEe3NuITFPZIYO9J9tUV5XKgiBfFhg#scrollTo=7RCLYGpn6nTQ



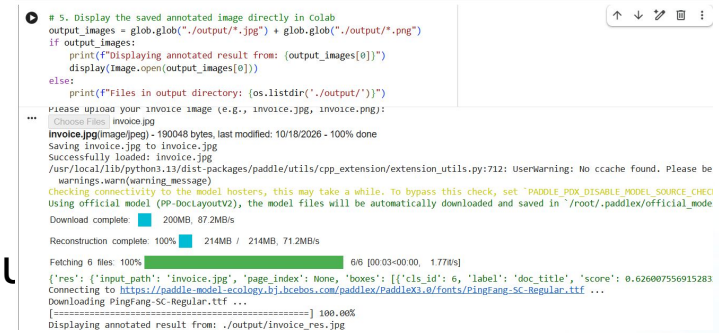
```
# 2. Upload your invoice image interactively
from google.colab import files
import os
import glob

print("Please upload your invoice image (e.g., invoice.jpg, invoice.png):")
uploaded = files.upload()

# Automatically grab the filename of the file you just uploaded
if uploaded:
    image_filename = list(uploaded.keys())[0]
    print(f"Successfully loaded: {image_filename}")
else:
    # Fallback if upload was skipped: search for any image in the directory
    images = glob.glob("*.jpg") + glob.glob("*.png") + glob.glob("*.jpeg")
    if images:
        image_filename = images[0]
        print(f"No new upload detected. Using existing file: {image_filename}")
    else:
        raise FileNotFoundError("No image file found! Please upload an image.")

# 3. Create output directory
os.makedirs("./output/", exist_ok=True)
```

2. Click “Runtime” -> “Run All”
3. User can upload your invoice.jpg
4. Load paddle OCR LLM model to condu

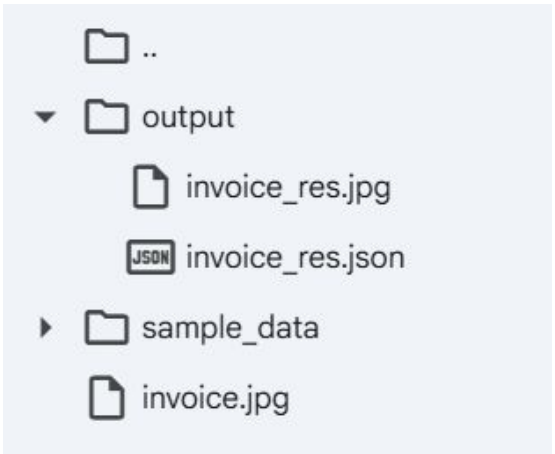


```
# 5. Display the saved annotated image directly in Colab
output_images = glob.glob("./output/*.jpg") + glob.glob("./output/*.png")
if output_images:
    print(f"Displaying annotated result from: {output_images[0]}")
    display(Image.open(output_images[0]))
else:
    print(f"Files in output directory: {os.listdir('./output/')}")
Please upload your invoice image (e.g., invoice.jpg, invoice.png):
Choose Files  invoice.jpg
invoice.jpg(image/jpeg) - 190048 bytes, last modified: 10/18/2026 - 100% done
Saving invoice.jpg to invoice.jpg
Successfully loaded: invoice.jpg
/usr/local/lib/python3.13/dist-packages/paddle/utils/cpp_extension/extension_utils.py:712: UserWarning: No ccache found. Please be
  warnings.warn(warning_message)
Checking connectivity to the model hosters, this may take a while. To bypass this check, set `PADDLE_PDX_DISABLE_MODEL_SOURCE_CHEC
Using official model (PP-DocLayoutV2), the model files will be automatically downloaded and saved in `/root/.paddlex/official_mode
Download complete: 200MB, 87.2MB/s
Reconstruction complete: 100% 214MB / 214MB, 71.2MB/s
Fetching 6 files: 100% 6/6 [00:03<00:00, 1.77it/s]
{'res': {'input_path': 'invoice.jpg', 'page_index': None, 'boxes': [{'cls_id': 6, 'label': 'doc_title', 'score': 0.626007556915283
Connecting to https://paddle-model-ecology.bj.bcebos.com/paddlex/PaddleX3.0/fonts/PingFang-SC-Regular.ttf ...
Downloading PingFang-SC-Regular.ttf ...
[==================================================] 100.00%
Displaying annotated result from: ./output/invoice_res.jpg
```

Python colab code to convert Image to Excel File

Procedure

5. An image with labelling and json file is generated.



invoice_res.jpg



invoice_res.json



invoice_res.json


```
{
  "input_path": "invoice.jpg",
  "page_index": null,
  "boxes": [
    {
      "cls_id": 6,
      "label": "doc_title",
      "score": 0.6260075569152832,
      "coordinate": [
        502,
        238,
        944,
        323
      ],
      "order": 1
    },
    {
      "cls_id": 12,
      "label": "header",
      "score": 0.9598553776741028,
      "coordinate": [
        1280,
```

Python anaconda code to convert Image to Excel File

Step 1 : Open Anaconda Prompt and enter “pip install paddleocr paddlepaddle”
Step 2: Open Jupyter notebook and enter following code

```
#Paddle OCR Layout Analysis Module
#https://www.paddleocr.ai/main/en/version3.x/module_usage/layout_analysis.html#iii-quick-integration
from paddleocr import LayoutDetection

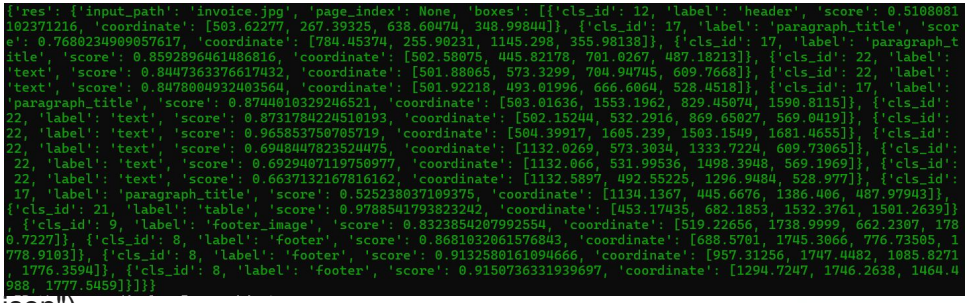
# Initialize the LayoutDetection model
model = LayoutDetection(model_name="PP-DocLayoutV2") #Better than PP-DocLayout_plus-L

# Predict layout from the image
output = model.predict("invoice.jpg", batch_size=1, layout_nms=True)

# Prepare to save results
results = []

# Process each result
for res in output:
    # Print the result details
    res.print()

    # Save the annotated image
    res.save_to_img(save_path="./output/")
    res.save_to_json(save_path="./output/invoice_res.json")
```

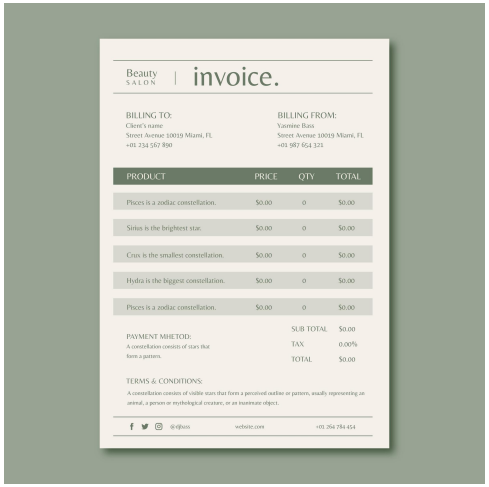


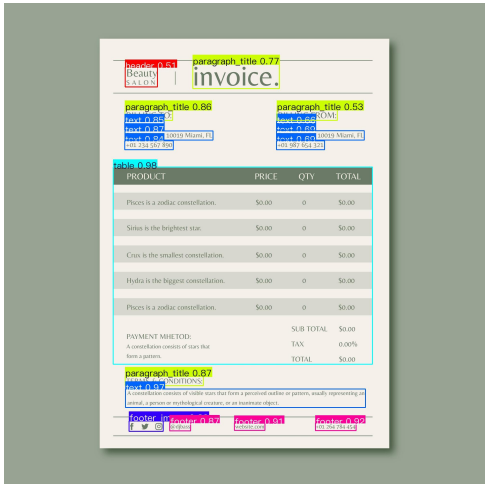
Python anaconda code to convert Image to Excel File

```
#Display original image and labelled image
from IPython.display import Image, display

# Specify the path to your image file
orig_image_path = "invoice.jpg"
label_image_path = "./output/invoice_res.jpg"

# Display the images with specified resolution
display(Image(filename=orig_image_path, width=640, height=480))
display(Image(filename=label_image_path, width=640, height=480))
```





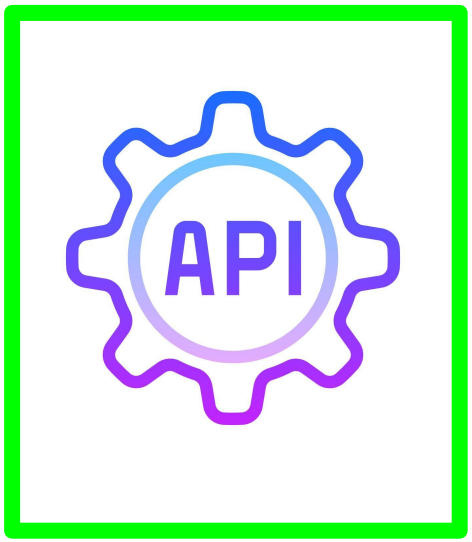
Machine Learning Pipeline



Structured Data Collection Method

Main Data collection are get from Google Sheets, API and Web Scraping







kaggle

https://www.kaggle.com/

開放數據平台
DATA.GOV.HK

https://data.gov.hk/en/

What is Data Dictionary from API

Data Dictionary from API = single source of truth that tells developers what each field means, its type, format, rules, and example - so API JSON can be validated and stored correctly.

Definition

structured metadata describing API data fields, types, formats.

- Describes each data field available in the API response
- Defines data type, format, validation, and requirement rules
- Ensures consistent usage across developers and systems

Example — /users Endpoint

Field Name	Data Type	Description	Required	Sample Value
id	integer	Unique identifier for the user	Yes	12345
name	string	Full name of the user	Yes	Jane Doe
email	string (email)	User's email address for contact	Yes	jane.doe@example.com

Purpose: Provides a single source of truth for API data structure to improve clarity, validation, and onboarding

API Documentation • Data Governance

Data Dictionary - Real API JSON Response Mapping

API Endpoint: **GET /api/products**
— Retrieves a list of products from the catalog via REST API

Data Dictionary Table

Field	Type	Format	Description	Example
product_id	string	UUID	Unique identifier for the product	550e8400-e29b-41d4-a716-446655440001
name	string	–	Product name	
price	number	decimal (2dp)	Product price in USD, 2 decimal places	199.99
published_date	date	ISO YYY-MM-DD	Date the product was published in ISO 8601	2024-03-12

API → Data Dictionary Flow



Validation Rules: UUID format, 2 decimal precision for price, ISO 8601 date

Mapping Purpose: Align API JSON fields to internal data model

Example corresponds to a single product object from the response array

** Note: Actual API response returns a JSON array of product objects. Example shown is one item for clarity.*

Data Collection - Reading HKO Weather Data from API

Below are the Flow Chart with the Process of Getting Weather Data from HKO API



Following Code contain 4 steps with the hypothesis
"Read Weather Forecasting Data from HKO API and Save in CSV File"

Step 1. Import the required libraries:

```
#import necessary library
import requests
import pandas as pd
```

Data Collection - Reading HKO Weather Data from API

Step 2: Load the weather forecasting data from 'https://data.weather.gov.hk/weatherAPI/opendata/'
Before writing code, please take a look on API documentation at Hong Kong Observatory
Study the Weather API data dict Documentation in
https://www.hko.gov.hk/en/weatherAPI/doc/files/HKO_Open_Data_API_Documentation.pdf

1. Weather Information API

Dataset

- 9-day Weather Forecast
- Current Weather Report
- Local Weather Forecast
- Weather Warning Information
- Weather Warning Summary
- Special Weather Tips

API URL

https://data.weather.gov.hk/weatherAPI/opendata/weather.php

Please include valid parameters in API request. For valid parameters, please refer to Request table in this section.

HTTP Request Method

GET

Return Type

JSON

Request Example

https://data.weather.gov.hk/weatherAPI/opendata/weather.php?dataType=flw&lang=en

Parameter

1. **dataType**
2. **lang**

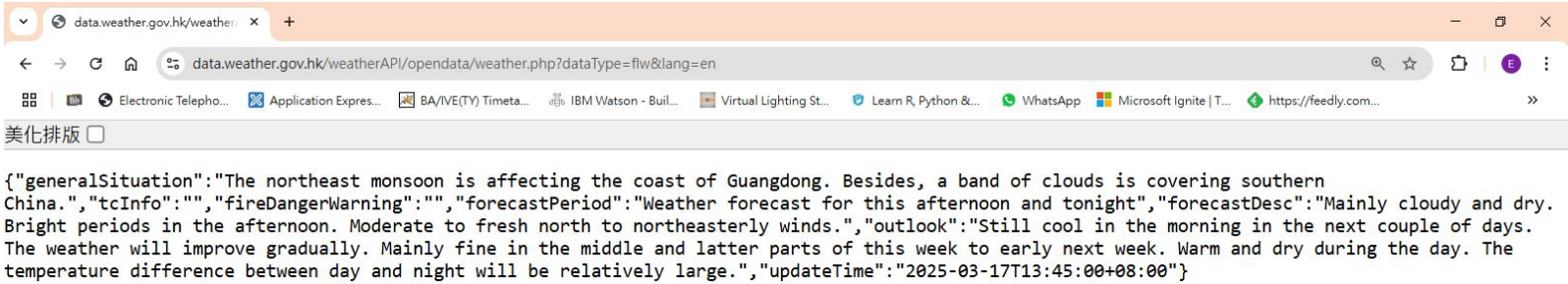
Request

Parameter	Accepted values	Description
dataType	flw	flw: Local Weather Forecast
	fnd	fnd: 9-day Weather Forecast
	rhrread	rhrread: Current Weather Report
	warnsum	warnsum: Weather Warning Summary
	warningInfo	warningInfo: Weather Warning Information
	swt	swt: Special Weather Tips
lang	en	en: English
	tc	tc: Traditional Chinese
	sc	sc: Simplified Chinese
		Default language: en

4

Data Collection - Reading HKO Weather Data from API

When browser load https://data.weather.gov.hk/weatherAPI/opendata/weather.php?dataType=flw&lang=en



generalSituation	tcInfo	fireDangerWarning	forecastPeriod	forecastDesc	outlook	updateTime
":"The northeast monsoon is affecting the coast of Guangdong. Besides, a band of clouds is covering southern China			Weather forecast for this afternoon and tonight	Mainly cloudy and dry. Bright periods in the afternoon. Moderate to fresh north to northeasterly winds	Still cool in the morning in the next couple of days. The weather will improve gradually. Mainly fine in the middle and latter parts of this week to early next week. Warm and dry during the day. The temperature difference between day and night will be relatively large.	2025-03-17T13:45:00+08:00

Data Collection - Reading HKO Weather Data from API

#Continue from previous page

Step 2: Load the weather forecasting data from 'https://data.weather.gov.hk/weatherAPI/opendata/'

```
# Initialize Parameters for the API
datatype = 'fnd'
lang = 'en'
apiurl = f'https://data.weather.gov.hk/weatherAPI/opendata/weather.php?dataType={datatype}&lang={lang}'

# Fetching the JSON data
response = requests.get(apiurl)
response.raise_for_status()  # Check for request errors

# Converting JSON to a DataFrame
json_data = response.json()
```

Step 3: Converting JSON Data to a DataFrame

```
weatherforcast_df = pd.DataFrame(json_data['weatherForecast'])

# Display the DataFrame
print(weatherforcast_df.head())
```

```
  forecastDate       week                                        forecastWind  \
0     20250115  Wednesday                                    North force 4 to 5.
1     20250116   Thursday      North to northeast force 4, force 5 at first.
2     20250117     Friday            Northeast force 4, east force 5 later.
3     20250118   Saturday  East to northeast force 4 to 5, occasionally f...
4     20250119     Sunday                  East to northeast force 2 to 3.

                                  forecastWeather  \
0  Mainly cloudy with a few light rain patches at...
1  Fine. Rather cool in the morning. Very dry dur...
2  Fine. Rather cool in the morning. Very dry dur...
3                          Dry with sunny periods.
4                             Mainly fine and dry.

            forecastMaxtemp            forecastMintemp  \
0  {'value': 20, 'unit': 'C'}  {'value': 15, 'unit': 'C'}
1  {'value': 19, 'unit': 'C'}  {'value': 13, 'unit': 'C'}
2  {'value': 19, 'unit': 'C'}  {'value': 13, 'unit': 'C'}
3  {'value': 20, 'unit': 'C'}  {'value': 15, 'unit': 'C'}
4  {'value': 21, 'unit': 'C'}  {'value': 16, 'unit': 'C'}

                     forecastMaxrh                     forecastMinrh  \
0  {'value': 80, 'unit': 'percent'}  {'value': 40, 'unit': 'percent'}
1  {'value': 65, 'unit': 'percent'}  {'value': 35, 'unit': 'percent'}
2  {'value': 65, 'unit': 'percent'}  {'value': 30, 'unit': 'percent'}
3  {'value': 70, 'unit': 'percent'}  {'value': 40, 'unit': 'percent'}
4  {'value': 70, 'unit': 'percent'}  {'value': 40, 'unit': 'percent'}

   ForecastIcon  PSR
0            51  Low
1            92  Low
2            50  Low
3            51  Low
4            50  Low
```

Data Collection - Reading HKO Weather Data from API

#Continue from previous page

Step 4. Saving Weather Forecasting DataFrame to a CSV

```
csv_file = 'weatherForecast.csv'  # Desired output CSV file
weatherforcast_df.to_csv(csv_file, index=False)

# Output successful message for saving weather forecasting data.
print(f"Data has been successfully saved to {csv_file}")
```

A	B	C	D	E	F	G	H	I	J
forecastDate	week	forecastWind	forecastWeather	forecastMaxtemp	forecastMintemp	forecastMaxrh	forecastMinrh	ForecastIcon	PSR
20250115	Wednesday	North force 4 to 5.	Mainly cloudy with a fe	{'value': 20, 'unit':	{'value': 15, 'unit':	{'value': 80, 'un:	{'value': 40, 'un	51	Low
20250116	Thursday	North to northeast force	Fine. Rather cool in the r	{'value': 19, 'unit':	{'value': 13, 'unit':	{'value': 65, 'un:	{'value': 35, 'un	92	Low
20250117	Friday	Northeast force 4, east	Fine. Rather cool in the r	{'value': 19, 'unit':	{'value': 13, 'unit':	{'value': 65, 'un:	{'value': 30, 'un	50	Low
20250118	Saturday	East to northeast force	Dry with sunny periods.	{'value': 20, 'unit':	{'value': 15, 'unit':	{'value': 70, 'un:	{'value': 40, 'un	51	Low
20250119	Sunday	East to northeast force	Mainly fine and dry.	{'value': 21, 'unit':	{'value': 16, 'unit':	{'value': 70, 'un:	{'value': 40, 'un	50	Low
20250120	Monday	East to northeast force	Fine and dry.	{'value': 22, 'unit':	{'value': 16, 'unit':	{'value': 70, 'un:	{'value': 40, 'un	50	Low
20250121	Tuesday	East to northeast force	Fine and dry.	{'value': 22, 'unit':	{'value': 17, 'unit':	{'value': 70, 'un:	{'value': 40, 'un	50	Low
20250122	Wednesday	East force 4, occasiona	Sunny periods.	{'value': 21, 'unit':	{'value': 17, 'unit':	{'value': 75, 'un:	{'value': 50, 'un	51	Low
20250123	Thursday	East force 4, occasiona	Sunny intervals.	{'value': 21, 'unit':	{'value': 17, 'unit':	{'value': 75, 'un:	{'value': 50, 'un	52	Low

Get Dictionary Attribute Values

- **API endpoint used**
  - URL: https://data.weather.gov.hk/weatherAPI/opendata/weather.php?dataType=flw&lang=tc
  - Purpose: retrieves **local weather forecast** (`flw`) in **Traditional Chinese** (`lang=tc`)
- **Sample response structure**
  - Example `weather_data` object:

```
weather_data = {
        "generalSituation": "Mainly cloudy with a few showers",
        "temperature": {
                "value": 28,
                "unit": "C"
        },
        "humidity": 85,
        "wind": "Moderate southerly winds",
        "updateTime": "2026-05-20 12:00"
}
print(weather_data["temperature"])
print(weather_data["temperature"]['value'])
print(weather_data["humidity"])
```

Dict Data Type

```
>>> print(weather_data["temperature"])
{'value': 28, 'unit': 'C'}
>>> print(weather_data["temperature"]['value'])
28
>>> print(weather_data["humidity"])
85
```

Convert Dictionary to DataFrame

1 pd.DataFrame (Flat Dict)

```
import pandas as pd
data = {"name': ["Alice",'Bob"], 'age': [25, 30]}
df = pd.DataFrame(data)
df
```

Best for: simple flat key-value dict → DataFrame

2 pd.DataFrame.from_records

```
records = {"'name':"Alice","age':25},
{"name':'Bob","age':30}]
df = pd.DataFrame.from_records(records)
df
```

Best for: list of dict records / table-like data

3 pd.json_normalize (Nested JSON)

```
from pandas import json_normalize
nested = {"user":{"id":1,"info":{"name":"Alice","city","NYC")}}
df = json_normalize(nested, sep"__")
df
```

Best for: nested dict → flattened columns using dot/separator

4 json_normalize + record_path

```
data = {"id":101,"orders"::[["id":A,"amt":50]}
{"id":B,"amt":75]}]
df = json_normalize(data, record_path="orders", meta=["id"]
df
```

Best for: nested lists/arrays → flatten with parent meta field

TIP: Use **"sep"_"** to control nested column names. Use **"meta_prefix"** to avoid column name clashes. Always check dtypes after normalization.

Pandas • Data Manipulation • Practical Python

Sequential Data Types

- **List**
  - "Mutable" object (something you can change)
  - Representation

```
school_list = ['hku', 'cuhk', 'hkust']
print(school_list)
print(type(school_list))
```

```
['hku', 'cuhk', 'hkust']
<class 'list'>
```

  - Indexing

```
print(school_list[0])      # First 1 item
print(school_list[0:2])    # First 2 items
print(school_list[-1])     # Last one
```

```
hku
['hku', 'cuhk']
polyu
```

Sequential Data Types

- **List**
  - Adding items

```
school_list = ['hku', 'cuhk', 'hkust'] + ['cityu', 'polyu']
school_list
```

```
['hku', 'cuhk', 'hkust', 'cityu', 'polyu']
```

```
school_list = ['hku', 'cuhk', 'hkust', 'cityu']
school_list.append('polyu')
school_list
```

```
['hku', 'cuhk', 'hkust', 'cityu', 'polyu']
```

Sequential Data Types

- **List**
  - Removing an item
    - By index ➔ pop()

```
school_list.pop(1)
school_list
```

```
['hku', 'hkust', 'cityu', 'polyu']
```

    - By element ➔ remove()

```
school_list.remove('cityu')
school_list
```

```
['hku', 'hkust', 'polyu']
```

Sequential Data Types

- **Common operations**
  - Membership Check ➔ "in"

```
# Membership check
school_list = ['hku', 'cuhk', 'hkust', 'cityu', 'polyu']
print('hku' in school_list)
print('hkbu' in school_list)
```

```
True
False
```

Sequential Data Types

- **Common operations**
  - Ordering ➔ max() / min() / reversed() / sorted()

```
number_list = [1, 5, 6, 2, 3]
print(max(number_list))                # Max
print(min(number_list))                # Min
print(list(reversed(number_list)))     # Reverse order
print(sorted(number_list))             # Sort by ascending order
```

```
6
1
[3, 2, 6, 5, 1]
[1, 2, 3, 5, 6]
```

Sequential Data Types

- **String (again)**
  - string = sequential!




```
# Index:        0123456789
school_name = 'HKU School of Professional and Continuing Education'

print(len(school_name))          # There is a total of 51 letters
print(school_name.index('c'))    # First c is the 6th letter (counting 1 space)
print(school_name.count('o'))    # There are 7 O's
print('HKU' in school_name)      # Substring matching
print(school_name[4:10])         # Substring extraction
```

```
51
5
7
True
School
```

Sequential Data Types

- **String (again)**
  - Conversions

```
# Conversion: String -> list
print(list('hello'))
```

```
['h', 'e', 'l', 'l', 'o']
```

```
# Conversion: List -> String
print(''.join(['h', 'e', 'l', 'l', 'o']))
print(','.join(['Welcome', 'Jackie!']))
```

```
hello
Welcome,Jackie!
```

```
# Conversion: String <-> int?
print(ord('a'))
print(chr(98))
```

```
97
b
```

Data Type - Dictionary

- **Motivation**
  - Business problems often involve key-value pairs
  - e.g., Employee ID ➔ Full Name / Job Title

	A	B	C	D
1	Employee ID	Full Name	Job Title	Department
2	E02002	Kai Le	Controls Engineer	Engineering
3	E02003	Robert Patel	Analyst	Sales
4	E02004	Cameron Lo	Network Administrator	IT
5	E02005	Harper Castillo	IT Systems Architect	IT
6	E02006	Harper Dominguez	Director	Engineering
7	E02007	Ezra Vu	Network Administrator	IT
8	E02008	Jade Hu	Sr. Analyst	Accounting
9	E02009	Miles Chang	Analyst II	Finance
10	E02010	Gianna Holmes	System Administrator	IT
11	E02011	Jameson Thomas	Manager	Finance
12	E02012	Jameson Pena	Systems Analyst	IT
13	E02013	Bella Wu	Sr. Analyst	Finance

  - Looking up a keyword
    ➔ like finding words in **dictionaries**



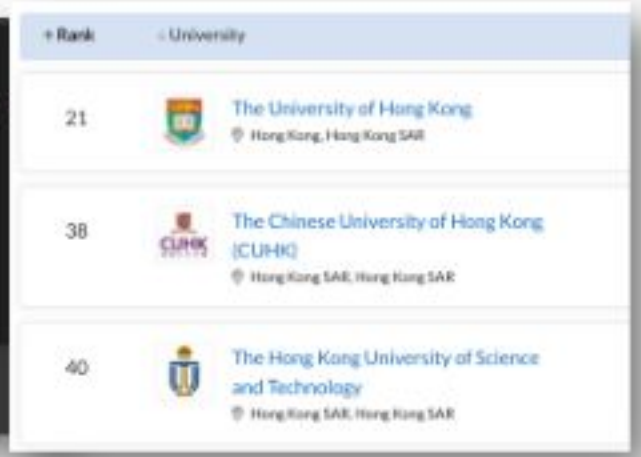
Data Type - Dictionary

- Representation

```
# Representation: option 1
school_rank_dict = {'hku': 21, 'cuhk': 38, 'hkust': 40}
print(type(school_rank_dict))

# Representation: option 2
school_rank_dict = dict(hku=21, cuhk=38, hkust=40)
print(school_rank_dict)
```

```
<class 'dict'>
{'hku': 21, 'cuhk': 38, 'hkust': 40}
```



- Lookup ➔ get()

```
# Lookup
print(school_rank_dict['cuhk'])              # Direct lookup
print(school_rank_dict.get('hkust'))         # get(): Success
print(school_rank_dict.get('test'))          # get(): Failed
print(school_rank_dict.get('test', 'N/A'))   # get(): Error handling
```

```
38
40
None
N/A
```

Data Type - Dictionary

- Adding an item

```
# Adding an item
school_rank_dict = {'hku': 21, 'cuhk': 38, 'hkust': 40}
school_rank_dict['cityu'] = 54
print(school_rank_dict)
```

```
{'hku': 21, 'cuhk': 38, 'hkust': 40, 'cityu': 54}
```

- Removing an item

```
# Removing an item
school_rank_dict = {'hku': 21, 'cuhk': 38, 'hkust': 40, 'cityu': 54}
school_rank_dict.pop('cityu')
print(school_rank_dict)
```

```
{'hku': 21, 'cuhk': 38, 'hkust': 40}
```

Data Type - Dictionary

- Listing out the details

```
# Show all the keys
print(f'keys: {list(school_rank_dict.keys())}')

# Show all the values
print(f'values: {list(school_rank_dict.values())}')

# Show all the items
print(f'items: {list(school_rank_dict.items())}')
```

```
keys: ['hku', 'cuhk', 'hkust']
values: [21, 38, 40]
items: [('hku', 21), ('cuhk', 38), ('hkust', 40)]
```

- Membership Check

```
# Membership Check
print('hku' in school_rank_dict)
```

```
True
```

Data Type - Dictionary

- **Conversion**
  - Cast from a list of tuples

	A	B	C
1	Employee ID	Full Name	Job Title
2	E02002	Kai Le	Controls Engineer
3	E02003	Robert Patel	Analyst
4	E02004	Cameron Lo	Network Administrator

```
# Conversion: list => dict
employee_name_list = [('E02002', 'Kai Le'), ('E02003', 'Robert Patel')]
print(dict(employee_name_list))
```

```
{'E02002': 'Kai Le', 'E02003': 'Robert Patel'}
```

Data Type - Dictionary

- **Conversion**
  - zip() can be used to create tuples

	A	B	C
1	Employee ID	Full Name	Job Title
2	E02002	Kai Le	Controls Engineer
3	E02003	Robert Patel	Analyst
4	E02004	Cameron Lo	Network Administrator

```
employee_ids = ['E02002', 'E02003']
full_names = ['Kai Le', 'Robert Patel']
job_titles = ['Controls Engineer', 'Analyst']

print( list(zip(employee_ids, full_names)) )
print( dict(zip(employee_ids, full_names)) )
print( dict(zip(employee_ids, job_titles)) )
```

```
[('E02002', 'Kai Le'), ('E02003', 'Robert Patel')]
{'E02002': 'Kai Le', 'E02003': 'Robert Patel'}
{'E02002': 'Controls Engineer', 'E02003': 'Analyst'}
```

Conditional Statements

- If-else
  - Python tracks the scope using indentation (spaces)



VBA

```
Dim speed As Integer
speed = 70

If speed > 50 Then
   Debug.Print ("Exceeded limit")
Else
   Debug.Print ("OK")
End If
```

Python

```
speed = 70


if speed > 50:
    print('Exceeded limit')
else:
    print('OK')
```

Conditional Statements

- If-else
  - Value assignment can be conditional too!

Multiline

```
speed = 70

if speed > 50:
    print('Exceeded limit')
else:
    print('OK')
```

1 line

```
speed = 70

print ('Exceeded limit' if speed > 50 else 'OK')
```

Conditional Statements

- If-elif-else
  - elif ➔ “Else If”

```
mark = 89
grade = None

if mark >= 90:
  grade = 'A'
elif mark >= 80:
  grade = 'B'
elif mark >= 70:
  grade = 'C'
else:
  grade = 'F'

print(grade)
```

```
B
```

Conditional Statements

- If-elif-else
  - elif ➔ “Else If”
  - May be broken down to if-else

```
mark = 89
grade = None

if mark >= 90:
  grade = 'A'
elif mark >= 80:
  grade = 'B'
elif mark >= 70:
  grade = 'C'
else:
  grade = 'F'

print(grade)
```

```
B
```

```
mark = 89
grade = None

grade = 'A' if mark >= 90 else ('B' if mark >= 80 else ('C' if mark >= 70 else 'F'))

print(grade)
```

```
B
```

Conditional Statements

- If
  - "else" is optional

```
salary = 10000
salary_growth = 1.2
years_of_service = 3

# No increment for first year
if years_of_service > 1:
    salary = salary * salary_growth
print(salary)
```

```
12000.0
```

Conditional Statements

- If-elif-else
  - Unlike VBA, scope is NOT defined with "If ➔ End If"
  - Python tracks the scope using indentation (spaces)

VBA

```
Dim speed As Integer
speed = 70

If speed > 50 Then
   Debug.Print ("Exceeded limit")
Else
   Debug.Print ("Speed is okay")
End If
```

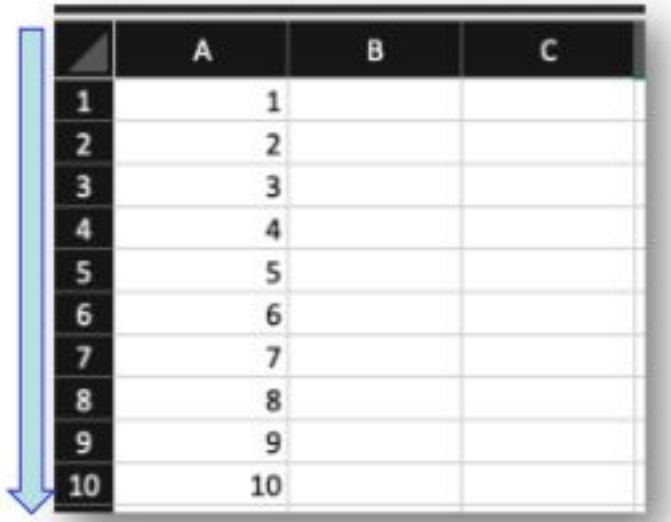
Python

```
speed = 70


if speed > 50:
    print('Exceeded limit')
else:
    print('Speed is okay')
```

Loops – Using "for"

- Iterating through a range



	A	B	C
1	1		
2	2		
3	3		
4	4		
5	5		
6	6		
7	7		
8	8		
9	9		
10	10		

VBA

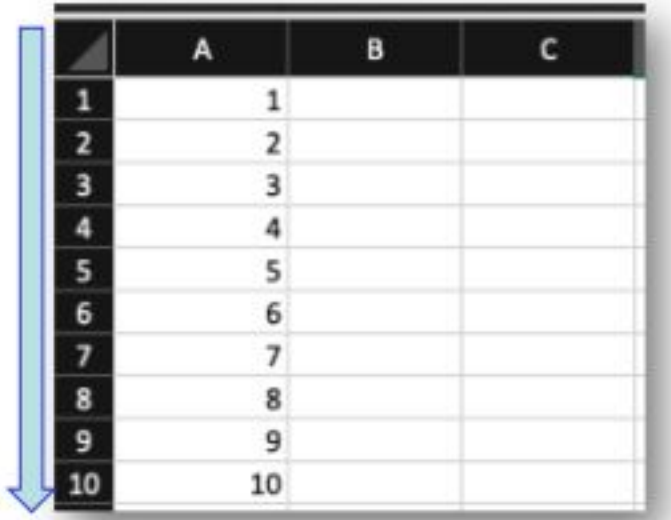
```
For i = 1 To 10
        Debug.Print (i)
Next i
```

Python

```
for i in range(1, 11):
    print(i)
```

Loops – Using "for"

- Different ways to specify ranges



	A	B	C
1	1		
2	2		
3	3		
4	4		
5	5		
6	6		
7	7		
8	8		
9	9		
10	10		

```
[11] for i in range(1, 11):
         print(i)  # 1, 2, ..., 10

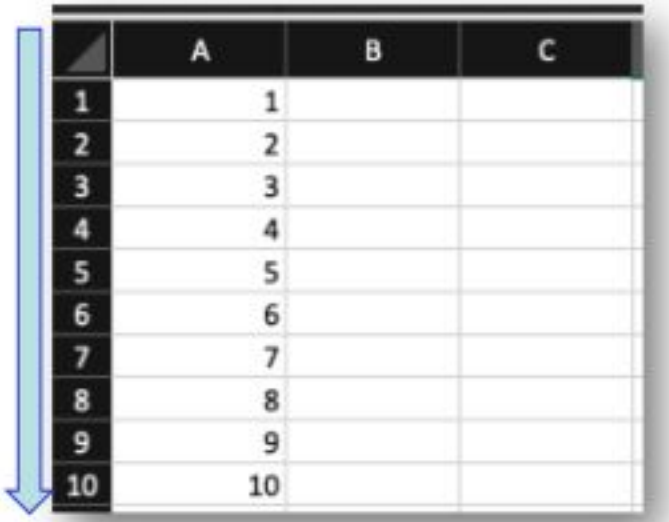
     1
     2
     3
     4
     5
     6
     7
     8
     9
     10
```

```
     for i in range(10):
         print(i + 1) # i = 0, 1, ..., 9

     1
     2
     3
     4
     5
     6
     7
     8
     9
     10
```

Loops – Using "for"

- Under the hood
  - Iterating through a sequence object




```
[21] print(range(10))

     range(0, 10)
```

```
list(range(10))  # In Python2, range() returns a list directly

[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

Loops – Using "for"

- Be careful…

```
item_prices = [200, 100]
total = 0
# Case 1: Apply $50 discount once
for price in item_prices:
    total = total + price
if total >= 200:
    total = total - 50
print(total)
```

```
250
```

```
item_prices = [200, 100]
total = 0
# Case 2: Apply $50 discount twice (accidentally..)
for price in item_prices:
    total = total + price
    if total >= 200:
        total = total - 50
print(total)
```

```
200
```

Loops – Using "for"

- Looping through a list

```
school_list = ['hku', 'cuhk', 'hkust']

for school in school_list:
    print(f'{school} is a good school in Hong Kong')
```

```
hku is a good school in Hong Kong
cuhk is a good school in Hong Kong
hkust is a good school in Hong Kong
```

Loops – Using "for"

- Looping through a dictionary

```
school_rank_dict = {'hku': 21, 'cuhk': 38, 'hkust': 40}

for school, rank in school_rank_dict.items():
    print(f'{school} is ranked {rank} in the world!')
```

```
hku is ranked 21 in the world!
cuhk is ranked 38 in the world!
hkust is ranked 40 in the world!
```

Loops – Using "for"

- Looping through a string
- Q: Why use "elif" instead of "else"?

```
# looping through a string
uppercase_letter_count = 0
lowercase_letter_count = 0

for letter in 'See you in Hong Kong':
  if letter.isupper():
    uppercase_letter_count = uppercase_letter_count + 1
  elif letter.islower():
    lowercase_letter_count = lowercase_letter_count + 1

print(f'Uppercase letters: {uppercase_letter_count}')
print(f'Lowercase letters: {lowercase_letter_count}')
```

```
Uppercase letters: 3
Lowercase letters: 13
```

Loops – Using "for"

- List comprehension
  ➔ compressing a loop in 1 line

```
print(f'8 % 2 = {8%2}')
print(f'9 % 2 = {9%2}')
```

```
8 % 2 = 0
9 % 2 = 1
```

```
# Traditional way
even_number_list = list()
for i in range(11):
    # Use the modulus operator to check for even number (Can be divded by 2)
    if i % 2 == 0:
        even_number_list.append(i)
print(even_number_list)
```

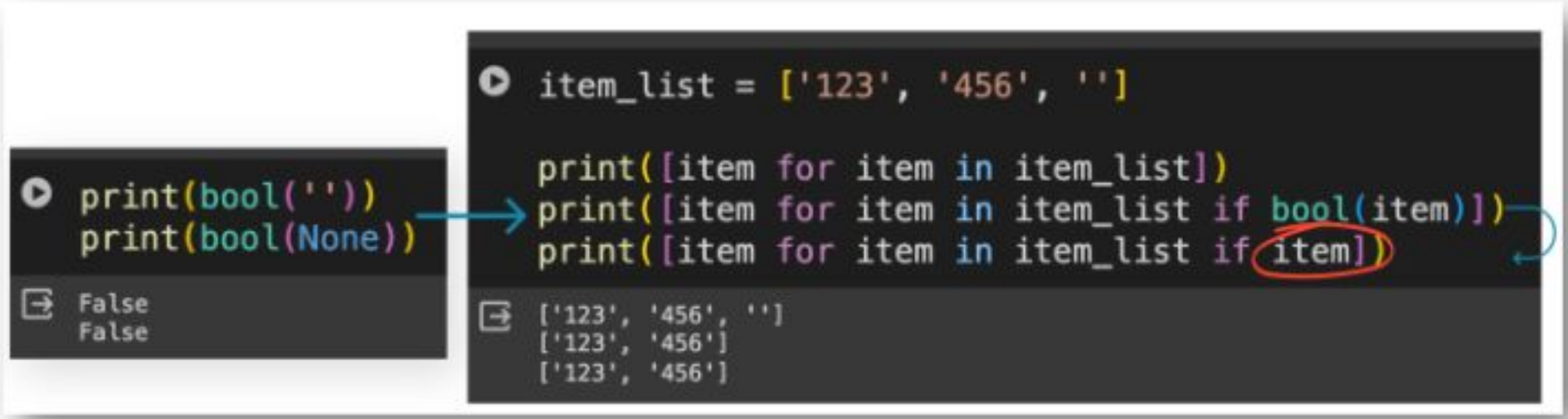
```
[0, 2, 4, 6, 8, 10]
```

```
# List comprehension
even_number_list = [i for i in range(11) if i % 2 == 0]
print(even_number_list)
```

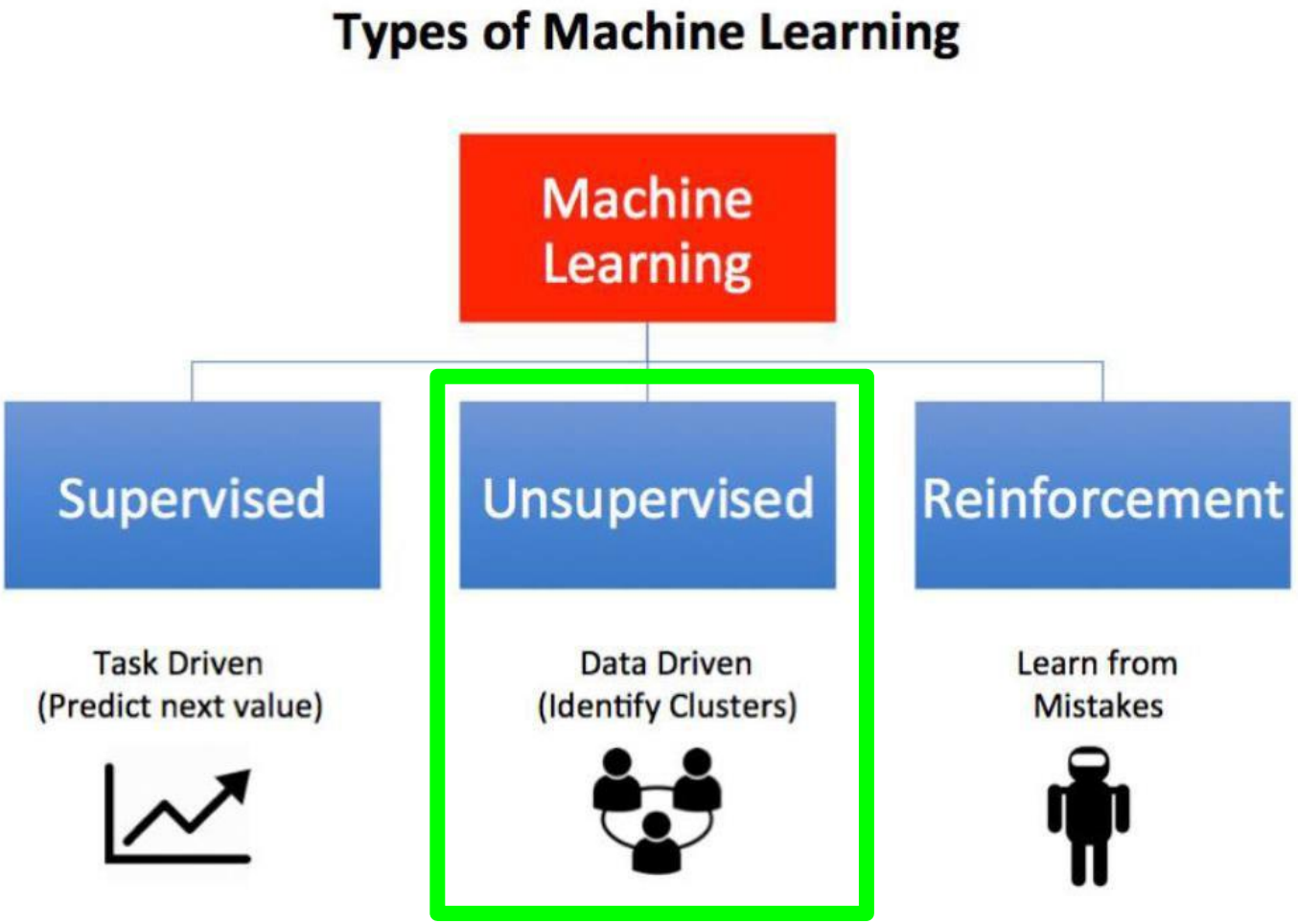
```
[0, 2, 4, 6, 8, 10]
```

Loops – Using "for"

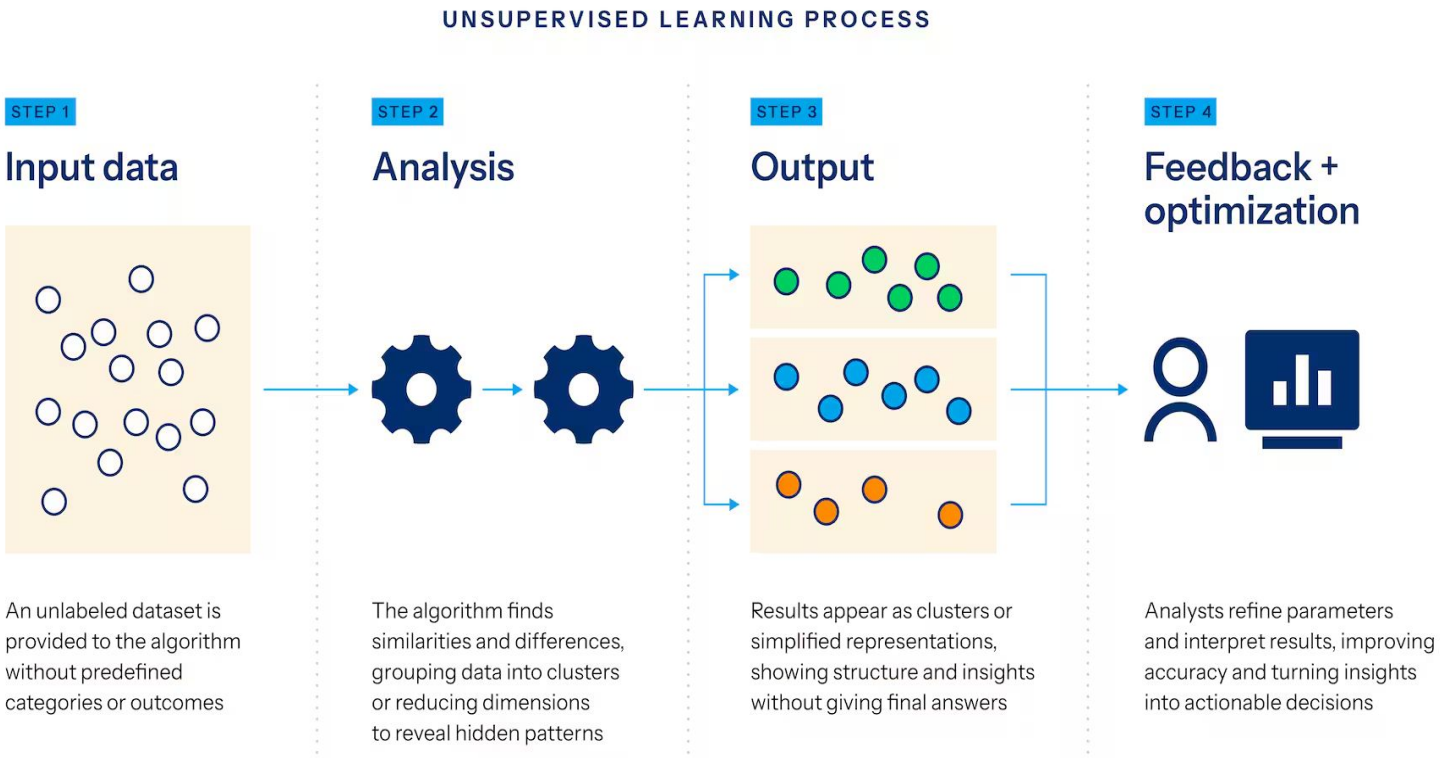
- List comprehension + Filtering



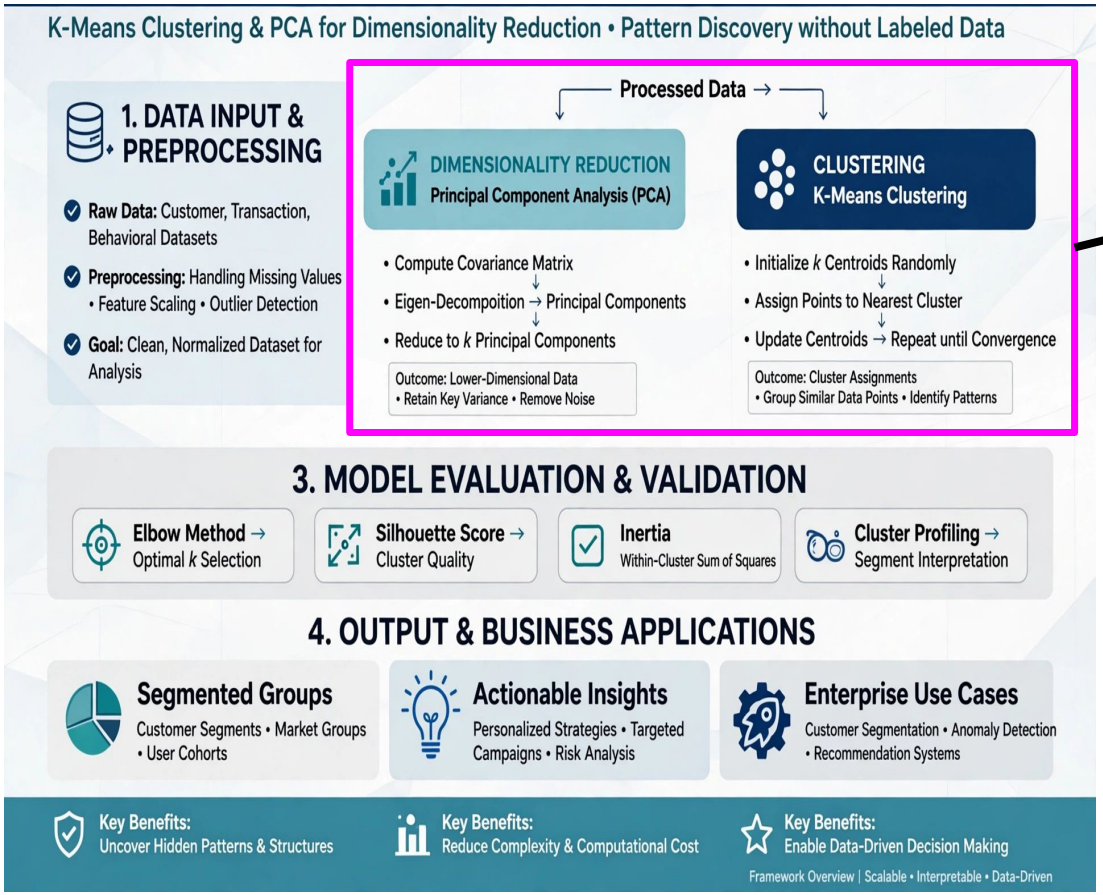
What is Business Analytics & Machine Learning?

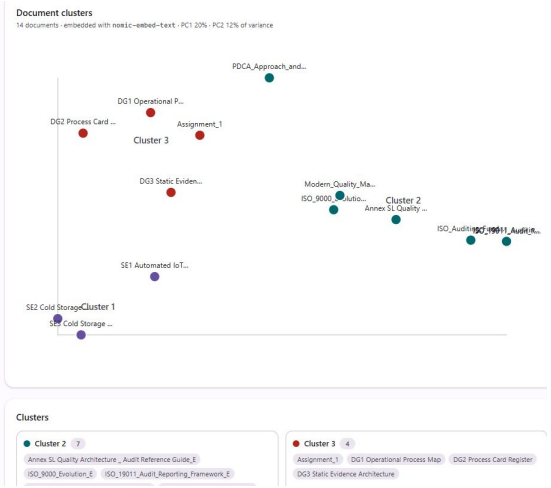


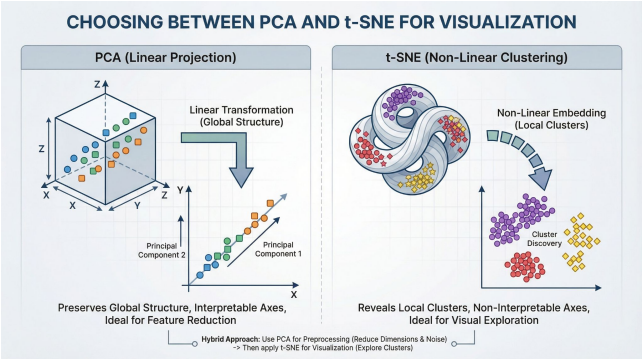
What is Unsupervised Learning?



What Unsupervised Learning Framework support?

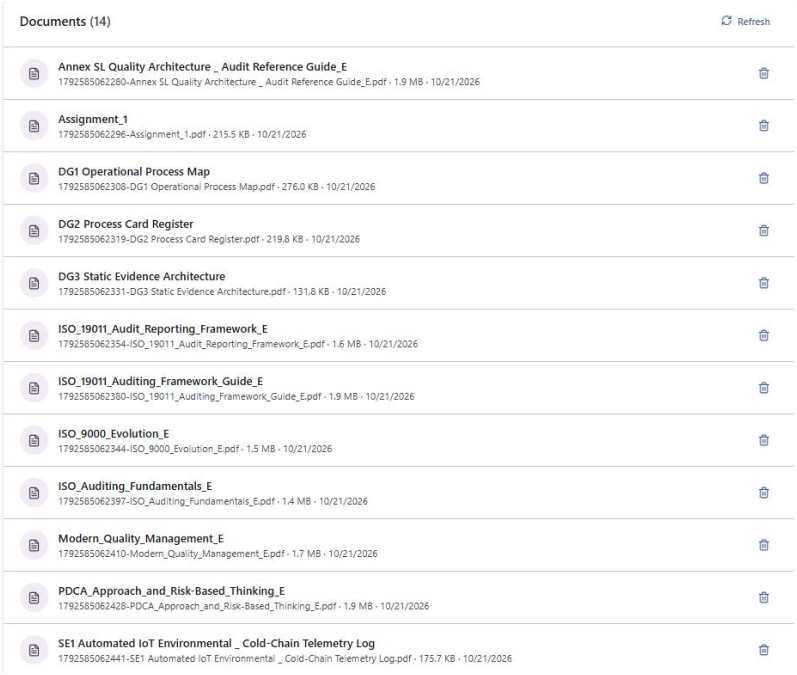


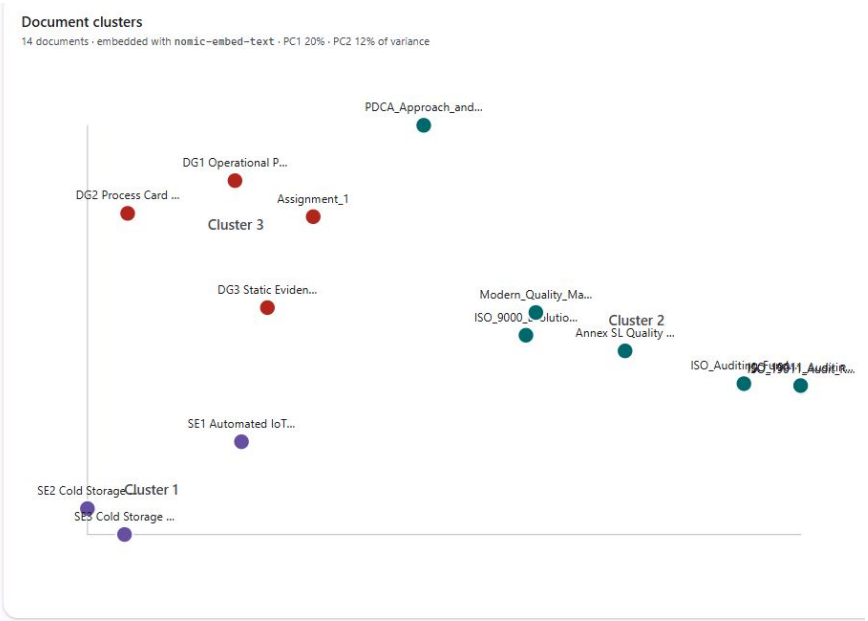




Document Clustering/Classification Examples

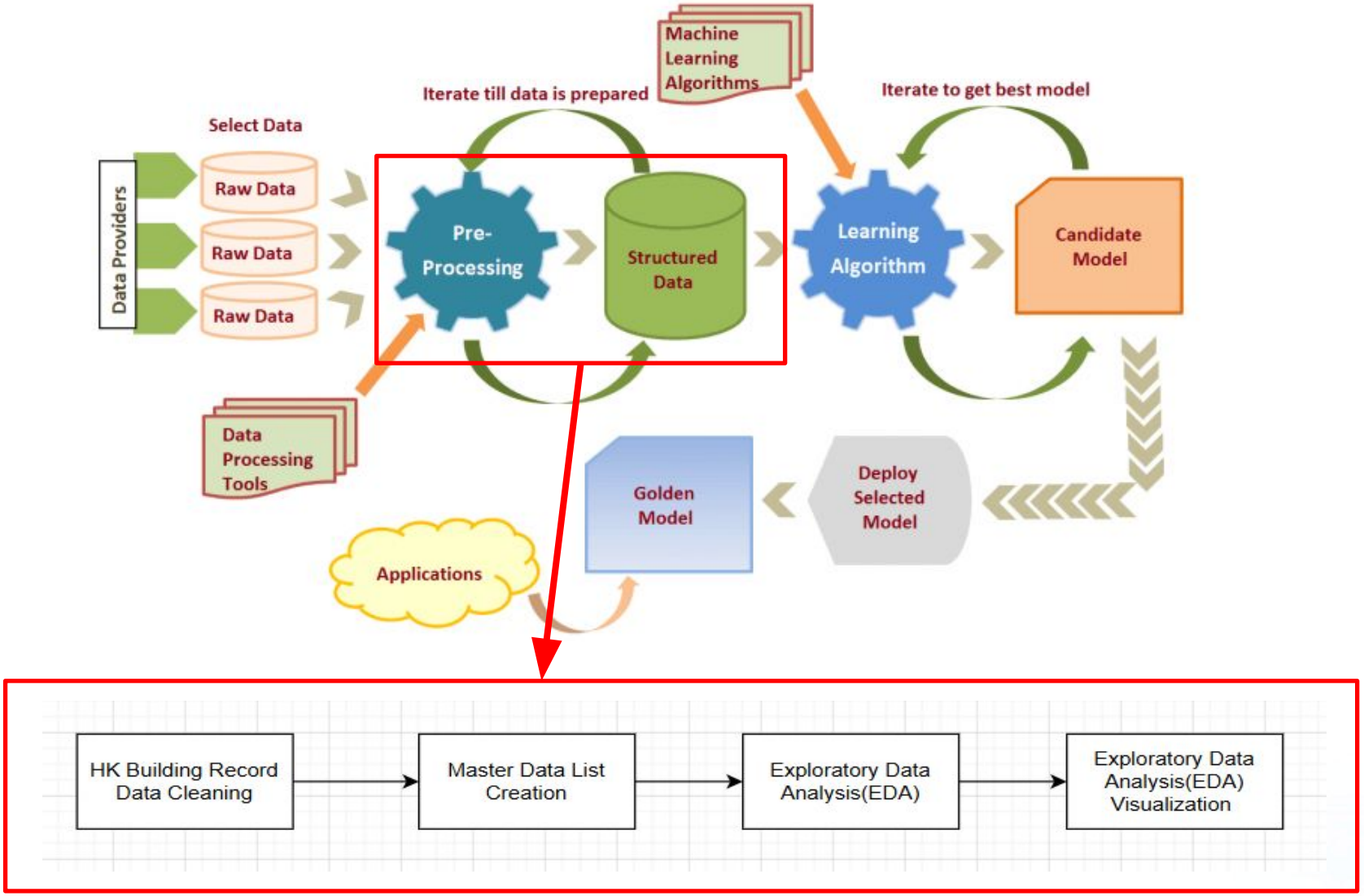
Classify multiple data set and group together and visualise in graph.







Applied Machine Learning Pipeline Overview



Applied Machine Learning Pipeline Overview

Structured Data Set (CSV)

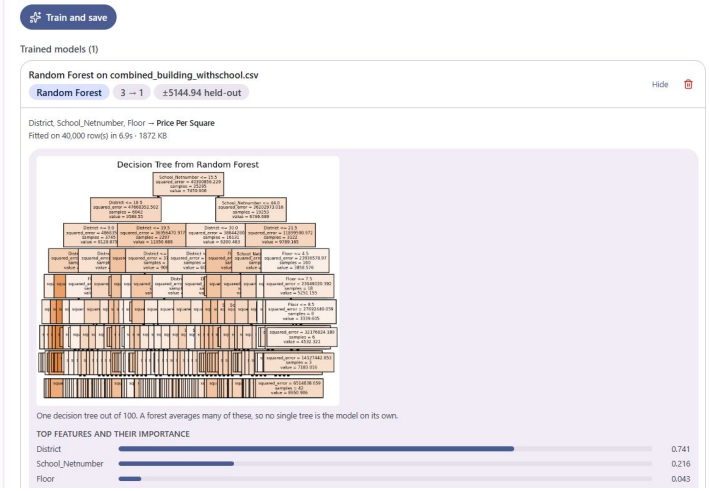
Dataset profile 20,000 rows 11 numeric 16 categorical 20000 missing cells

Column	Kind	Missing	Distinct	Summary
date	categorical	0	425	2020-05-27 (101), 2019-03-01 (95), 2019-03-05 (93)
id	numeric	0	20000	mean 668764.79355 · sd 9270.39635 · 652462 to 683339
Building Name	categorical	0	1732	Upper East (735), Mei Foo Sun Chuen (691), Whampoa Garden (561)
District	categorical	0	32	hung hom (2140), to kwa wan (1282), kowloon bay (1182)
url_father	categorical	0	32	https://data.28hse.com/en/kl/hung-hom (2140), https://data.28hse.com/en/kl/to-kwa-wan (1282), https://data.28hse.com/en/kl/kowloon-bay (1182)
url_cat	categorical	0	1820	https://data.28hse.com/en/kl/hung-hom/8639_upper-east (735), https://data.28hse.com/en/kl/mei-foo/2520_mei-foo-sun-chuen (691), https://data.28hse.com/en/kl/whampoa/3060_whampoa-garden (561)
contract	categorical	0	2	Agreement (17200), Assignment (2800)
memo	numeric	0	20000	mean 19588314673410.906 · sd 498818990737.454 · 19022800150022 to 20112702480015
price	categorical	0	1909	$6M (455), $4M (376), $10M (273)
price_value	numeric	0	3016	mean 8092319.18945 · sd 7909575.369751 · 150000 to 180000000
holddate	categorical	0	6489	-- (3032), 3 years 142 days (25), 3 years 145 days (23)
winloss_flag	numeric	0	3	mean 0.60005 · sd 0.779757 · -1 to 1
winloss	categorical	0	861	-- (3680), 30% (161), 33% (154)
act_area	categorical	0	911	-- (10721), 517ft² (294), 194ft² (112)
area	categorical	0	1154	-- (5936), 639ft² (185), 680ft² (165)

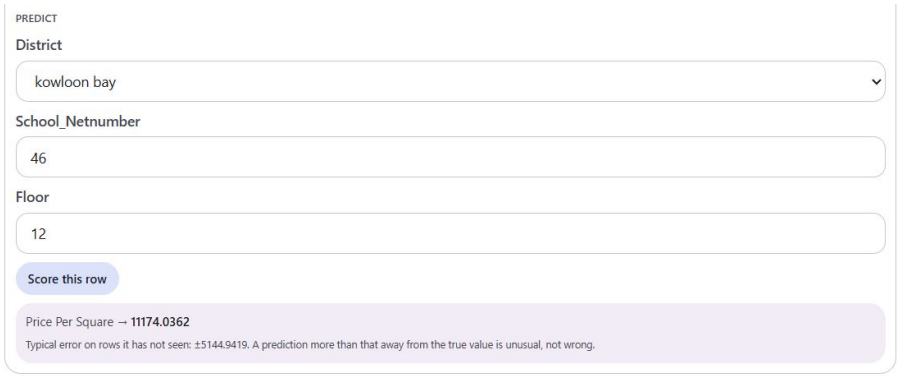
ETL Data Visualisation



Model Training (Random Forest)

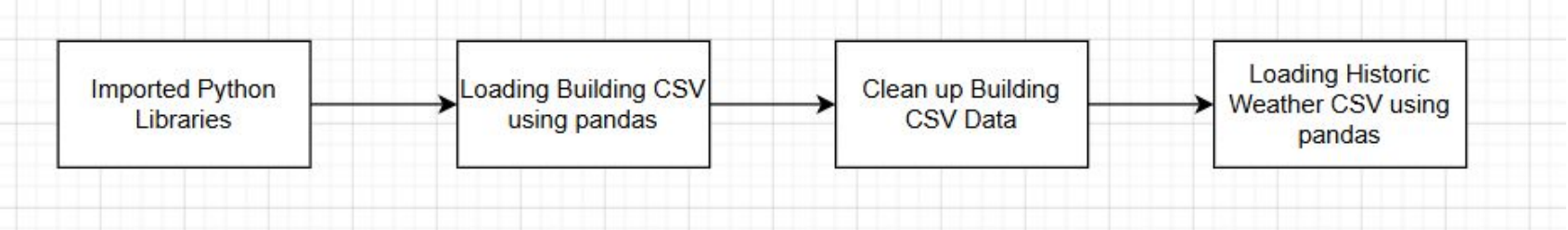


Prediction Result



HK Building Record Data Cleaning

Below are the Flow Chart with the Process of Data Collection to Data Cleaning on Historic Building Data



This Tasks is to conduct Data Collection and Cleaning

Step 1: Import the required libraries:

```
import pandas as pd
```

Step 2: Loading Historic Weather CSV using pandas

```
#Loading orginal CSV file containing historic property data:
hkbuildingdata = pd.read_csv('hkbuildinginfo.csv', encoding='ISO-8859-1')
hkbuildingdata.head(5)
```

HK Building Record Data Cleaning

Step 3: Clean up the data

```
# Check for missing values
print("\nMissing values in each column:")
print(hkbuildingdata.isnull().sum())

#Handle missing values by Dropping rows with missing values
hkbuildingdata.drop(columns=['withpre', 'rootid', 'fatherid', 'catid', 'source', 'block', 'state',
      'date_dm'],inplace=True, errors='ignore')

# Assuming 'floor' is the column causing the error
hkbuildingdata['floor'] = hkbuildingdata['floor'].astype(str)  # Convert to string type

# Now you can safely use the .str accessor
hkbuildingdata['floor'] = hkbuildingdata['floor'].str.replace(r'[^\d]', '', regex=True) #Remove non-numeric
      character

# Convert back to numeric if needed
hkbuildingdata['floor'] = pd.to_numeric(hkbuildingdata['floor'], errors='coerce')  # Convert to numeric

# Fill missing values for 'floor' and 'room'
median_value = hkbuildingdata['floor'].median()
hkbuildingdata['floor'] = hkbuildingdata['floor'].fillna(median_value )  # Fill with median for numeric data
mode_value = hkbuildingdata['room'].mode()[0]
hkbuildingdata['room'] = hkbuildingdata['room'].fillna(mode_value)   # Fill with mode for categorical data
```

```
Missing values in each column:
date                    0
withpre             23564
id                      0
rootid                  0
fatherid                0
catid                   0
catname                 0
catfathername           0
url_father              0
url_cat                 0
source              11556
contract                0
memo                    0
price                   0
price_value             0
holddate                0
winloss_flag            0
winloss                 0
act_area                0
area                    0
arearaw                 4
sq_price                0
sq_price_value          0
sq_actprice             0
sq_actprice_value       0
month                   0
day                     0
date_dm                 0
date_y                  0
block               32308
state               77003
floor                  32
room                  258
addr                    0
Unnamed: 34         80000
dtype: int64
```

HK Building Record Data Cleaning

#Continue from previous page

```
# Rename columns to more meaningful names
hkbuildingdata.rename(columns={'floor': 'Floor', 'sq_price_value': 'Price Per Square', 'catname': 'Building Name', 'catfathername':
        'District'}, inplace=True)

# Display the cleaned DataFrame and check for remaining missing values
print("\nCleaned DataFrame:")
print(hkbuildingdata.head())

print("\nRemaining Missing Values:")
print(hkbuildingdata.isnull().sum())
```

Step 4: Loading Weather CSV using pandas

```
#Loading orginal CSV file containing historic weather data:
weatherdata = pd.read_csv("historic_weatherdata.csv")
print(weatherdata.head(5))
weatherdata['date'] = pd.to_datetime(weatherdata['date']).dt.strftime('%Y-%m-%d')
```

```
Cleaned DataFrame:
         date      id       Building Name       District  \
0  2020-11-27  683333     Bel Air Heights   Diamond Hill
1  2020-11-27  683332       Fa Yuen Plaza       Mong Kok
2  2020-11-27  683331      Caldecott Hill  Yau Yat Tsuen
3  2020-11-27  683330    Pang Ching Court   Wong Tai Sin
4  2020-11-27  683329  Metro Harbour View   Tai Kok Tsui

                                        url_father  \
0   https://data.28hse.com/en/kl/diamond-hill
1       https://data.28hse.com/en/kl/mong-kok
2  https://data.28hse.com/en/kl/yau-yat-tsuen
3   https://data.28hse.com/en/kl/wong-tai-sin
4   https://data.28hse.com/en/kl/tai-kok-tsui

                                             url_cat   contract  \
0  https://data.28hse.com/en/kl/diamond-hill/2058...  Agreement
1  https://data.28hse.com/en/kl/mong-kok/2712_fa-...  Agreement
2  https://data.28hse.com/en/kl/yau-yat-tsuen/309...  Agreement
3  https://data.28hse.com/en/kl/wong-tai-sin/5035...  Agreement
4  https://data.28hse.com/en/kl/tai-kok-tsui/2564...  Agreement

            memo   price  price_value  ... Price Per Square  sq_actprice  \
0  20112702480015    $10M     10000000  ...          12106.54       $15221
1  20112702470044  $4.28M      4280000  ...          12334.29       $20000
2  20112702440018    $11M     11000000  ...           8580.34       $11100
3  20112702410017   $4.9M      4900000  ...           8404.80           --
4  20112702400032  $7.15M      7150000  ...          12016.81       $17354

  sq_actprice_value month day  date_y Floor  room                addr  \
0          15220.70   Nov  27    2020   2.0     E  BLOCK 2 2#/F Room E
1          20000.00   Nov  27    2020   1.0     C          1#/F Room C
2          11099.90   Nov  27    2020   7.0     E   TOWER 1 7/F Room E
3              0.00   Nov  27    2020   1.0    22         1#/F Room 22
4          17354.37   Nov  27    2020   1.0     B  BLOCK 8 1#/F Room B
```

HK School Net Number CSV File

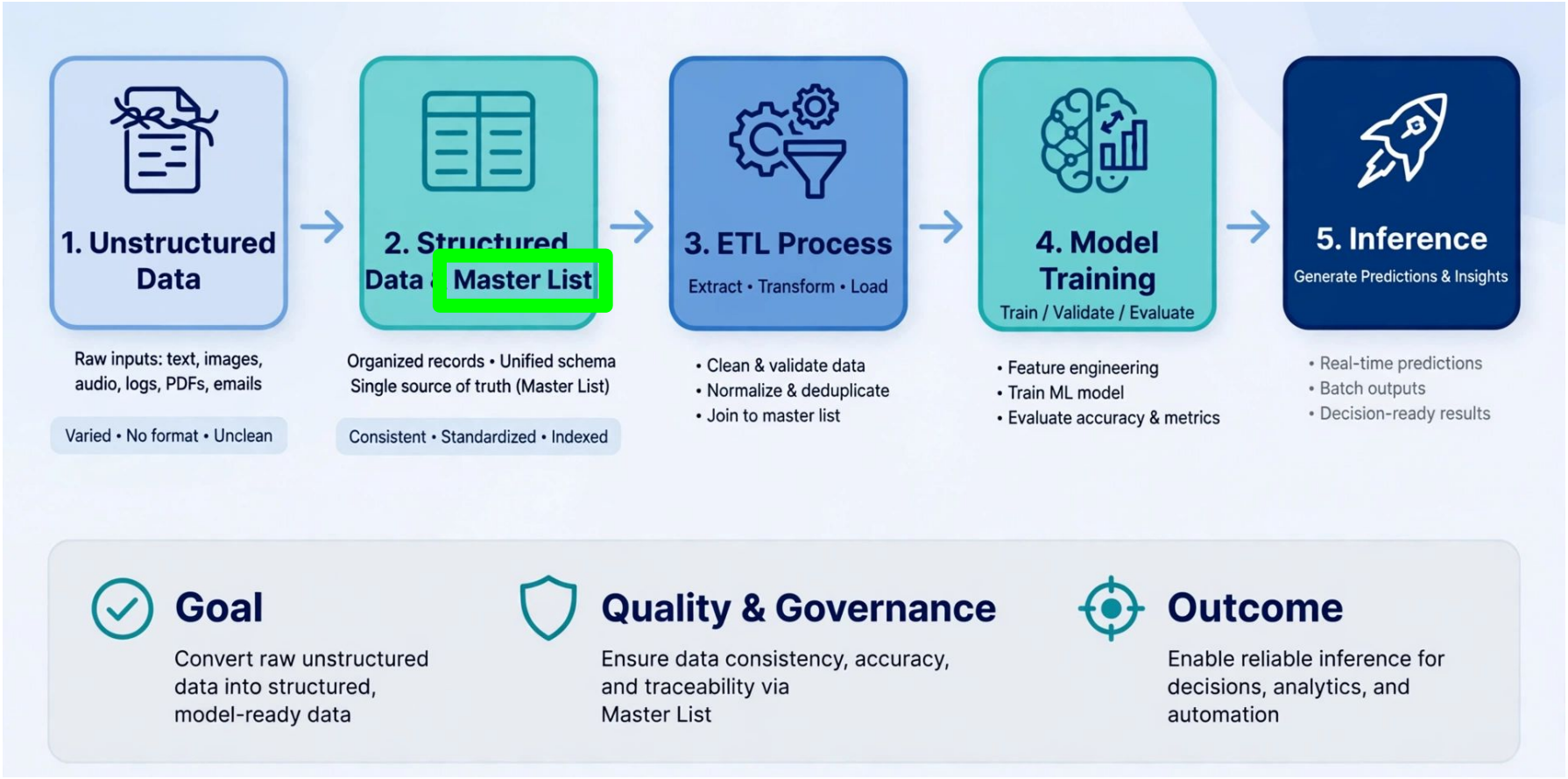
Data from Manual Input

District and POA School Net Look-up Table

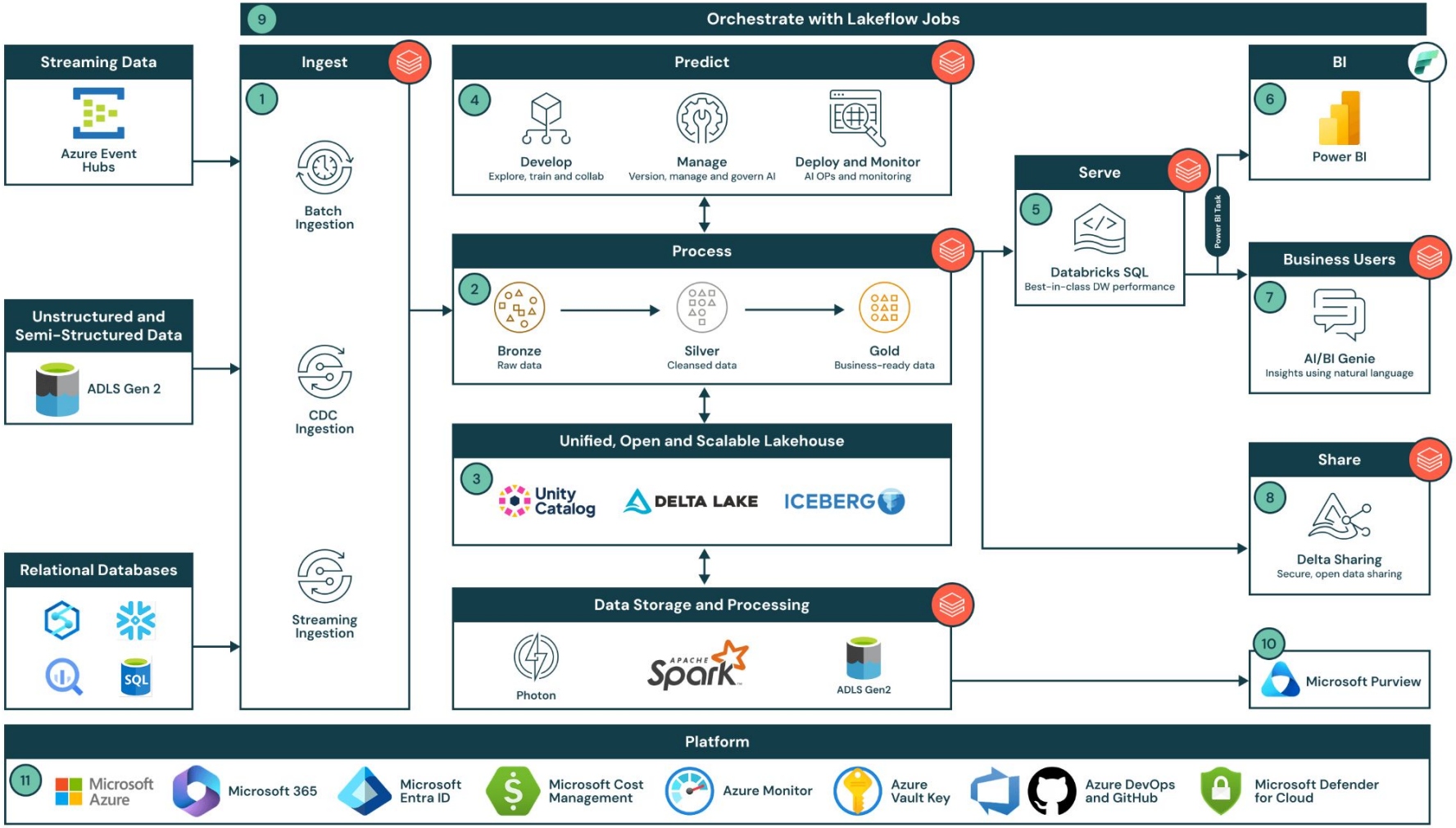
https://www.edb.gov.hk/attachment/en/edu-system/primary-secondary/spa-systems/primary-1-admission/school-lists/District_Net_LookUpTable.pdf

區域 District	小一學校網編號 POA School Net Number
中西區 Central & Western District	11
灣仔 Wan Chai	12
北角/筲箕灣/柴灣 North Point/ Shau Kei Wan/ Chai Wan	14, 16
薄扶林/香港仔/黃竹坑/赤柱 Pok Fu Lam/ Aberdeen/ Wong Chuk Hang/ Stanley	18
油麻地/尖沙咀/旺角 Yau Ma Tei/ Tsim Sha Tsui/ Mong Kok	31, 32
何文田/紅磡/九龍城 Ho Man Tin/ Hung Hom/ Kowloon City	34, 35, 41
深水埗 Sham Shui Po	40
黃大仙 Wong Tai Sin	43, 45
九龍灣/牛頭角/觀塘/油塘 Kowloon Bay/ Ngau Tau Kok/ Kwun Tong/ Yau Tong	46, 48
荃灣 Tsuen Wan	62
葵涌/荔景/大窩口/青衣 Kwai Chung/ Lai King/ Tai Wo Hau/ Tsing Yi	64, 65, 66
屯門 Tuen Mun	70, 71
天水圍/元朗 Tin Shui Wai/ Yuen Long	72, 73, 74
上水/粉嶺/北區 Sheung Shui/ Fanling/ North District	80, 81, 83
大埔 Tai Po	84
沙田 Shatin	88, 89, 91
西貢/將軍澳 Sai Kung/ Tseung Kwan O	95
離島/東涌 Islands/ Tung Chung	96, 97, 98, 99

Machine Learning Pipeline



Data Intelligence End-to-End Architecture



Master Data List Combination System Architecture

How to Merge Two DataFrames using Left Join

Matching on normalized District column • Using pandas • Output saved to CSV

Key: LEFT JOIN (how="left") • schoolnetnumber_df

1 • Normalize the District Column

- Convert both District columns to lovercase for matching

```
merged_df["District"] = merged_df["District"].str.lower()
schoolnetnumber_df["District"] = schoolnetnumber_df["District Areas()
```

Ensures consistent matching by lowercasing text

2 • Merge DataFrames (Left Join)

- Merge schoolnetnumber_df into merged_df using left join

```
merged_df = merged_df.merge(
schoolnetnumber_df["District", 'Sch_Net_No'].],
on="District", how="left")
```

Keeps all rows from merged_df; adds Sch_Net_No from schoolnetnumber_df

3 • Create & Fill School_Netnumber Column

- Fill missing values with 0 and convert to integer

```
merged_df["School_Netnumber"] = merged_df["Sch_Net_No"].
 .fillna(0).astupe(int)
```

0 = no record found in schoolnetnumber_df

4 • Save to CSV & Preview Results

- Export to CSV and view first 5 rows

```
merged_df.to_csv("combined_building_withweather_school.csv",
index=False)
merged_df.head(5)
```

Saved as: combined_building_withweather_school.csv

Tip: Left join keeps all rows from the left DataFrame (merged_df). Missing matches are filled as 0 in School_Netnumber.

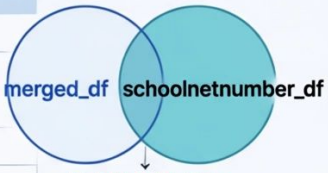
Master Data List Combination System Architecture

How to Merge Two DataFrames Using LEFT JOIN

Matching on District column • Keeping all rows from merged_df

merged_df (5 rows)

District	Temperature (°C)	Humidity (%)	Rainfall (mm)
Colombo	28	78	12
Kandy	24	82	8
Galle	29	80	15
Jaffna	31	70	2
Matara	27	85	10



schoolnetnumber_df (5 rows)

District Areas	Sch_Net_No
colombo	12
kandy	7
galle	3
negombo	9
anuradhapura	5

LEFT JOIN → keep all rows from merged_df, matching from schoolnetnumber_df

AFTER MERGE —
Resulting merged_df (5 rows)

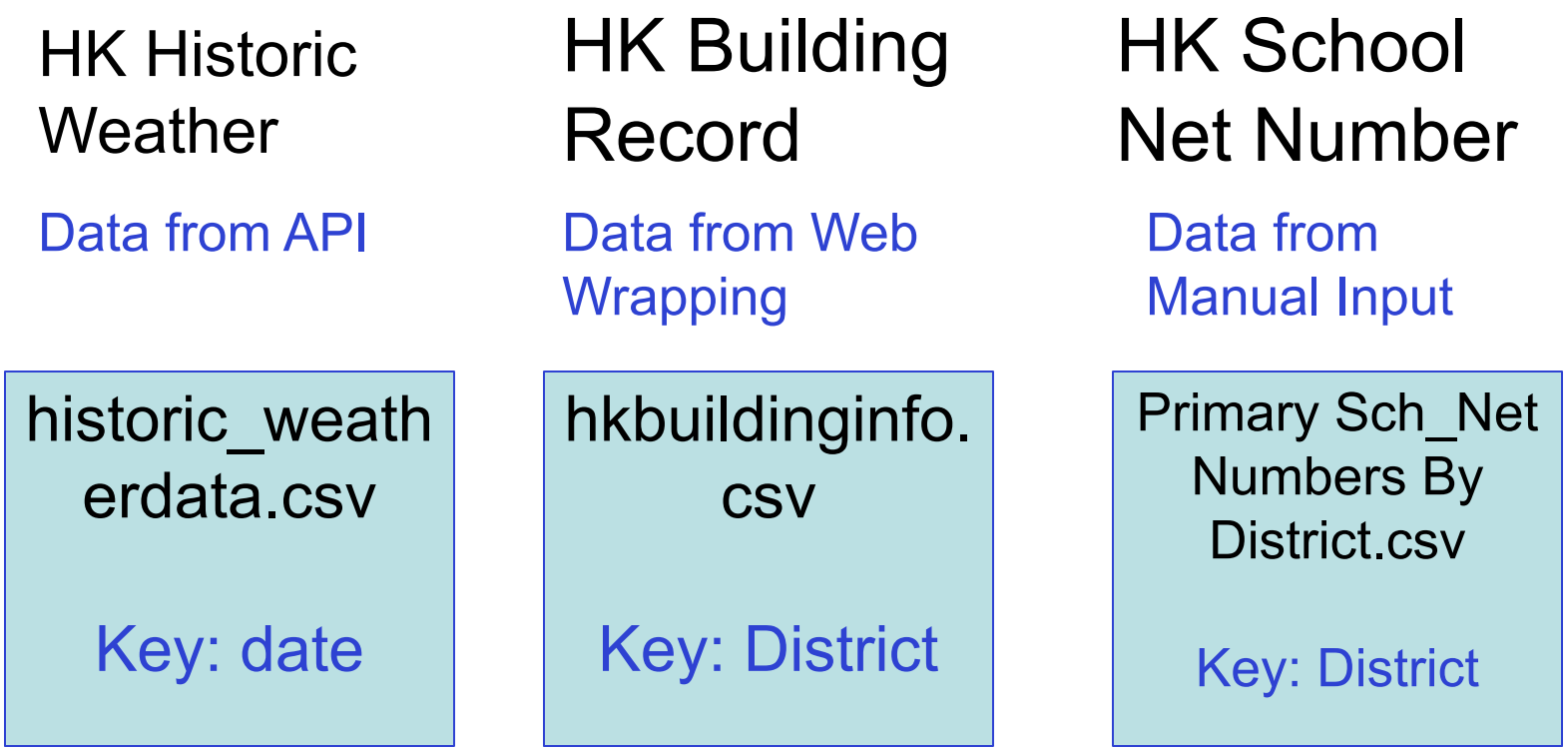
District	Temperature (°C)	Humidity (%)	Rainfall (mm)	Sch_Net_No	School_Netnumber
Colombo	28	78	12	12	12
Kandy	24	82	8	7	7
Galle	29	80	15	3	3
Jaffna	31	70	2	NaN	0
Matara	27	85	10	NaN	0

Key: LEFT JOIN keeps all rows from the left table (merged_df). Non-matching rows get NaN → fill with 0.

Note: Jaffna & Matara had no match → Sch_Net_No is NaN → replaced with 0 in School_Netnumber

```
Python Code
merged_df["District"] = merged_df["District"].str.lower()
schoolnetnumber_df["District"] = schoolnetnumber_df["District Areas"].str.lower() #Merge
# Merge_df = merged_df.merge(schoolnetnumber_df[["District", "Sch_Net_No"]], on=District", how=left")
# Create the "School_Netnumber" column and fill in 0 for no record found
merged_df["School_Netnumber"] = merged_df["Sch_Net_No"].fillna(0).astype(int)
merged_df.to.csv("combined_building_withweather_school.csv", index=False)
merged_df.head(5).
```

Master Data List Combination System Architecture



Master Data List Combination System Architecture

historic_weatherdata.csv

Key: date

date	wind	weather	maxtemp	mintemp	icon	psr
27/11/2020	North force 4 to 5.	Mainly cloudy with a few light rain patches at first. Becoming fin	20	15	51	Low
26/11/2020	North to northeast force 4, force 5 at first.	Fine. Rather cool in the morning. Very dry during the day.	19	13	92	Low
25/11/2020	Northeast force 4, east force 5 later.	Fine. Rather cool in the morning. Very dry during the day.	19	13	50	Low

Key: date

hkbuildinginfo.csv

date	withpre	id	rootid	fatherid	catid	catname	catfatherna	url_father	url_cat	source	contract	memo	price
27/11/2020	1	683333	2	79	2058	Bel Air Hei	Diamond H	https://data	https://data	Land Regis	Agreement	2.01E+13	$10M
27/11/2020	1	683332	2	71	2712	Fa Yuen Pl	Mong Kok	https://data	https://data	Land Regis	Agreement	2.01E+13	$4.28M
27/11/2020	1	683331	2	78	3094	Caldecott H	Yau Yat Ts	https://data	https://data	Land Regis	Agreement	2.01E+13	$11M
27/11/2020		683330	2	57	5035	Pang Ching	Wong Tai S	https://data	https://data	Land Regis	Agreement	2.01E+13	$4.9M
27/11/2020	1	683329	2	67	2564	Metro Harb	Tai Kok Ts	https://data	https://data	Land Regis	Agreement	2.01E+13	$7.15M
27/11/2020	1	683328	2	65	2521	Nob Hill	Mei Foo	https://data	https://data	Land Regis	Agreement	2.01E+13	$2.46M
27/11/2020	1	683327	2	62	5281	Chi Fung C	Sham Shui	https://data	https://data	Land Regis	Agreement	2.01E+13	$3.2M
27/11/2020	1	683326	2	56	2080	Telford Ga	Kowloon E	https://data	https://data	Land Regis	Agreement	2.01E+13	$7.68M
27/11/2020	1	683325	2	57	2104	Tsui Chuk	Wong Tai S	https://data	https://data	Land Regis	Agreement	2.01E+13	$3.67M
27/11/2020	1	683324	2	77	2548	Grand wate	To Kwa W	https://data	https://data	Land Regis	Agreement	2.01E+13	$9M
27/11/2020	1	683323	2	77	3073	Sun Cheon	To Kwa W	https://data	https://data	Land Regis	Agreement	2.01E+13	$4.2M
27/11/2020	1	683322	2	77	5461	Siu Fung C	To Kwa W	https://data	https://data	Land Regis	Agreement	2.01E+13	$4.9M
27/11/2020	1	683321	2	71	8587	SKYPARK	Mong Kok	https://data	https://data	Land Regis	Agreement	2.01E+13	$7.35M
27/11/2020	1	683320	2	127	5263	Aria	Ngau Chi W	https://data	https://data	Land Regis	Agreement	2.01E+13	$8.25M
27/11/2020	1	683319	2	60	5948	One Victor	Ho Man Ti	https://data	https://data	Land Regis	Agreement	2.01E+13	$8.83M
27/11/2020	1	683318	2	77	22478	Wan Tung	To Kwa W	https://data	https://data	Land Regis	Agreement	2.01E+13	$5.43M
27/11/2020		683317	2	56	2082	Richland G	Kowloon E	https://data	https://data	Land Regis	Agreement	2.01E+13	$5.67M
27/11/2020		683316	2	62	2429	Cheung Ho	Sham Shui	https://data	https://data	Land Regis	Agreement	2.01E+13	$3.73M
27/11/2020	1	683315	2	60	2243	Dragon Vie	Ho Man Ti	https://data	https://data	Land Regis	Agreement	2.01E+13	$9.8M

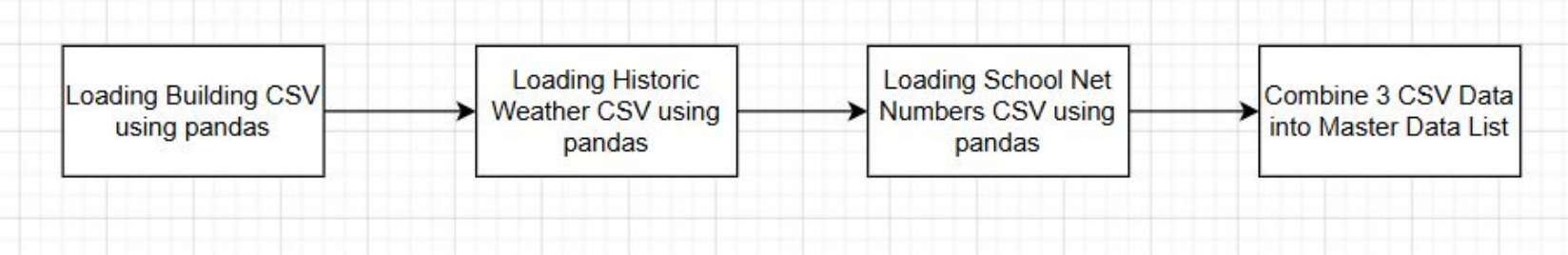
Primary Sch_Net Numbers By District.csv

Key: District

Sch_Net_No	District Areas	Districts
1	CENTRAL	Hong Kong Island
1	THE PEAK	Hong Kong Island
1	SHEUNG WAN	Hong Kong Island
1	SAI YING PUN	Hong Kong Island
1	SHEK TONG TSUI	Hong Kong Island
1	KENNEDY TOWN	Hong Kong Island
1	WAN CHAI	Hong Kong Island
1	CAUSEWAY BAY	Hong Kong Island
1	SO KON PO	Hong Kong Island
1	HAPPY VALLEY	Hong Kong Island
1	TAI HANG	Hong Kong Island
1	JARDINE?? LOOKOUT	Hong Kong Island
1	NORTH POINT	Hong Kong Island
1	QUARRY BAY	Hong Kong Island
1	LEI KING WAN	Hong Kong Island
1	SHAU KEI WAN	Hong Kong Island
1	SAI WAN HO	Hong Kong Island
1	SHEK O	Hong Kong Island

Master Data List Combination System Architecture

Below are the Flow Chart with the Process of Combining 3 CSV Data to a master list



#Combine Weather Data with Building Data

```
hkbuildingdata['date'] = pd.to_datetime(hkbuildingdata['date']).dt.strftime('%Y-%m-%d')
merged_df = hkbuildingdata.merge(weatherdata[['date', 'wind', 'maxtemp', 'mintemp']], on='date', how='left')
```

#Combine Combined Data with School Net Number Data

```
#Loading orginal CSV file from school net number data
schoolnetnumber_df = pd.read_csv("Primary Sch_Net Numbers By District.csv")
schoolnetnumber_df.head(5)
```

Combining Weather, Building Information and School Net Numbers

Normalize the 'District' column in both DataFrames to lowercase

```
merged_df['District'] = merged_df['District'].str.lower()
schoolnetnumber_df['District'] = schoolnetnumber_df['District Areas'].str.lower()
```

Merge schooldistrict_df with districtcode_df on the normalized 'District' column

```
merged_df = merged_df.merge(schoolnetnumber_df[['District', 'Sch_Net_No']], on='District', how='left')
```

Create the 'School_Netnumber' column and fill in 0 for not record found

```
merged_df['School_Netnumber'] = merged_df['Sch_Net_No'].fillna(0).astype(int)
```

Save the merged_df into csv with name "combined_building_withweather_school.csv"

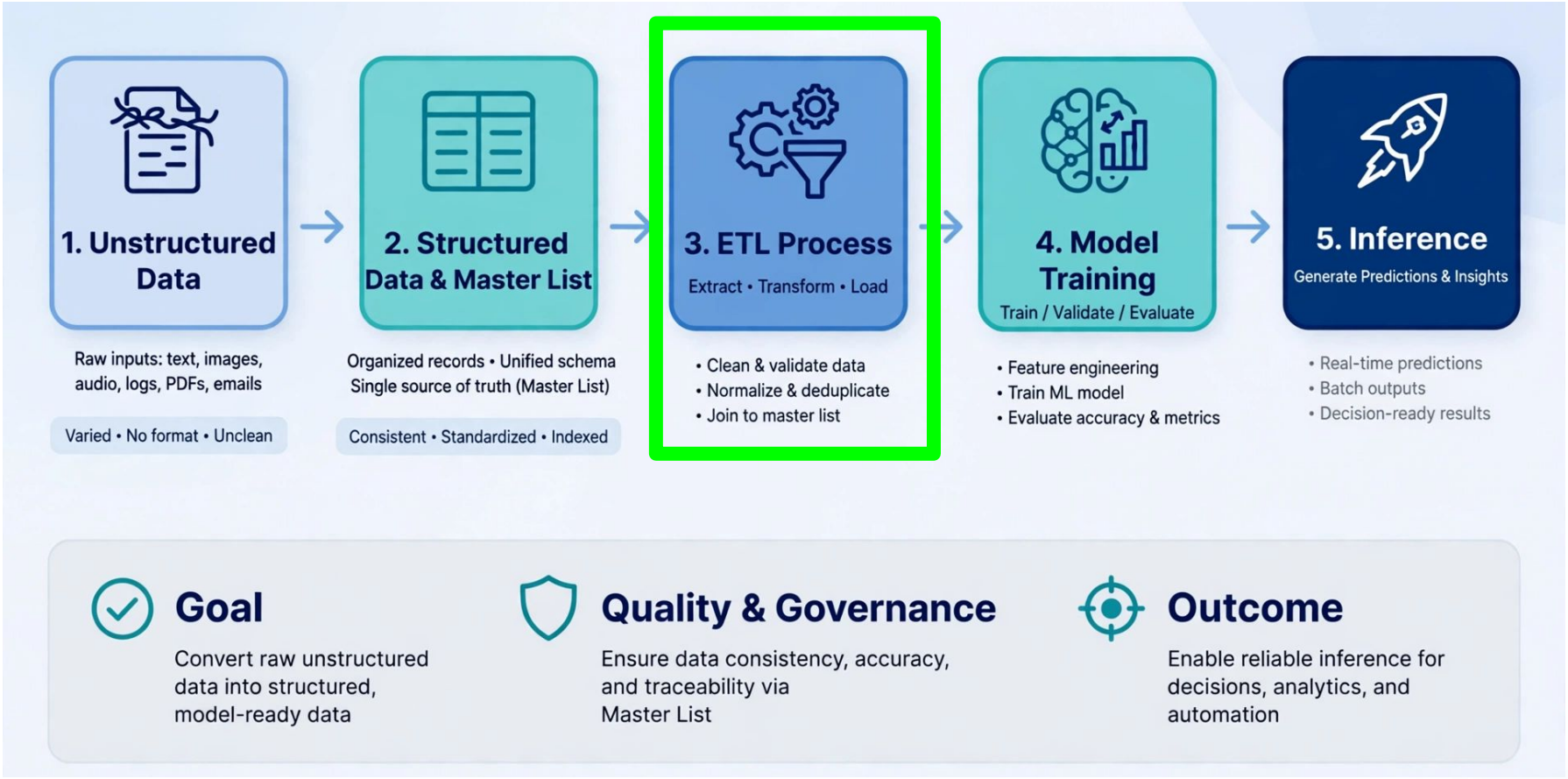
```
merged_df.to_csv('combined_building_withweather_school.csv', index=False)
merged_df.head(5)
```

[27]:

	date	id	Building Name	District	url_father	url_cat	contract	memo	price	price_va
0	2020-11-27	683333	Bel Air Heights	diamond hill	https://data.28hse.com/en/kl/diamond-hill	https://data.28hse.com/en/kl/diamond-hill/2058...	Agreement	20112702480015	$10M	10000(
1	2020-11-27	683332	Fa Yuen Plaza	mong kok	https://data.28hse.com/en/kl/mong-kok	https://data.28hse.com/en/kl/mong-kok/2712_fa-...	Agreement	20112702470044	$4.28M	4280(
2	2020-11-27	683331	Caldecott Hill	yau yat tsuen	https://data.28hse.com/en/kl/yau-yat-tsuen	https://data.28hse.com/en/kl/yau-yat-tsuen/309...	Agreement	20112702440018	$11M	11000(
3	2020-11-27	683330	Pang Ching Court	wong tai sin	https://data.28hse.com/en/kl/wong-tai-sin	https://data.28hse.com/en/kl/wong-tai-sin/5035...	Agreement	20112702410017	$4.9M	4900(
4	2020-11-27	683329	Metro Harbour View	tai kok tsui	https://data.28hse.com/en/kl/tai-kok-tsui	https://data.28hse.com/en/kl/tai-kok-tsui/2564...	Agreement	20112702400032	$7.15M	7150(

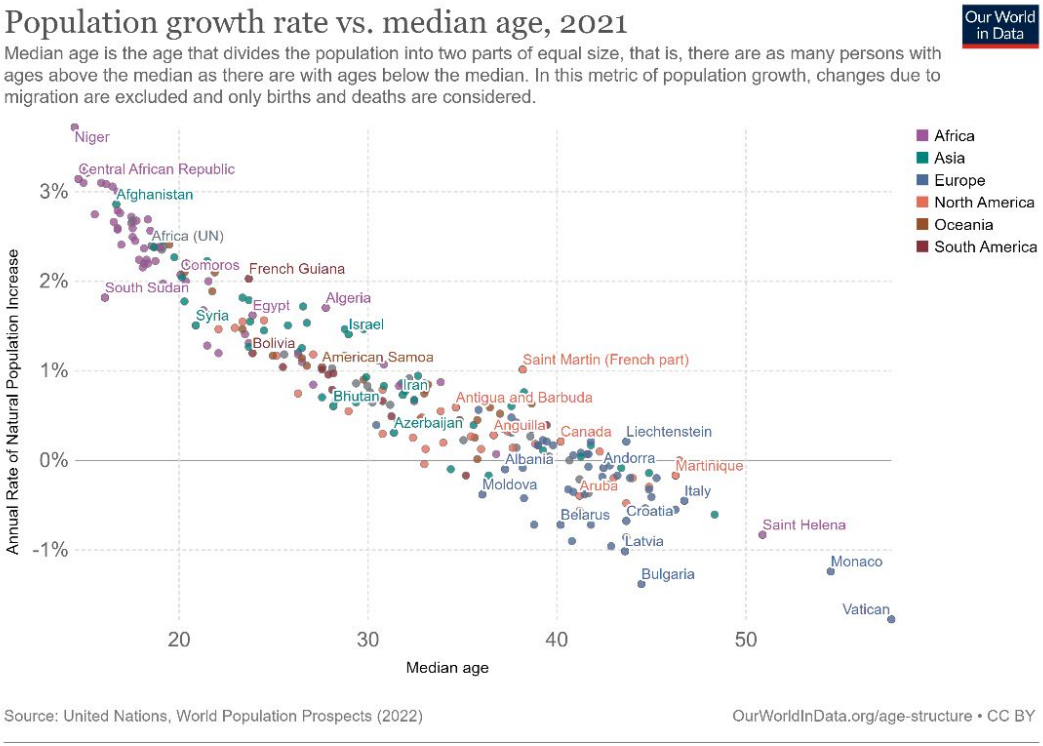
5 rows × 29 columns

Machine Learning Pipeline



Exploratory Data Analysis(EDA) and Data Visualization

- **EDA is the first step in understanding a dataset.** It uses summary statistics and visualizations to explore patterns, spot anomalies, and understand relationships between variables.
- **Purpose:** find trends, and unexpected behavior before building models or making decisions.
- **Common techniques**: use scatter plots to show the relationship between two variables



Exploratory Data Analysis(EDA) and Data Visualization

Exploring your data is a crucial step in data analysis. It involves:

Organising the data set

Plotting aspects of the data set

Maybe producing some numerical summaries; central tendency and spread, etc.

"Exploratory data analysis can never be the whole story, but nothing else can serve as the foundation stone."
- John Tukey.

Exploratory Data Analysis(EDA) and Data Visualization

Why?

- EDA encompasses the "*explore* data" part of the data science process
- EDA is crucial but often overlooked:
  - If your data is bad, your results will be bad
  - Conversely, understanding your data well can help you create smart, appropriate models

Exploratory Data Analysis(EDA) and Data Visualization

What?

1. Store data in data structure(s) that will be convenient for exploring/processing
   (Memory is fast. Storage is slow)
2. Clean/format the data so that:
   - Each row represents a single object/observation/entry
   - Each column represents an attribute/property/feature of that entry
   - Values are numeric whenever possible
   - Columns contain atomic properties that cannot be further decomposed*

* Unlike food waste, which can be composted.
Please consider composting food scraps.

What? (continued)

3. Explore **global** properties: use histograms, scatter plots, and aggregation functions to summarize the data
4. Explore **group** properties: group like-items together to compare subsets of the data (are the comparison results reasonable/expected?)

   This process transforms your data into a format which is easier to work with, gives you a basic overview of the data's properties, and likely generates several questions for you to follow-up in subsequent analysis.

HK Building Data - Filtering Methodology & 2-Year Camparison

Methodology: Master DataFrame & Two Filter Criteria

Master DataFrame

hkbuildingdata

- Created full_date column: pd.to_datetime(hkbuildingdata['month') + ' ' + hkbuildingdata['day').astupe(str) + ' '
- ' ' + hkbuildingdata['date_y').astupe(str))

Criteria 1: 2020 COVID Period

```
start_date = 2020-01-01
end_date = 2020-12-31
covidfiltered_data = hkbuildingdata[(hkbuildingdata["full_date"] >=
start_date) & (hkbuildingdata["full_date"] <= end_date)]
```

Purpose: Analyze visitor data during COVID-19 year

Records: 5,432

Criteria 2: 2019 Baseline (Pre-COVID)

```
start_date = 2019-01-01
end_date = 2019-12-31
filtered_data = hkbuildingdata[(hkbuildingdata["full_date"] >= start_date)
& (hkbuildingdata["full_date"] <= end_date)]
```

Purpose: Baseline comparison for prior year

Records: 5,210

Results — 2×2 Data Point Charts Comparison

2020 Avg Daily Visitors — Trend Over Time



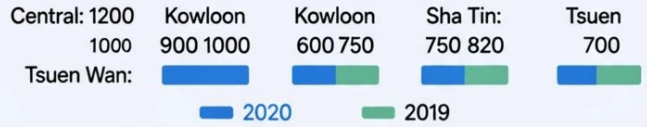
Peak: 1,240 visits (Jan) • Low: 520 visits (Apr)

2019 Avg Daily Visitors — Trend Over Time

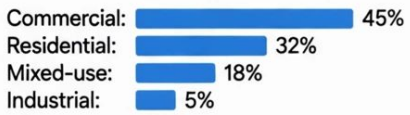


Peak: 1,180 visits (Oct) • Average: 980 visits

YoY Comparison by District — 2020 vs 2019



Building Type Breakdown (2020)



Method:
Pandas DataFrame Filtering

Time Range:
2019 & 2020 Yearly Comparison

Key Insight: 2020 shows ~18% average drop in visitors during Apr–Jun due to COVID-19

Exploratory Data Analysis(EDA) and Data Visualization

Exploratory Data Analysis and Visualization

#filter the data with period "2020-1-1" to "2020-12-31'

Create a full date column from 'month', 'day', and 'date_y'

hkbuildingdata['full_date'] = pd.to_datetime(hkbuildingdata['month'] + ' ' + hkbuildingdata['day'].astype(str) + ' ' +
hkbuildingdata['date_y'].astype(str))

Filter the DataFrame for the date range "2020-01-01" to "2020-12-31"

start_date = '2020-01-01'

end_date = '2020-12-31'

covidfiltered_data = hkbuildingdata[(hkbuildingdata['full_date'] >= start_date) & (hkbuildingdata['full_date'] <= end_date)]

Display the first few rows of the filtered DataFrame

print(covidfiltered_data.head(5))



	date	id	Building Name	District	url_father	url_cat	contract
0	2020-11-27	683333	Bel Air Heights	diamond hill	https://data.28hse.com/en/kl/diamond-hill	https://data.28hse.com/en/kl/diamond-hill/2058...	Agreement
1	2020-11-27	683332	Fa Yuen Plaza	mong kok	https://data.28hse.com/en/kl/mong-kok	https://data.28hse.com/en/kl/mong-kok/2712_fa-...	Agreement
2	2020-11-27	683331	Caldecott Hill	yau yat tsuen	https://data.28hse.com/en/kl/yau-yat-tsuen	https://data.28hse.com/en/kl/yau-yat-tsuen/309...	Agreement
3	2020-11-27	683330	Pang Ching Court	wong tai sin	https://data.28hse.com/en/kl/wong-tai-sin	https://data.28hse.com/en/kl/wong-tai-sin/5035...	Agreement
4	2020-11-27	683329	Metro Harbour View	tai kok tsui	https://data.28hse.com/en/kl/tai-kok-tsui	https://data.28hse.com/en/kl/tai-kok-tsui/2564...	Agreement

Filter the DataFrame for the date range "2019-01-01" to "2019-12-31"

start_date = '2019-01-01'

end_date = '2019-12-31'

filtered_data = hkbuildingdata[(hkbuildingdata['full_date'] >= start_date) & (hkbuildingdata['full_date'] <= end_date)]

Display the first few rows of the filtered DataFrame

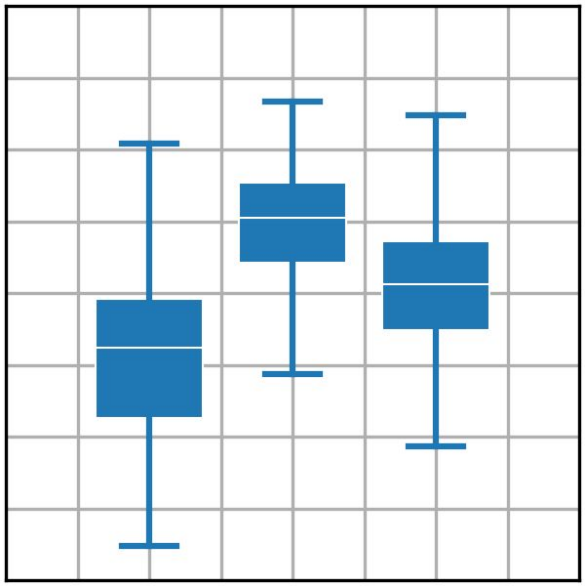
print(filtered_data.head(5))

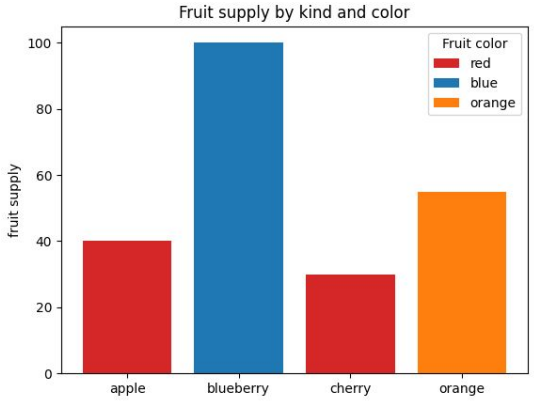
Matplotlib: Visualization with Python

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. Matplotlib makes easy things easy and hard things possible.

Features:

1. Create publication quality plots.
2. Make interactive figures that can zoom, pan, update.
3. Customize visual style and layout.
4. Export to many file formats.
5. Embed in JupyterLab and Graphical User Interfaces.
6. Use a rich array of third-party packages built on Matplotlib.





https://matplotlib.org/

Exploratory Data Analysis(EDA) and Data Visualization

```
import matplotlib.pyplot as plt

# Create a scatter plot using matplotlib
plt.figure(figsize=(10, 6))

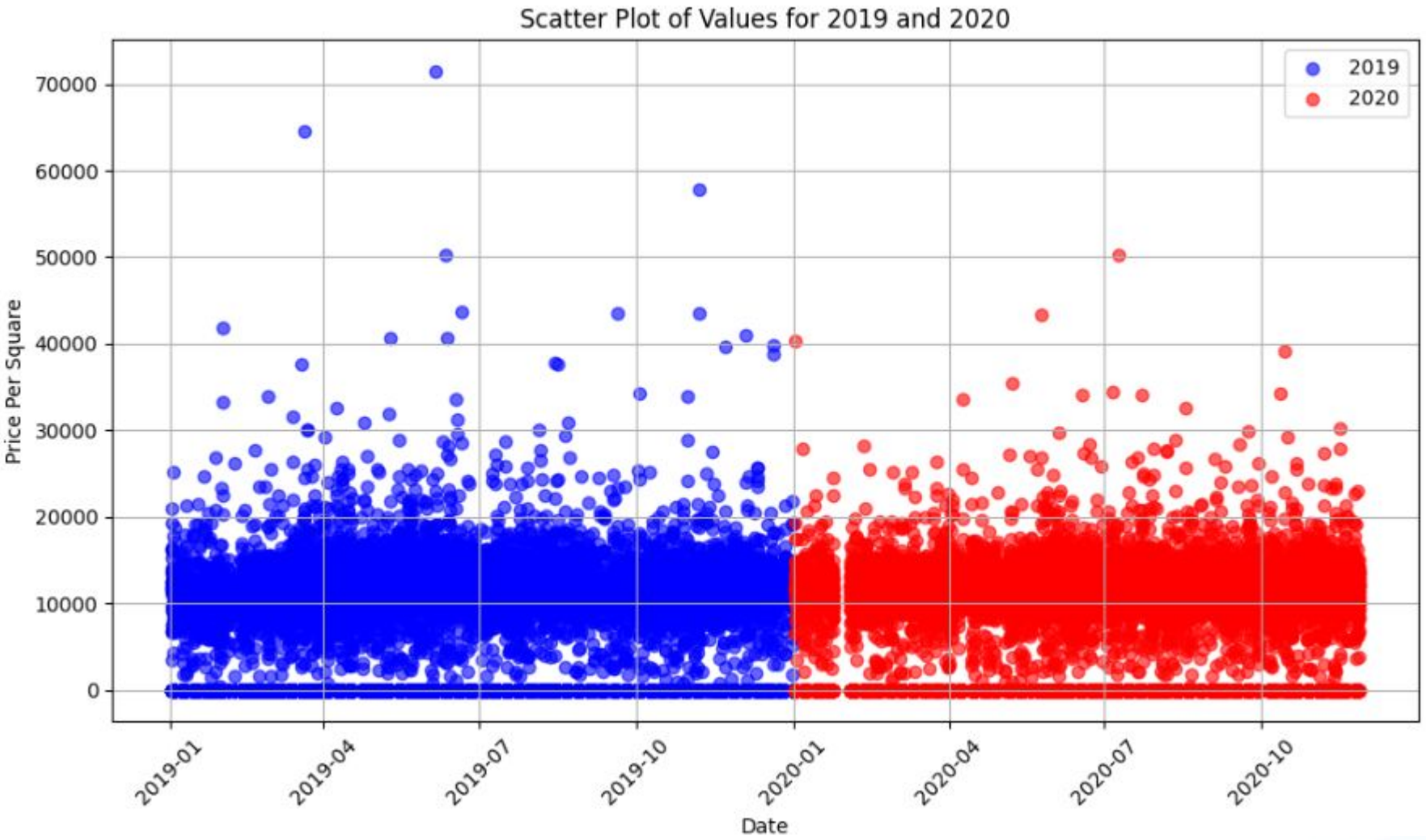
# Plot for 2019
xaxis_value = filtered_data['full_date']
yaxis_value = filtered_data['Price Per Square']
plt.scatter(xaxis_value, yaxis_value, color='blue', label='2019', alpha=0.6)

# Plot for 2020
xaxis_value = covidfiltered_data['full_date']
yaxis_value = covidfiltered_data['Price Per Square']
plt.scatter(xaxis_value, yaxis_value, color='red', label='2020', alpha=0.6)

# Adding titles and labels
plt.title('Scatter Plot of Values for 2019 and 2020')
plt.xlabel('Date')
plt.ylabel('Price Per Square')
plt.xticks(rotation=45)
plt.legend()
plt.grid()

# Show the plot
plt.tight_layout()
plt.show()
```

Exploratory Data Analysis(EDA) and Data Visualization



Exploratory Data Analysis(EDA), Scatter Chart

Understanding the Scatter Plot Chart Components Breakdown

Chart Components Explained

Title
Describes the chart's overall purpose
Scatter Plot of Values for 2019 and 2020. Tells the viewer what the chart is about at a glance.

Legend
Identifies each data series
Blue = 2019 data points. Red = 2020 data points.
Helps distinguish the two datasets.

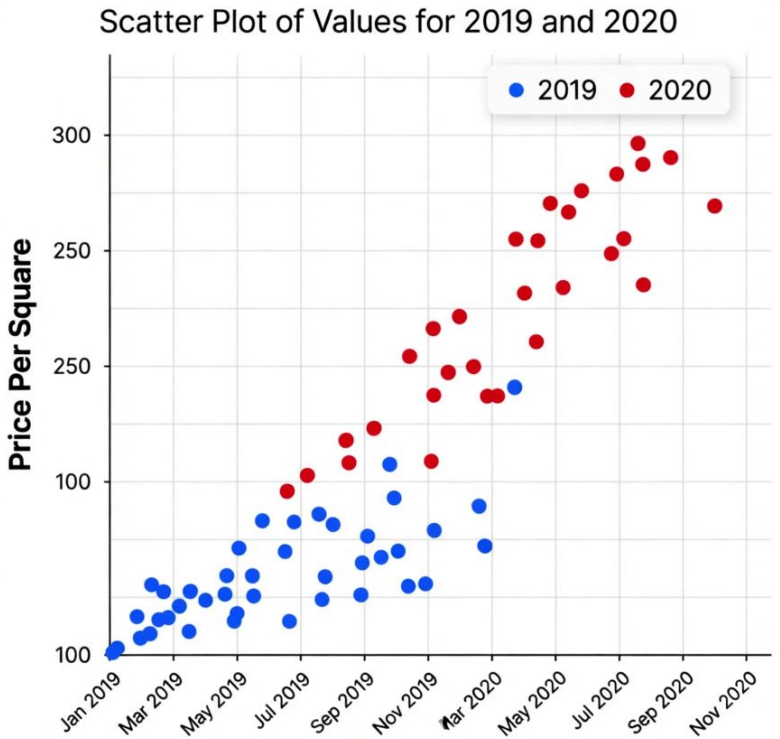
X-Axis (Horizontal)
Date
Represents time, shown as dates from 2019 to 2020.
Labeled as "**Date**" and rotated for readabielly.

Y-Axis (Vertical)
Price Per Square
Represents the measured value.
Labeled as "**Price Per Square**", showing the nuneric scale.

Data Points & Grid
Observations & Reference
Blue and red statter points represent individual observations for each year. Gridlines help read values easily.

Python Code Used:

```
plt.scatter(x, y, color="blue", label="2019",
alpha=0.6). # 2019 data
plt.scatter(x, y, color="red", label="2020",
alpha=0.6) # 2020 data
plt.title'Scatter Plot of Values for 2019 and 2020
pltchabeld Date
plt.legend".Price Per Square
```

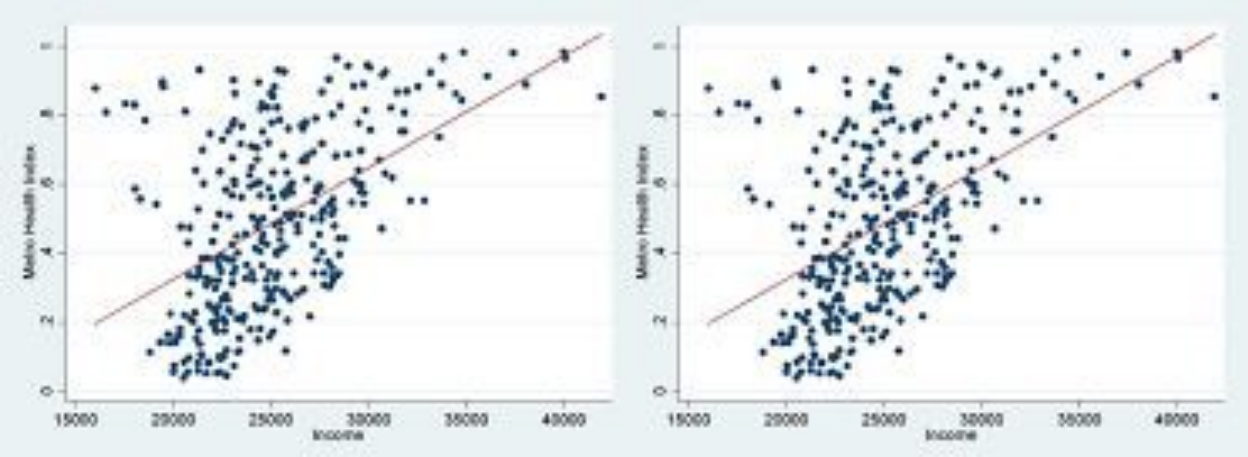


Key Insight: Comparing 2019 vs 2020 shows how Price Per Square increased in 2020, visualized by the higher cluster of red points.

Exploratory Data Analysis(EDA) and Data Visualization

Tasks 1 - Housing Analysis

1. Change to different period from 2019-2020 and 2020-2021 to 2014-2017, 2018-2021
2. Filter the data with school net number “34”, “35”, or others
3. Visualise the result in Scatter Plot
4. Change the chart title, axis name, color, legend.



Common Panda functions

Grouping/Splitting/Aggregating:

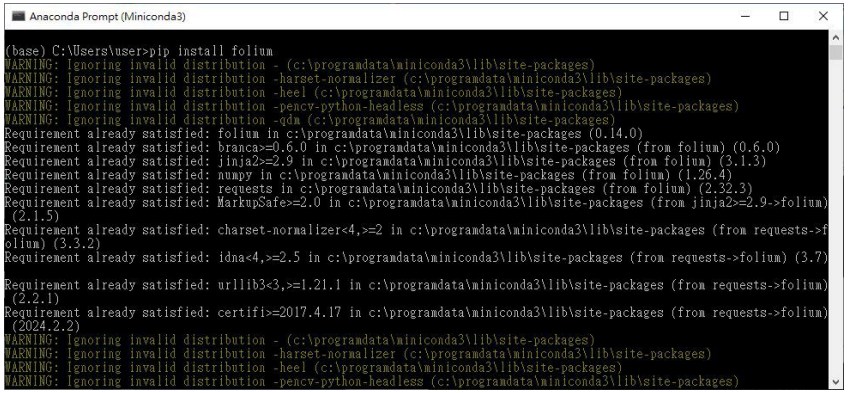
- groupby(), .get_groups()
- .merge()
- .concat()
- .aggegate()
- .append()

Exploratory Data Analysis(ETA), Dynamic Data Visualization

1. Open tools “Anaconda Prompt” which will direct you to the command prompt.



1. Install python package requests by entering “pip install pygwalker”

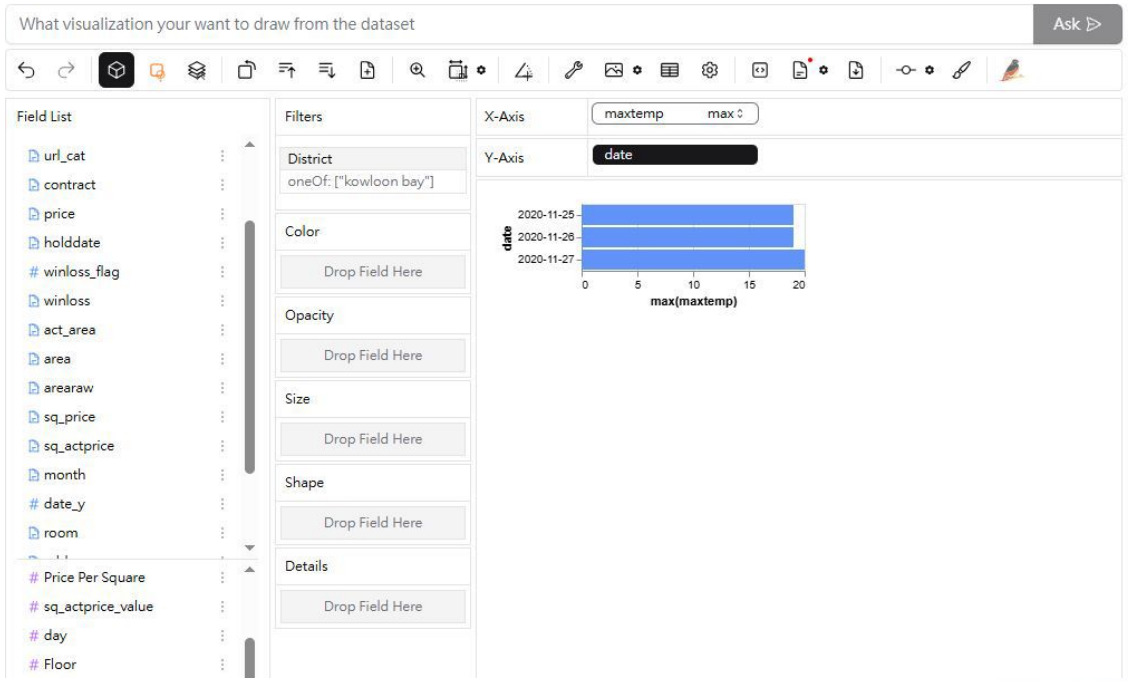


Exploratory Data Analysis(ETL), Dynamic Data Visualization

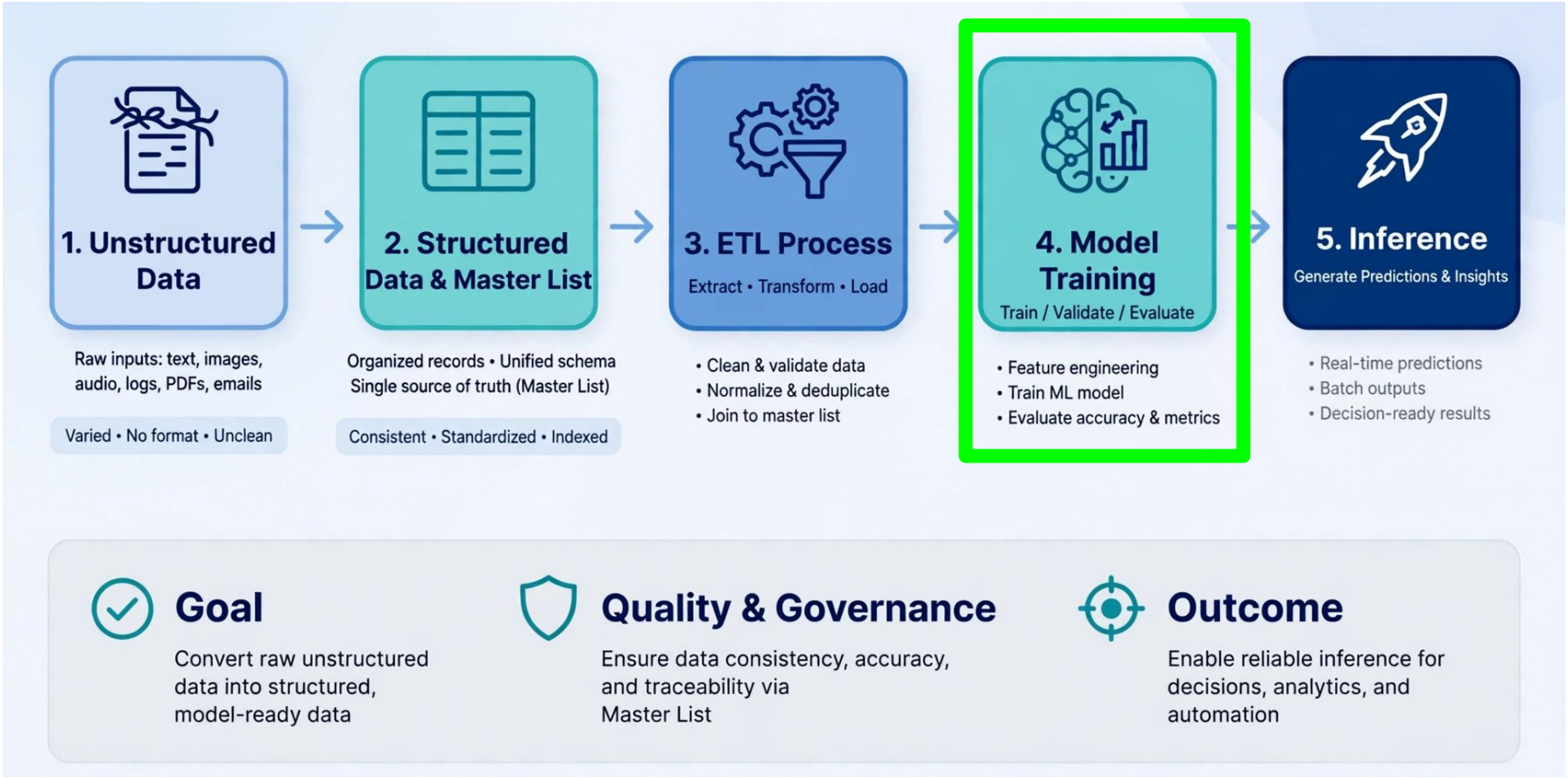
We mainly use PyGWalker (Python Library) for Dynamic Data Visualization, similar to use PowerBI or Tableau.
Official Information can be found in https://kanaries.net/pygwalker

Write python code

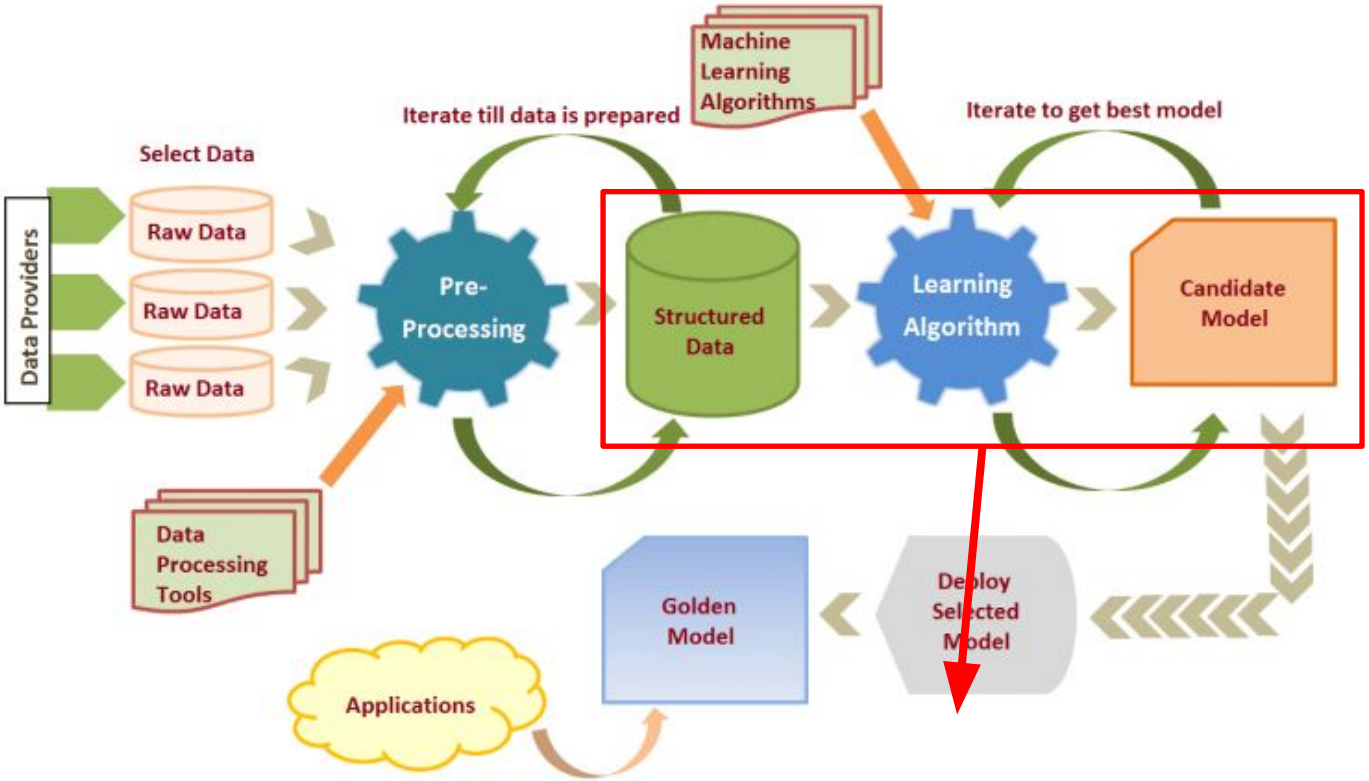
```
import pandas as pd
import pygwalker as pyg
df = pd.read_csv("combined_building_withweather_school.csv")
pyg.walk(df)
```

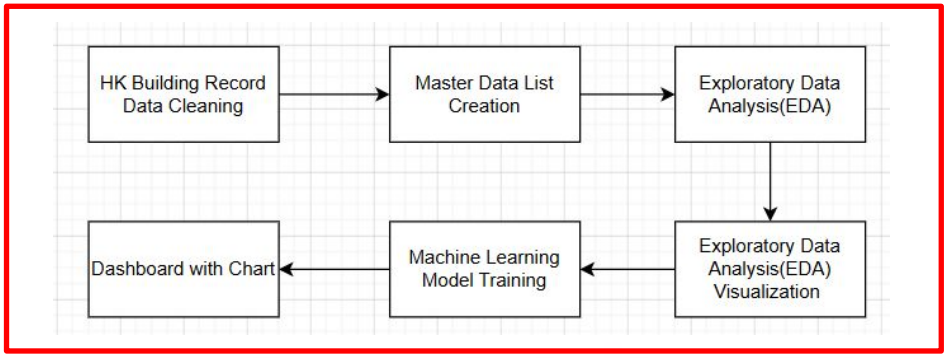


Machine Learning Pipeline



Applied Machine Learning Pipeline Overview





How To Do Data Mining: Machine Learning Techniques

Features / Attributes

Are used to train Learning algorithm

- Sales Price
- Saleable Area
- Number of Bedrooms
- Year built
- (Refurbished) Schools in the area
- Transportation (MTR, buses)

Address: 270-280 QUEEN'S ROAD WEST
Price per sqft.: 17,841
Qty transaction: 2
Year: 2017



Using a Home Valuation Prediction Model as an example

How To Do Data Mining: Machine Learning Techniques

Machine Learning

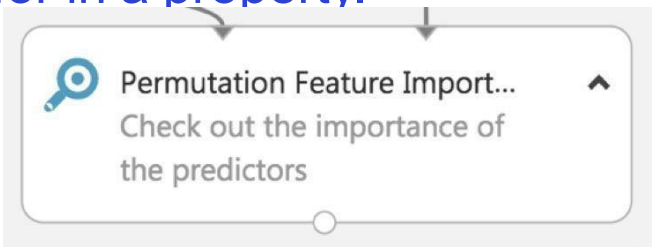
Feature	Score
SaleableArea	0.858545
Eff_Build_Ages	0.166877
Bus_Route	0.076519
Parks	0.019011

Choosing a 'useful' or ***significant*** feature (predictors) ***makes*** or ***breaks*** a data science project.

Domain experts, property agents will tell you what property buyers value in a home.

Data scientists would ***NOT*** have a clue what home buyers look for in a property.

But not all is lost!



How To Do Data Mining: Machine Learning Techniques

Features / Attributes

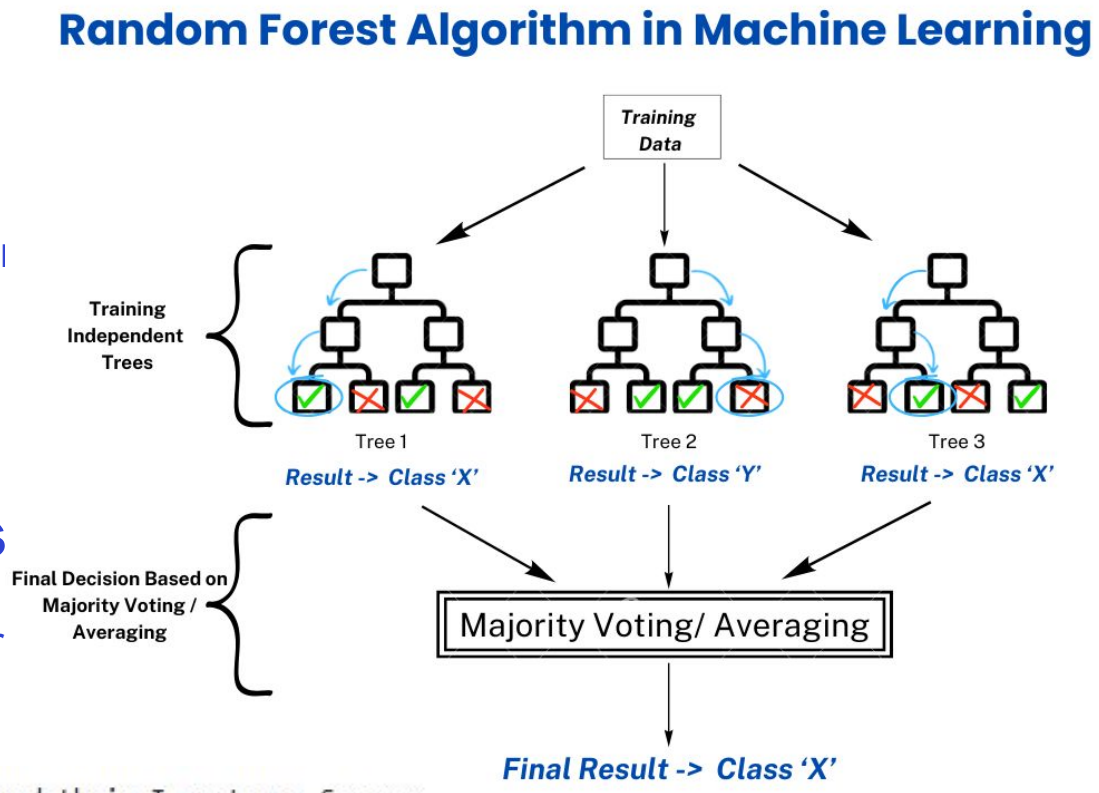
Set X using RandomForestRegressor

Parameter

1. **District**
2. **Floor**
3. **School Net Numbers**

Set y using RandomForestRegressor

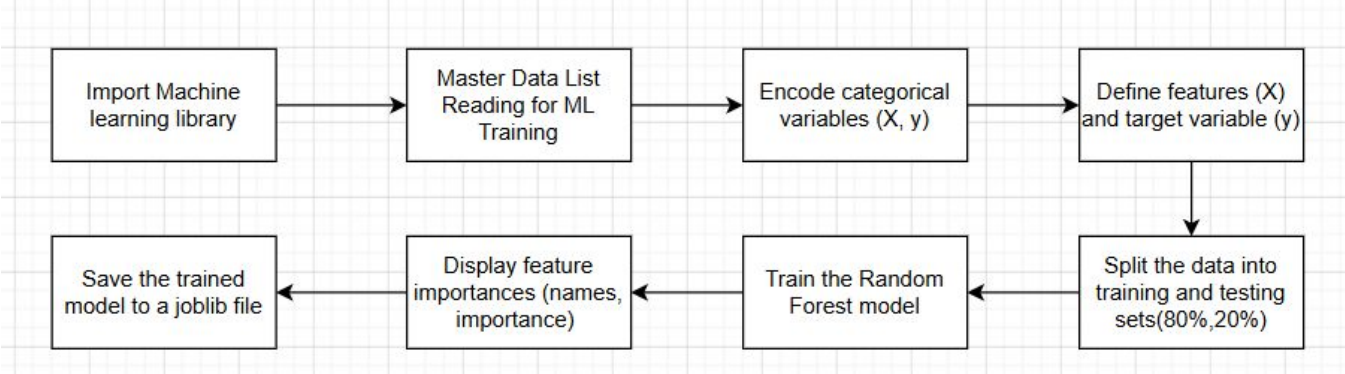
Outcome - Price Per Square




```
Top 5 Features and their Importance Scores:
           Feature  Importance
0         District    0.728459
2  School_Netnumber    0.199190
1            Floor    0.072351
```

Machine Learning Random Forest Model training

Below are the Flow Chart with the Process of Machine Learning Random Forest Model Training



#This code is going to train an random forest model to find top important features

#Step 1 : Import necessary library

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestRegressor
from sklearn.tree import plot_tree
import joblib  # Import joblib for saving the model
```

Machine Learning Random Forest Model training

#Step 2 : Master Data List Reading for ML Training

```
data = pd.read_csv("combined_building_withweather_school.csv")
```

#Step 3 : Encode categorical variables (X, y)

```
label_encoder = LabelEncoder()
data['District'] = label_encoder.fit_transform(data['District'])
data['School_Netnumber'] = label_encoder.fit_transform(data['School_Netnumber'])
```

#Step 4: Define features and target variable

```
X = data[['District', 'Floor', 'School_Netnumber']]  # Features
y = data['Price Per Square']  # Target variable

# Check the distribution of the target variable
print(y.value_counts())
print(X)

# Ensure the target variable is continuous
print(f"Target variable type: {y.dtype}")
```

```
Price Per Square
0.00        30508
10000.00      217
8333.33       109
8000.00        81
12500.00       70
             ...
8554.05          1
13764.04         1
10443.86         1
15421.30         1
4672.90          1
Name: count, Length: 31331, dtype: int64
       District  Floor  School_Netnumber
0             1    2.0                 8
1            16    1.0                 2
2            31    7.0                 5
3            28    1.0                 7
4            24    1.0                 2
...         ...    ...               ...
79995         6    2.0                 9
79996         9    9.0                 6
79997         3    4.0                 4
79998         6    1.0                 9
79999        26    3.0                 1

[80000 rows x 3 columns]
```

Machine Learning Random Forest Model training

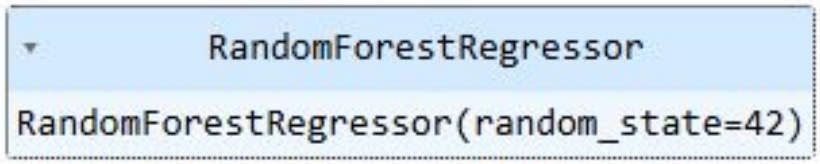
#Step 5: Split the dataset into training and testing sets

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

#Step 6: Train the Random Forest model

```
rf_model = RandomForestRegressor(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)
```

```
#Display a decision tree from the Random Forest with reduced size
plt.figure(figsize=(6, 6))  # Reduced by 10%
plot_tree(rf_model.estimators_[5], feature_names=X.columns.tolist(), filled=True)
plt.title("Decision Tree from Random Forest")
plt.show()
```



Machine Learning Random Forest Model training

#Step 8: Display feature importances

```
importances = rf_model.feature_importances_
feature_names = X.columns.tolist()
feature_importances = pd.DataFrame({'Feature': feature_names, 'Importance': importances})

# Sort by importance and display the top 5 features
top_features = feature_importances.sort_values(by='Importance', ascending=False).head(5)
print("Top 5 Features and their Importance Scores:")
print(top_features)
```



#Step 9: Save the trained model to a joblib file

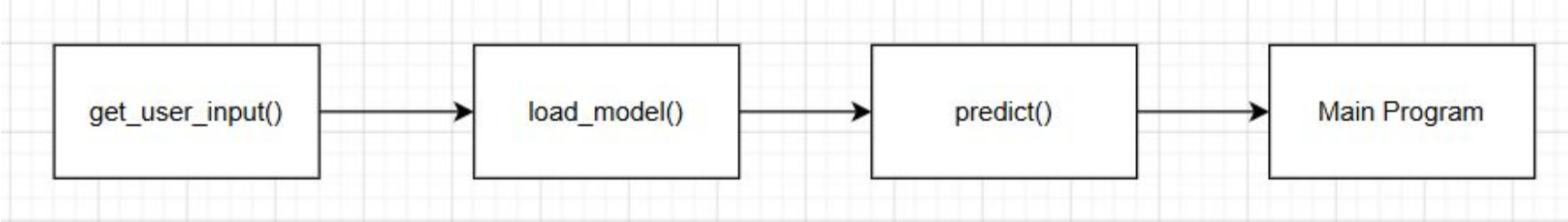
```
model_filename = 'housing_valuation_model.joblib'  # Specify the filename
joblib.dump(rf_model, model_filename)  # Save the model
print(f'Model saved to {model_filename}')
```

```
Top 5 Features and their Importance Scores:
            Feature  Importance
0          District    0.728459
2  School_Netnumber    0.199190
1             Floor    0.072351
```

Machine Learning Pipeline



Machine Learning Random Forest Inference Model : Display the predicted housing price per square




```
#add necessary library
import joblib

#Define function to get user input
def get_user_input():
    district = float(input("Enter District Number : "))
    schoolnet_no = float(input("Enter School Net Number: "))
    floor = float(input("Enter Floor "))
    input_list = [[district, schoolnet_no, floor]] #assign in data type {list}
    return input_list
```

Machine Learning Random Forest Inference Model : Display the predicted housing price per square

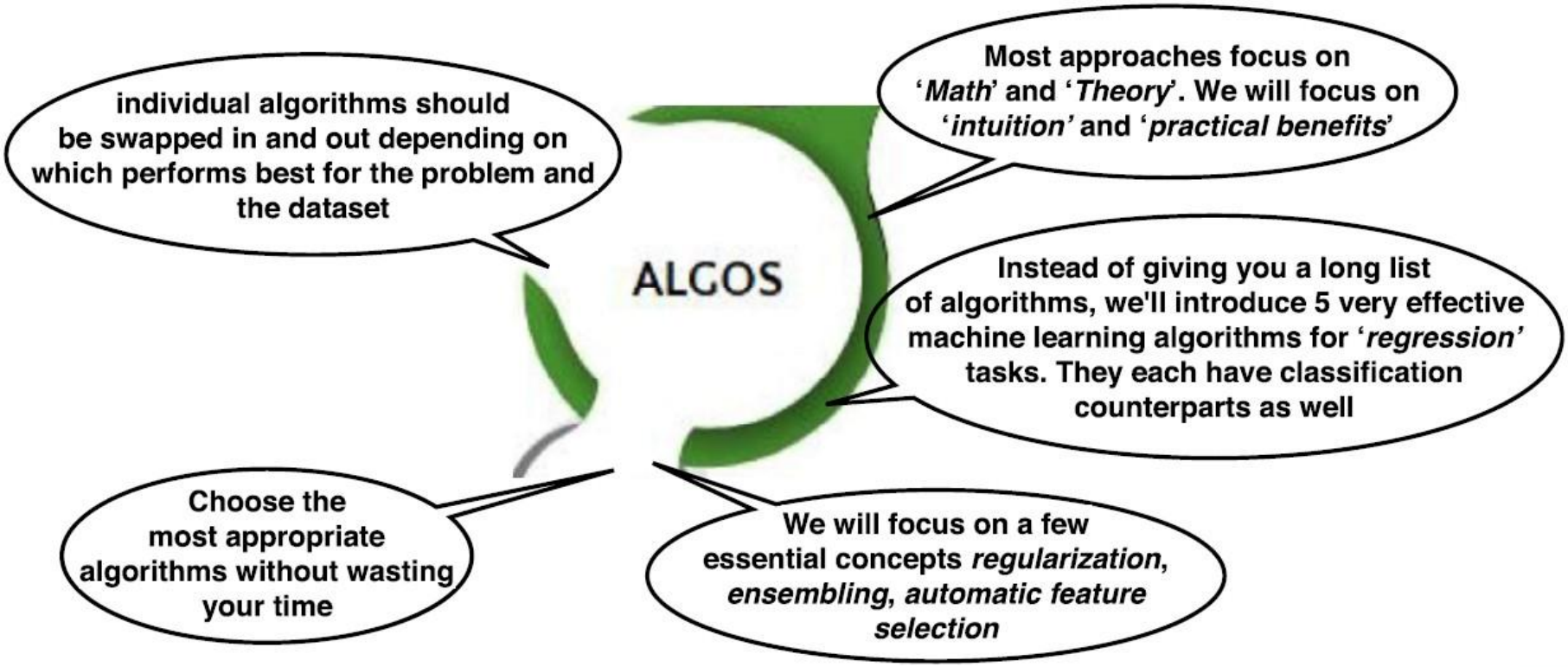
```
#Define function for loading random forest model
def load_model(model_path):
    return joblib.load(model_path)

#Define function to predict housing price per square based on 3 data input
def predict(model, input_data):
    return model.predict(input_data

#Main Program to display the predicted housing price per square
input_data = get_user_input()
model_path = 'housing_valuation_model.joblib'
model = load_model(model_path)
result = predict(model, input_data)
print(f"Predicted Hosing Price Per Square is : HKD ${round(result[0],2)}")
```

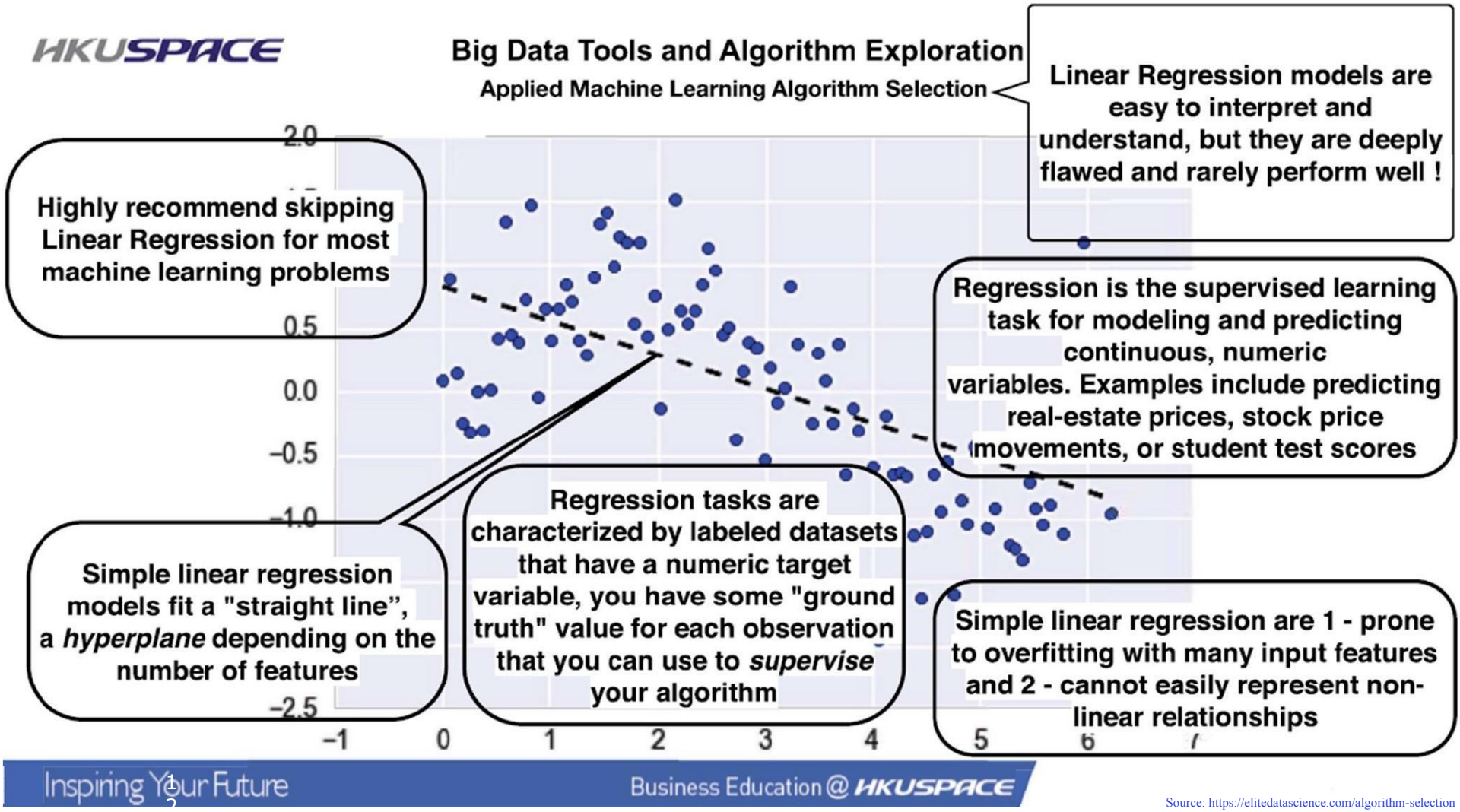
```
Enter District Number :  12
Enter School Net Number:  24
Enter Floor  3
Predicted Hosing Price Per Square is : HKD $10839.77
```

Big Data Tools and Algorithm Exploration
Applied Machine Learning Algorithm Selection Overview



Source: https://elitedatascience.com/algorithm-selection

Big Data Tools and Algorithm Exploration
Applied Machine Learning Algorithm Selection



Source: https://elitedatascience.com/algorithm-selection

Big Data Tools and Algorithm Exploration
Applied Machine Learning Algorithm Selection

Random forests train a large number of "strong" decision trees and combine their predictions through bagging

Each tree is only trained on a random subset of observations (a process called resampling)



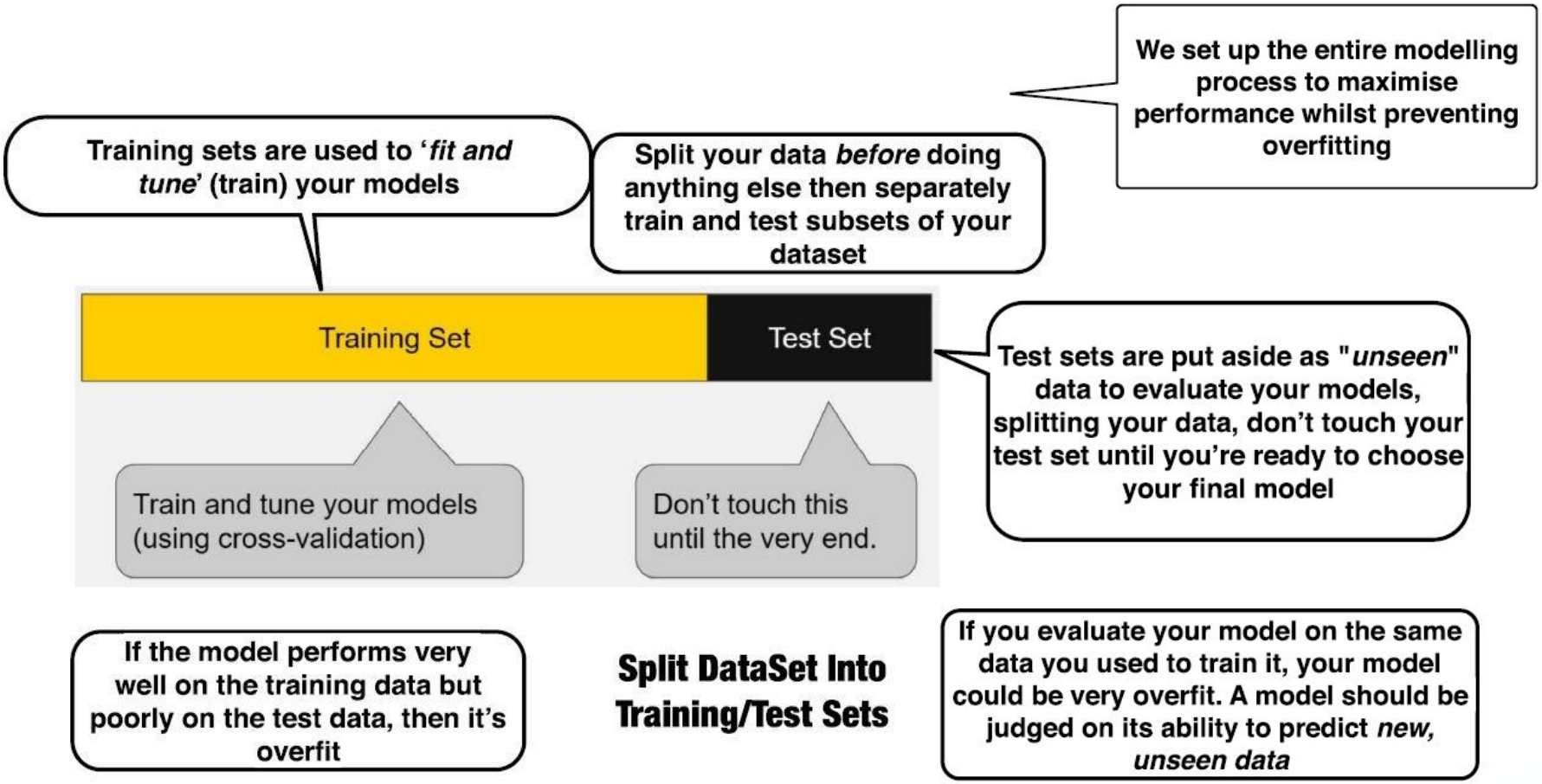
Random Forests

Each tree is only allowed to choose from a random subset of features to split on (leading to feature selection)

random forests tend to perform very well right out of the box often beat many other models that take up to weeks to develop, and are the perfect "*swiss-army-knife*" algorithm that almost always gets good results

Source: https://elitedatascience.com/algorithm-selection

Big Data Tools and Algorithm Exploration
Model Training

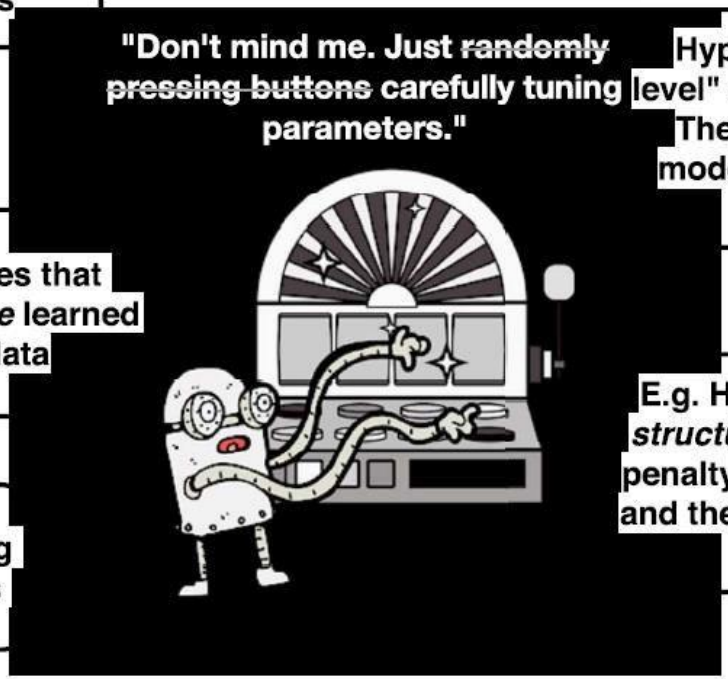


Source: https://elitedatascience.com/model-training

Big Data Tools and Algorithm Exploration
Model Training

Examples of Model parameters include regression coefficients, decision tree split locations

Hyperparameters: When we talk of tuning models, we specifically mean tuning hyperparameters



Hyperparameters express "higher-level" *structural settings* for algorithms. They are decided before fitting the model because they can't be learned from the data

Model parameters are attributes that define *individual models, and are* learned directly from the training data

E.g. Hyperparameters "higher-level" *structural settings* are strength of the penalty used in regularized regression and the number of trees to include in a random forest

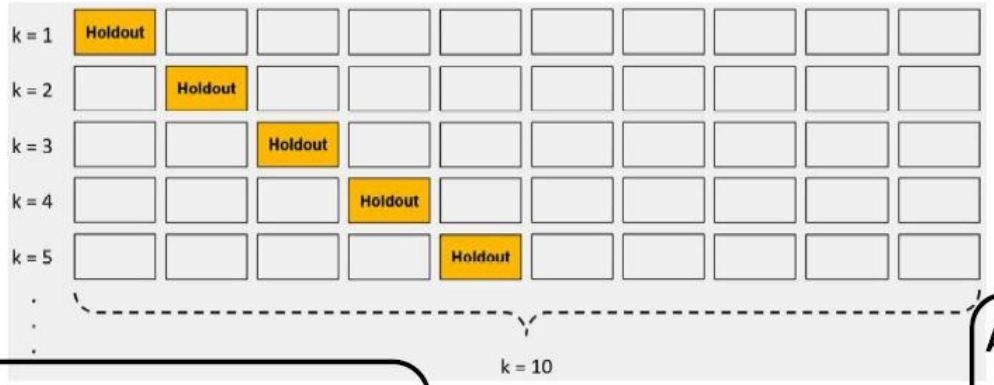
There are two types of parameters in machine learning algorithms: Model Parameters and hyperparameters

Source: https://elitedatascience.com/model-training

Big Data Tools and Algorithm Exploration
Model Training

Cross-validation is a tuning method for getting a reliable estimate of model performance using only training data

1- Split your data into 10 equal parts, or "folds", 2- Train your model on 9 folds (e.g. the first 9 folds)



3 - Evaluate it on the 1 remaining "hold-out" fold, Perform steps (2) and (3) 10 times, each time holding out a different fold

10-fold cross-validation, breaks your training data into 10 equal parts (or *folds*), essentially creating 10 miniature train/test splits

Average the performance across all 10 hold-out folds, average performance across the 10 hold-out folds is your final performance estimate, also called your *cross-validated* score

Source: https://elitedatascience.com/model-training

Big Data Tools and Algorithm Exploration
Model Training

We've split our dataset into training and test sets, and we've learned about hyperparameters and cross-validation, we're ready fit and tune our models

perform the entire cross-validation loop detailed above on each set of hyperparameter values using a high-level pseudo-code

```
For each algorithm (i.e. regularized regression, random forest, etc.):
  For each set of hyperparameter values to try:
    Perform cross-validation using the training set.
    Calculate cross-validated score.
```

At the end of this process, you will have a cross-validated score for each set of hyperparameter values... for each algorithm

Elastic-Net		
Penalty Ratio	Penalty Strength	CV-Score
75/25	0.01	0.63
75/25	0.05	0.64
75/25	0.10	**0.67**
50/50	0.01	0.62
50/50	0.05	0.63
50/50	0.10	0.66

Source: https://elitedatascience.com/model-training

Big Data Tools and Algorithm Exploration
Model Training

By now, you'll have 1 "best" model *for each algorithm* that has been tuned through cross-validation. Most importantly, you've only used the training data so far

Keep the set of hyperparameter values with best cross-validated score. Re-train the algorithm on the entire training set (without cross-validation)

For classification tasks, we recommend Area Under ROC Curve (AUROC). *Higher values are better*

Because you've saved your test set as a truly unseen dataset, you can now use it get a reliable estimate of each models' performance.There are a variety of performance metrics you could choose from

For regression tasks, we recommend Mean Squared Error (MSE) or Mean Absolute Error (MAE). *(Lower values are better)*



like the Hunger Games... each algorithm sends its own "representatives" (i.e. model trained on the best set of hyperparameter values) to the final selection

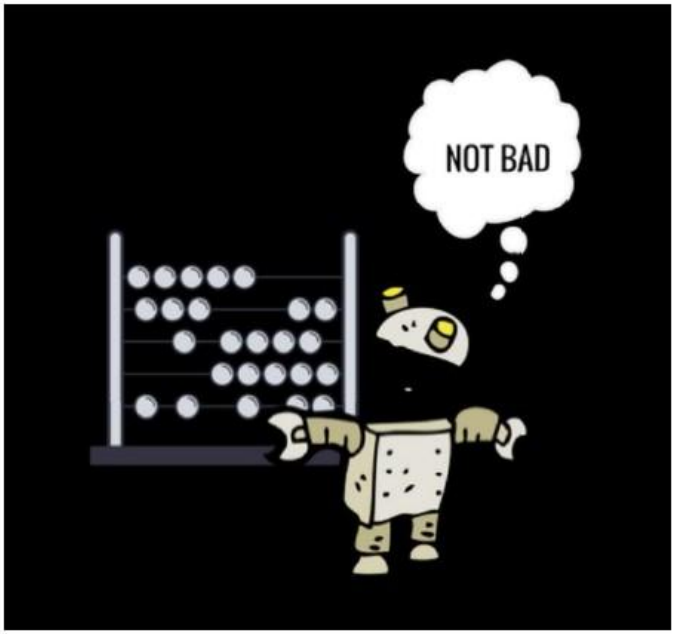
Source: https://elitedatascience.com/model-training

Big Data Tools and Algorithm Exploration
Model Training Conclusion

Pick the winning model by asking: Which model had the best performance on the test set? (performance)

Calculate performance metrics using those predictions and the "ground truth" target variable from the test set

For each of your models, make predictions on your test set



Pick the winning model by asking: Does it perform well across various performance metrics? (robustness)

Pick the winning model by asking: Did it also have (one of) the best cross-validated scores from the training set? (consistency)

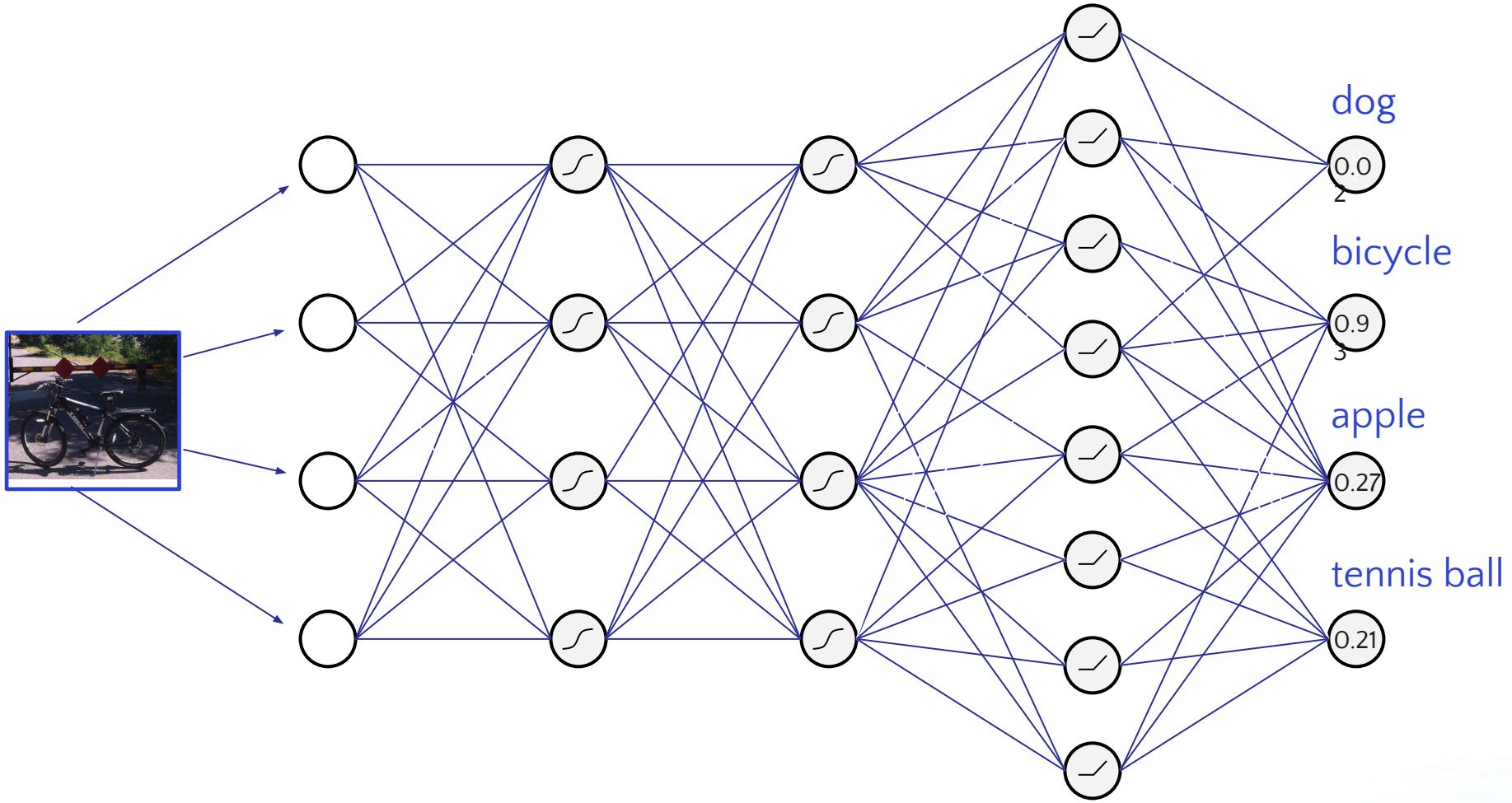
Pick the winning model by asking: Does it solve the original business problem? (win condition)

Source: https://elitedatascience.com/model-training

Modelling Techniques

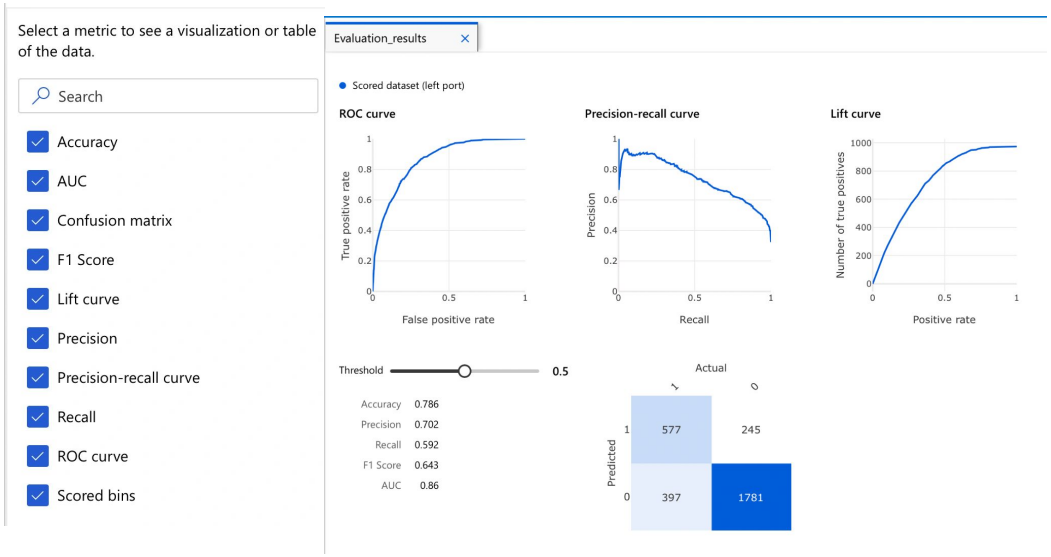
NLP	Machine Learning	Deep Learning
Word2vec	Linear Regression	Deep Neural Network - DNN
Fasttext	Neural Network - NN	Convolutional Neural Network - CNN
Bert	SVM	Region CNN - R-CNN
elmo	Xgboost	Long-short term memory - LSTM
BoW	Random forest	Recurrent neural networks - RNN
Flair	Artificial Neural Network - ANN	Autoregressive integrated moving average - arima

Trained Convolutional NN



Machine Learning Model Training Visualization

Machine learning model training visualization involves representing various aspects of the training process and the resulting model through graphical or interactive means. This aims to enhance understanding, facilitate debugging, and improve interpretability for both technical and non-technical stakeholders



Score bin ↓	Positive exam...	Negative exam...	Fraction above thres...	Accura...	F1 Score	Precisi...	Recall	Negative precis...	Negative recall	Cumulative AUC
(0.900,1.000]	95	11	0.035	0.703	0.176	0.896	0.098	0.696	0.995	0.000
(0.800,0.900]	149	21	0.092	0.746	0.390	0.884	0.251	0.732	0.984	0.002

https://www.analyticsvidhya.com/blog/2021/09/a-comprehensive-guide-on-using-azure-machine-learning/

Machine Learning Model Training Visualization

Key Concepts in Machine Learning Model Training Visualization:

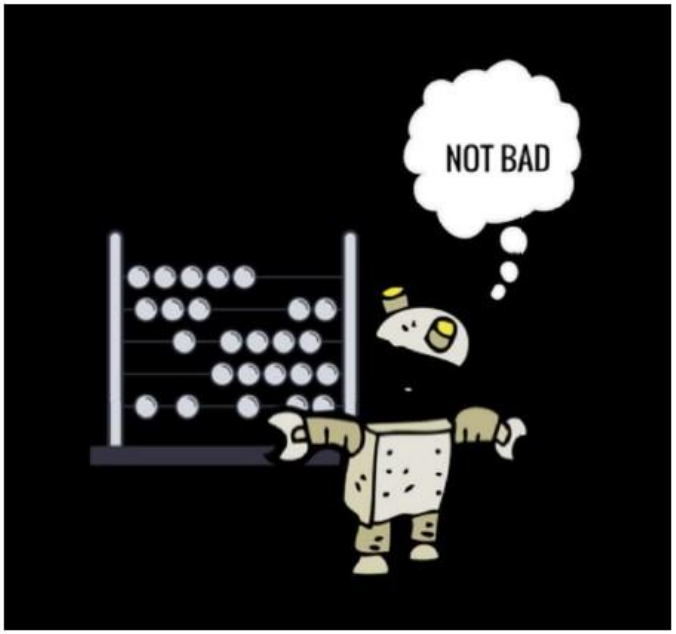
- Monitoring Training Progress:
  - Loss Curves: Plots showing the training and validation loss over epochs or iterations, indicating convergence and potential overfitting/underfitting.
    - Interpretation: Both decreasing and converging, Training loss decreasing, validation loss increasing.
  - Metric Curves: Visualizations of performance metrics (e.g., accuracy, precision, recall, F1-score) on training and validation sets, providing insight into model effectiveness.
    - Interpretation: High and stable validation metric, Significant gap between training and validation metrics, Low metrics on both sets
  - Learning Rate Schedules: Graphs illustrating how the learning rate changes during training, especially with adaptive optimizers or learning rate schedulers
    - Interpretation: Visualizing the schedule, Impact on convergence, Tuning

Machine Learning Model Training Conclusion

Pick the winning model by asking: Which model had the best performance on the test set? (performance)

Calculate performance metrics using those predictions and the "ground truth" target variable from the test set

For each of your models, make predictions on your test set



Pick the winning model by asking: Does it perform well across various performance metrics? (robustness)

Pick the winning model by asking: Did it also have (one of) the best cross-validated scores from the training set? (consistency)

Pick the winning model by asking: Does it solve the original business problem? (win condition)

Source: https://elitedatascience.com/model-training

Financial Forecasting with Machine Learning using Python

Key Features of ML in Financial Forecasting

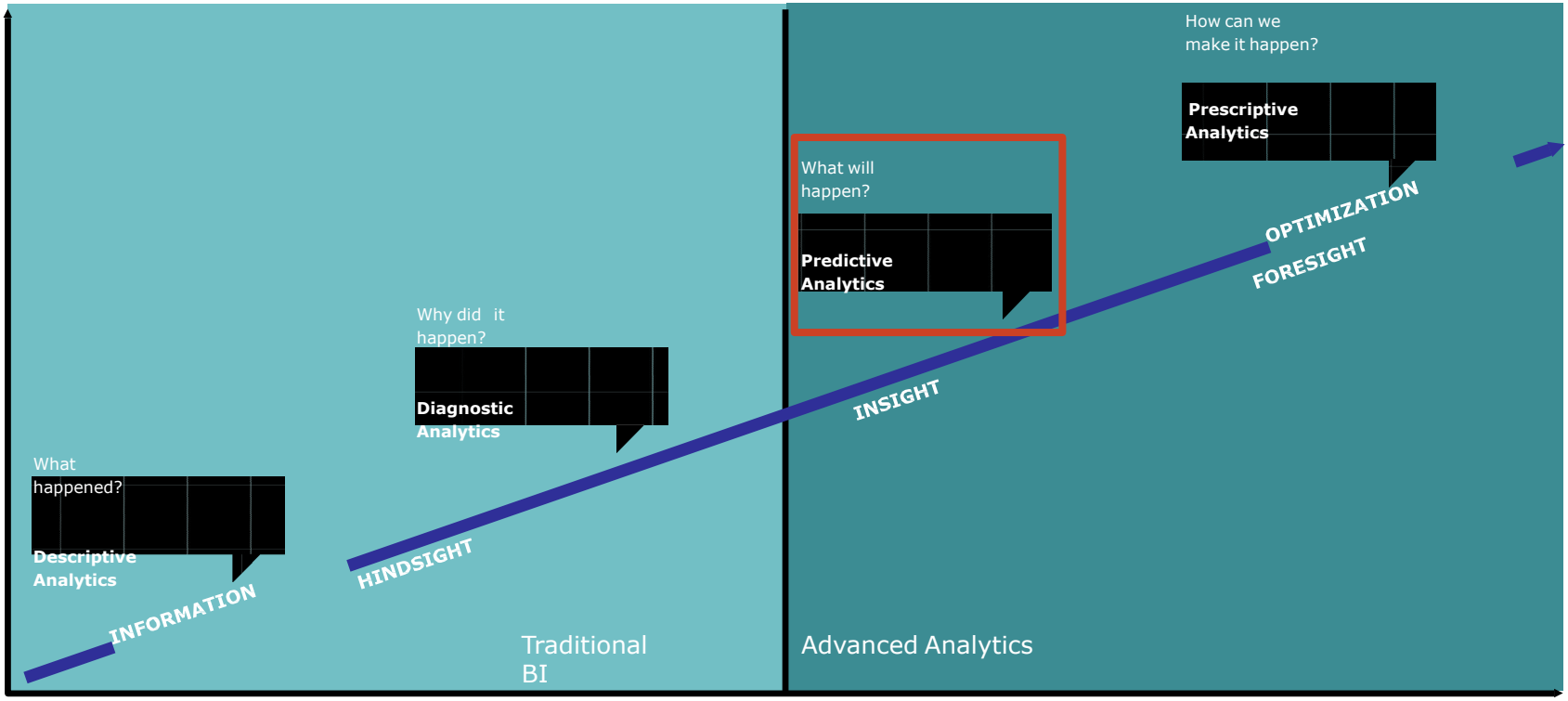
- Pattern recognition of non-linear relationships in prices, volumes, and macro indicators that traditional linear models miss
- Support for multiple forecast targets: prices, returns, volatility, default probability, credit risk, and cash-flow or revenue trajectories
- Integration of unstructured data (text, sometimes images) so that sentiment and qualitative information can influence quantitative forecasts (https://www.netsuite.com.hk/portal/hk/resource/articles/financial-management/financial-forecast-machine-learning.shtml)

Core Machine Learning Technologies Used

- Time-series models, including ARIMA and LSTM, to capture temporal dependencies in financial data
- Tree-based methods such as decision trees, random forests, and gradient boosting (XGBoost, LightGBM, CatBoost) for credit scoring, risk forecasting, and tabular financial data.

Tutorial - https://medium.com/@aditib259/predicting-stock-prices-using-lstms-time-series-forecasting-a-step-by-step-guide-a70ebb04bbb8

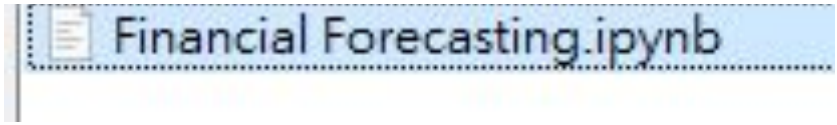
Financial Forecasting with Machine Learning using Python



Extra Lab - Financial Forecasting with Machine Learning using Python (Numpy, Pandas, Matplotlib and Scikit-learn)

Open Google Colab

Step 1. Import the required libraries:



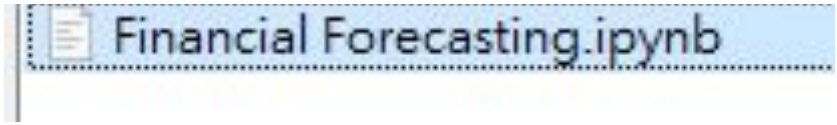

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing  import MinMaxScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error
from keras.models  import Sequential
from keras.layers import Dense, LSTM
import requests
```

Step 2: Load the finance data from https://www.alphavantage.co:

```
api_key = 'demo'
symbol = 'IBM'
url =
f'https://www.alphavantage.co/query?function=TIME_SERIES_DAILY_ADJUSTED&symbol={symbol}&outputsize=full&a
pikey={api_key}'
response = requests.get(url)
data = response.json()
df = pd.DataFrame(data['Time Series (Daily)']).transpose()
df.index = pd.to_datetime(df.index)
df = df.sort_index()
```

Financial Forecasting with Machine Learning using Python (Numpy, Pandas, Matplotlib and Scikit-learn)

Step 3 – Preprocess the data:




```
# Extract the closing prices
y = df['4. close'].values.astype(float)
# Normalize the closing prices
scaler = MinMaxScaler(feature_range=(0, 1))
y = scaler.fit_transform(y.reshape(-1, 1))
# Create the feature matrix
X = []
for i in range(60, len(df)):
      X.append(y[i-60:i, 0])
X = np.array(X)# Split the data into training and validation sets
X_train, X_val, y_train, y_val = train_test_split(X, y[60:], test_size=0.2, shuffle=False)
```

Step 4 – Define the model:

```
model = Sequential()
model.add(LSTM(units=50, return_sequences=True, input_shape=(X_train.shape[1], 1)))
model.add(LSTM(units=50))
model.add(Dense(units=1))
model.compile(optimizer='adam', loss='mean_squared_error')
```

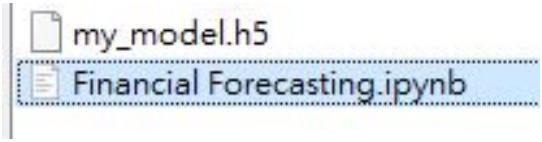
Financial Forecasting with Machine Learning using Python (Numpy, Pandas, Matplotlib and Scikit-learn)

Step 5 – Train the model with 100 steps:

```
model.fit(X_train, y_train, epochs=100, batch_size=32, validation_data=(X_val, y_val))
```

Step 6 – Save the model in h5 format:

```
model.save('my_model.h5')  # Save the model in HDF5 format
```

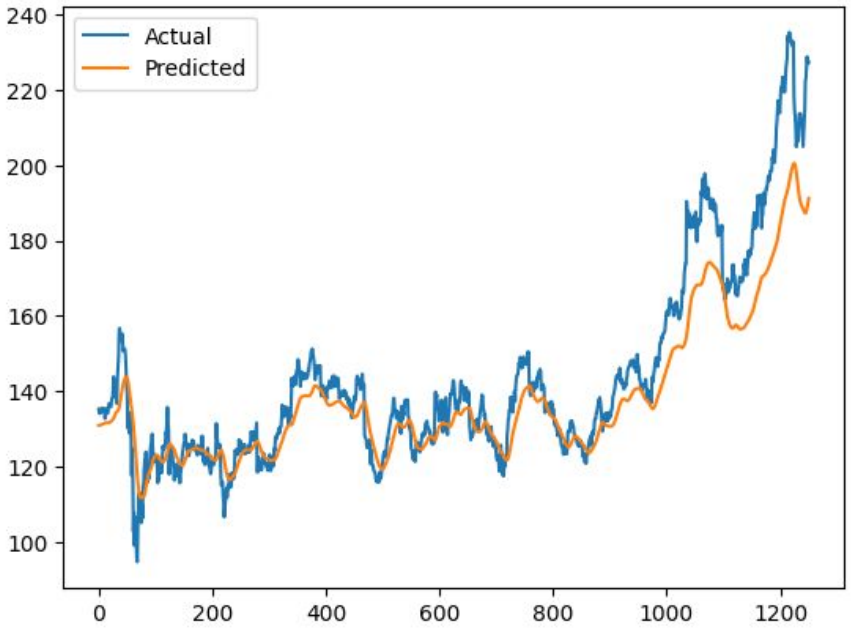


Step 7 – Evaluate the model:

```
# Evaluate the model on the validation set
y_pred = model.predict(X_val)
rmse = np.sqrt(mean_squared_error(y_val, y_pred))
print('Root Mean Squared Error:', rmse)
```

Step 8 – Visualize the results:

```
# Visualize the results
y_pred = scaler.inverse_transform(y_pred)
y_val = scaler.inverse_transform(y_val)
plt.plot(y_val, label='Actual')
plt.plot(y_pred, label='Predicted')
plt.legend()
plt.show()
```



Extra Lab - Google Sheet Content Update

1. Open Anaconda Prompt, `enter "pip install gspread google-auth google-api-python-client"` to install google library
2. Open Jupyter notebook and type following python code

```
import gspread
from google.oauth2.service_account import Credentials
from googleapiclient.discovery import build

def update_values(sheet_id, range_name, value_input_option, values):
    SCOPES = ['https://www.googleapis.com/auth/spreadsheets']
    SERVICES_ACCOUNT_FILE = "graceful-disk-484104-a2-152ebd9c692e.json"

    credentials = Credentials.from_service_account_file(SERVICES_ACCOUNT_FILE,
                                scopes=SCOPES)
    service = build('sheets', 'v4', credentials=credentials)

    body = {"values": values}
```

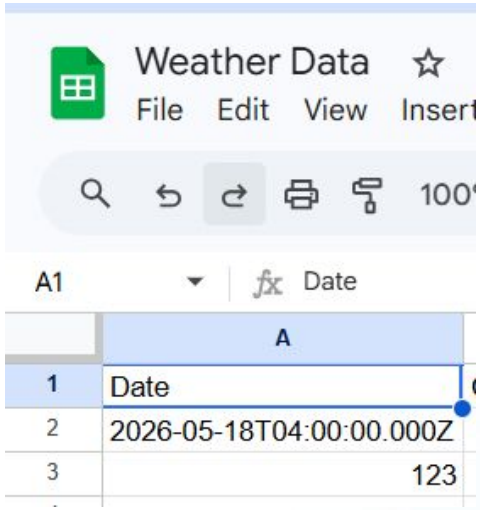
Google Sheet Content Update

```
    # Update the range to just the starting cell for appending
     result = service.spreadsheets().values().update(
         spreadsheetId=sheet_id,
         range=range_name,  # Change range here
         valueInputOption=value_input_option,
         body=body
     ).execute()

     print(f"{result.get('updatedCells')} cells updated.")
     return result

sheet_id = '1mqrxumbU1Si5aGmkg2jpffu7zn9uWxFfvXX1Wnuwkas'
range_name = "Sheet1!A3"
value_input_option = 'USER_ENTERED'
values = [['123']]

update_values(sheet_id, range_name, value_input_option, values)
```



Visualization using Programming

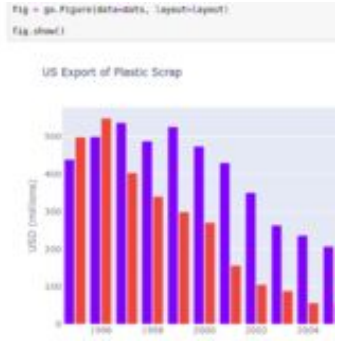
- Python
  - **matplotlib**
  - seaborn
  - **plotly(mainly)**
  - pylab
- R
  - graphics
  - ggplot2

Plotly: Visualization with Python

Plotly's Python graphing library makes interactive, publication-quality graphs.

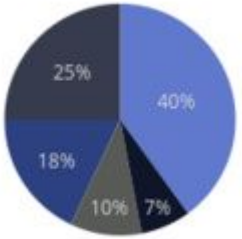
Examples of how to make line plots, scatter plots, area charts, bar charts, error bars, box plots, histograms, heatmaps, subplots, multiple-axes, polar charts, and bubble charts.
Plotly.py is free and open source..

Basic Chart: https://plotly.com/python/basic-charts/



Features:

1. Create publication quality plots.
2. Make interactive figures that can zoom, pan, update.
3. Customize visual style and layout.
4. Export to many file formats.
5. Embed in JupyterLab and Graphical User Interfaces.
6. Use a rich array of third-party packages built on Matplotlib.



https://plotly.com/python/

Plotly: Visualization with Python

The main modules in Plotly are:

- plotly.plotly
- **plotly.graph.objects**
- plotly.express

- plotly.plotly acts as the interface between the local machine and Plotly. It contains functions that require a response from Plotly's server.

- **plotly.graph_objects module contains the objects (Figure, layout, data, and the definition of the plots like scatter plot, line chart) that are responsible for creating the plots.**

- plotly.express module can create the entire Figure at once. It uses the graph_objects internally and returns the graph_objects.Figure instance.

Plotly Exercise 1

```
#using plotly, display histgram chart and download chart
import numpy as np
import plotly.graph_objects as go

# Generate sample data for the sine wave
x = np.linspace(0, 10, 100)  # 100 points from 0 to 10
y = np.sin(x)                # y is the sine of x

# Generate sample data for the histogram
data = np.random.normal(loc=0, scale=1, size=1000)
random values from a normal distribution

# Create a plotly figure
fig = go.Figure()

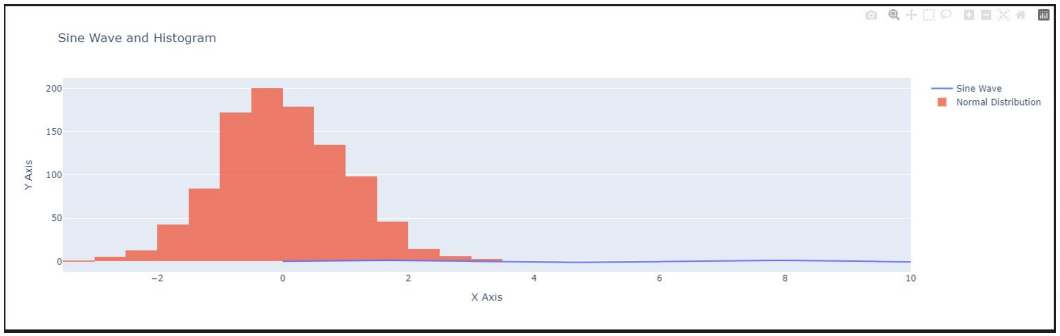
# Add a line plot for the sine wave
fig.add_trace(go.Scatter(x=x, y=y, mode='lines',
    name='Sine Wave'))

# Add a histogram
fig.add_trace(go.Histogram(x=data, name='Normal
Distribution', opacity=0.75, nbinsx=30))

# Customize layout
fig.update_layout(title='Sine Wave and Histogram',
                  xaxis_title='X Axis',
                  yaxis_title='Y Axis',
                  barmode='overlay')  # Overlay histograms

# Show the plot
fig.show()
# Show the plot
fig.show()

# Save the figure as a PNG filep
io.write_image(fig, 'sine_wave_histogram.png')
```



Plotly Example - Line Chart

https://plotly.com/python/line-charts/

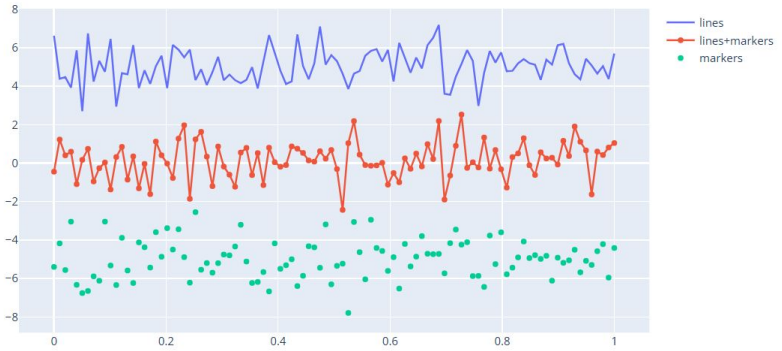
```
import plotly.graph_objects as go

# Create random data with numpy
import numpy as np
np.random.seed(1)

N = 100
random_x = np.linspace(0, 1, N)
random_y0 = np.random.randn(N) + 5
random_y1 = np.random.randn(N)
random_y2 = np.random.randn(N) - 5

# Create traces
fig = go.Figure()
fig.add_trace(go.Scatter(x=random_x, y=random_y0,
                         mode='lines',
                         name='lines'))
fig.add_trace(go.Scatter(x=random_x, y=random_y1,
                         mode='lines+markers',
                         name='lines+markers'))
fig.add_trace(go.Scatter(x=random_x, y=random_y2,
                         mode='markers', name='markers'))

fig.show()
```



Plotly Example - Bar Chart

https://plotly.com/python/bar-charts/

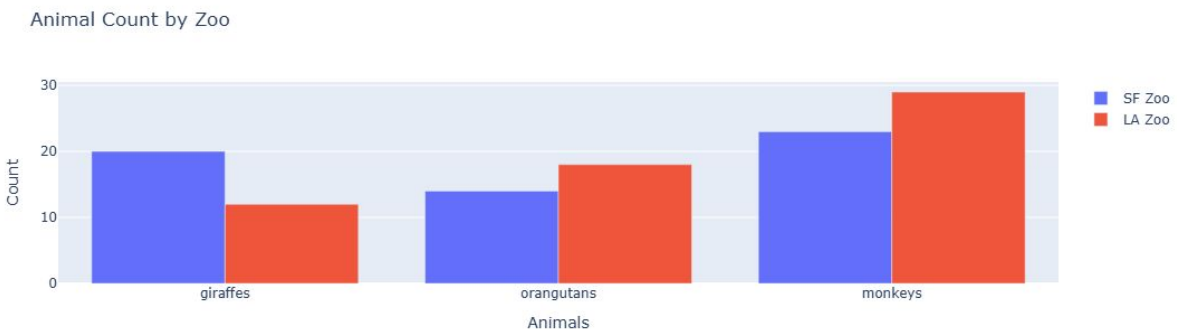
```
import plotly.graph_objects as go

animals = ['giraffes', 'orangutans', 'monkeys']

fig = go.Figure(data=[
    go.Bar(name='SF Zoo', x=animals, y=[20, 14, 23]),
    go.Bar(name='LA Zoo', x=animals, y=[12, 18, 29])
])

# Change the bar mode
fig.update_layout(
    barmode='group',
    title='Animal Count by Zoo',
    xaxis_title='Animals',
    yaxis_title='Count'
)

fig.show()
```



Plotly Example - Scatter Chart

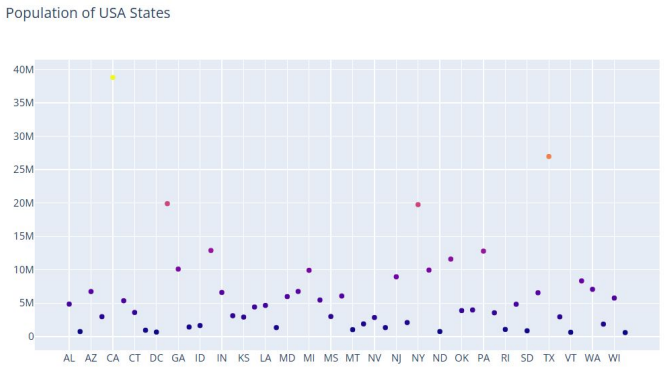
https://plotly.com/python/line-and-scatter/

```
import plotly.graph_objects as go
import pandas as pd

data= pd.read_csv("https://raw.githubusercontent.com/plotly/datasets/master/2014_usa_states.csv")

fig = go.Figure(data=go.Scatter(x=data['Postal'],
                                y=data['Population'],
                                mode='markers',
                                marker_color=data['Population'],
                                text=data['State'])) # hover text goes here

fig.update_layout(title=dict(text='Population of USA States'))
fig.show()
```



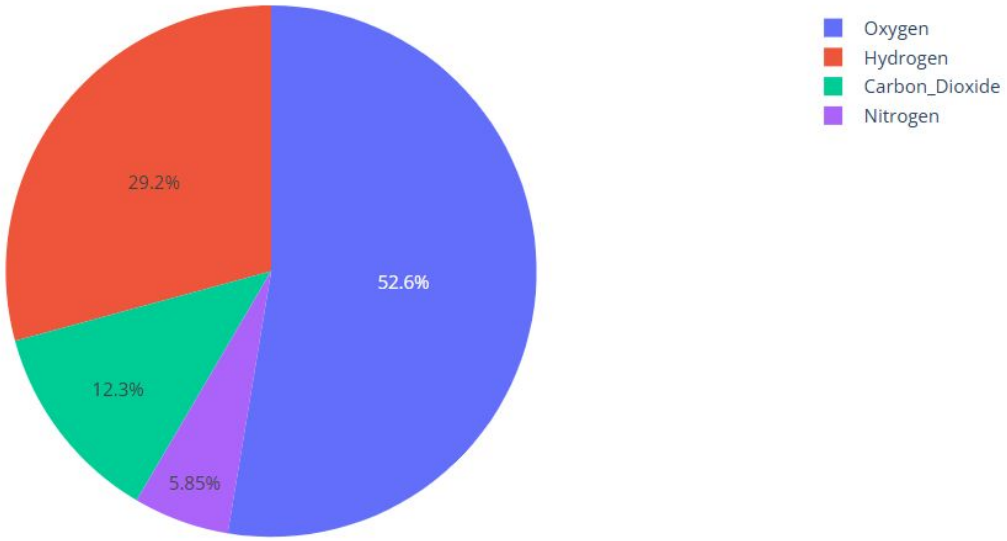
Plotly Example - Pie Chart

https://plotly.com/python/pie-charts/

```
import plotly.graph_objects as go

labels = ['Oxygen','Hydrogen','Carbon_Dioxide','Nitrogen']
values = [4500, 2500, 1053, 500]

fig = go.Figure(data=[go.Pie(labels=labels, values=values)])
fig.show()
```



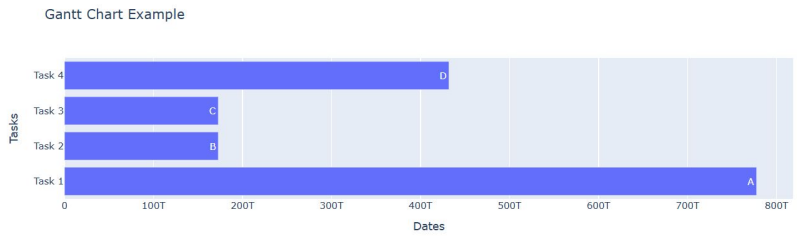
Plotly Example - Gantt Chart

https://plotly.com/python/gantt/

```
Import plotly.graph_objects as go
import pandas as pd

# Sample data for Gantt Chart
data = {
    'Task': ['Task 1', 'Task 2', 'Task 3', 'Task 4'],
    'Start': ['2023-01-01', '2023-01-13', '2023-01-18', '2023-01-25'],
    'Finish': ['2023-01-10', '2023-01-15', '2023-01-20', '2023-01-30'],
    'Resource': ['A', 'B', 'C', 'D']
}

# Convert to DataFrame
df = pd.DataFrame(data)
```




```
# Create Gantt chart
fig = go.Figure(data=[
    go.Bar(
        y=df['Task'],
        x=pd.to_datetime(df['Finish']) - pd.to_datetime(df['Start']),
        base=pd.to_datetime(df['Start']),
        orientation='h',
        text=df['Resource'],
        hoverinfo='text'
    )
])

# Update layout
fig.update_layout(
    title='Gantt Chart Example',
    yaxis_title='Tasks',
    xaxis_title='Dates',
    barmode='stack',
)

# Show the figure
fig.show()
```

Data Type Example

	Data Type	Example	Mutable?	Ordered?	Use Case	Example Code
1	**list** list	[1, 2, 3, 2]	✓ Yes	✓ Yes	General ordered collection; allows duplicates	my_list = [1, 2, 3]
2	**tuple** tuple	(1, 2, 3)	✗ No	✓ Yes	Immutable ordered collection; hashable	my_tuple = (1, 2, 3)
3	**set** set	{1, 2, 3}	✓ Yes	No No	Unique items; fast membership tests	my_set = {1, 2, 3}
4	**dict** dict	~."a": 1, "b": 2}	✓ Yes	Yes (3.7+)	Key-value pairs; fast lookups	my_dict = {"key": "value")
5	**pandas.Series** Series	Index → Value	✓ Yes	✓ Yes	1D labeled array; like a column	pd.Series([10,20], index{"a","b")
6	**pandas.DataFrame** pandas	Table of rows/columns	✓ Yes	✓ Yes	2D tabular data; spreedsheet-like	pd.DataFrame("col": [1,2]))
6	**numpy.nndraff** numpy	array([1, 2, 3])	✓ Yes	✓ Yes	Numerical array; vectorized ops, fast math	np.array([1,2,3])
8	**json** json	~."name": "Alice", "age": 25}	No* (text)	No* (dict order)	Data interchange format; string/text	json.loads("[k": 1")

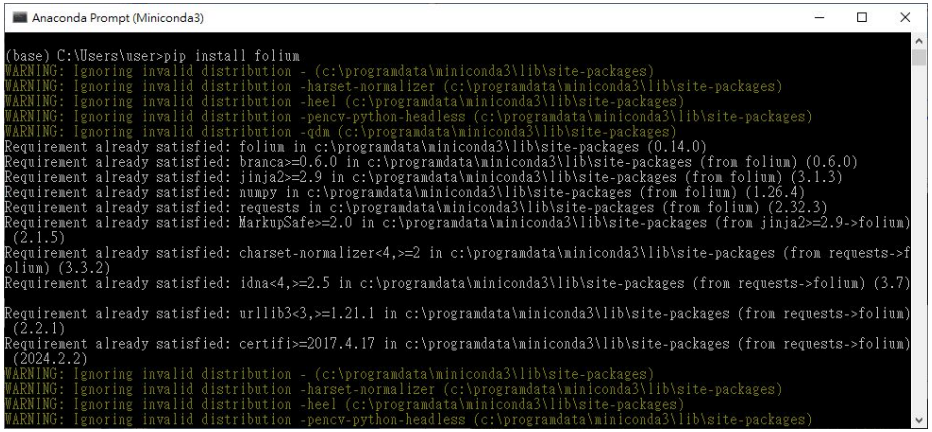
Tip: Use *list/set/dict* for general Python. Use NumPy for numerical work. Use pandas for tabular/data analysis. JSON is for storage/transport, not a native Python type.

Python Google Map Tutorial

1. Open tools “Anaconda Prompt” which will direct you to the command prompt.



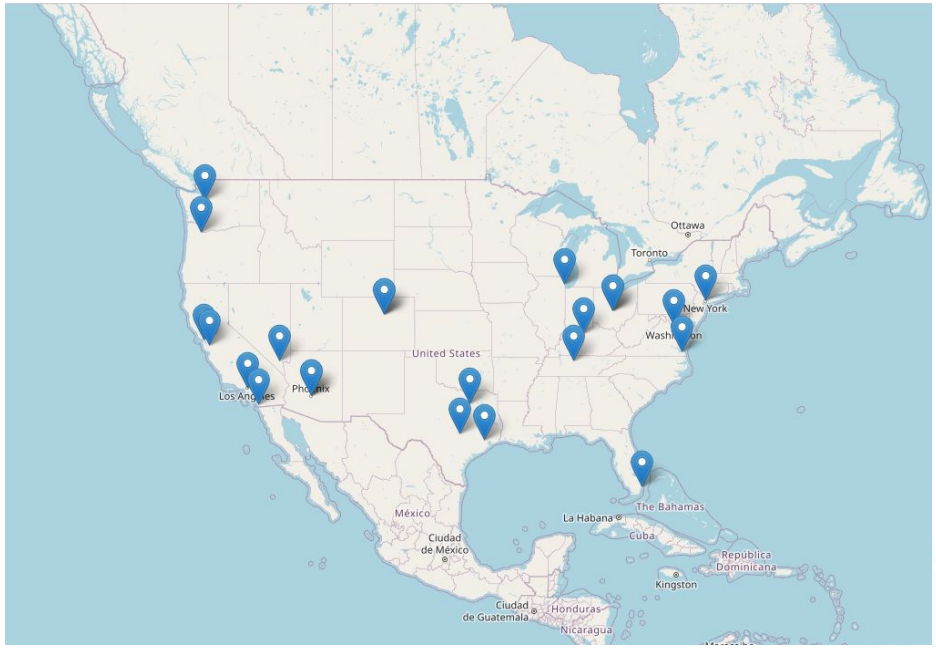
2. Install python package requests by entering “pip install folium”



Python Google Map Tutorial

```
import folium

# List of 23 coordinates (latitude, longitude)
coordinates = [
    (37.7749, -122.4194),  # San Francisco, CA
    (34.0522, -118.2437),  # Los Angeles, CA
    (40.7128, -74.0060),   # New York, NY
    (41.8781, -87.6298),    # Chicago, IL
    (29.7604, -95.3698),    # Houston, TX
    (33.4484, -112.0740),   # Phoenix, AZ
    (39.7392, -104.9903),   # Denver, CO
    (47.6062, -122.3321),   # Seattle, WA
    (38.9072, -77.0369),    # Washington, D.C.
    (39.9612, -82.9988),    # Columbus, OH
    (30.2672, -97.7431),    # Austin, TX
    (32.7767, -96.7970),    # Dallas, TX
    (37.3382, -121.8863),   # San Jose, CA
    (39.7392, -104.9903),   # Denver, CO
    (36.1699, -115.1398),    # Las Vegas, NV
    (25.7617, -80.1918),    # Miami, FL
    (38.2527, -85.7585),    # Louisville, KY
    (39.9612, -82.9988),    # Columbus, OH
    (36.8508, -76.2859),    # Norfolk, VA
    (33.4484, -112.0740),   # Phoenix, AZ
    (45.5155, -122.6793),   # Portland, OR
    (36.1627, -86.7816),    # Nashville, TN
    (32.7157, -117.1611),   # San Diego, CA
]
```



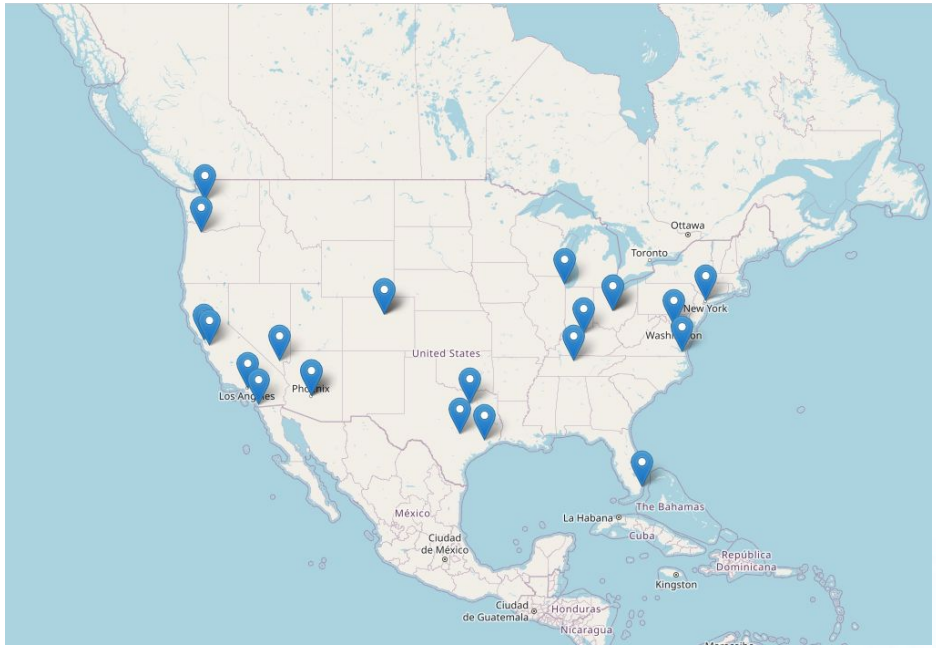
Python Google Map Tutorial

```
# Create a map centered at the first coordinate
my_map = folium.Map(location=coordinates[0],
    zoom_start=4)

# Add markers for each coordinate
for lat, lon in coordinates:
    folium.Marker(location=(lat,
    lon)).add_to(my_map)

# Save the map to an HTML file
my_map.save("my_map.html")

# If you're using Jupyter Notebook, you can
    display the map inline
my_map
```



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THANK YOU

